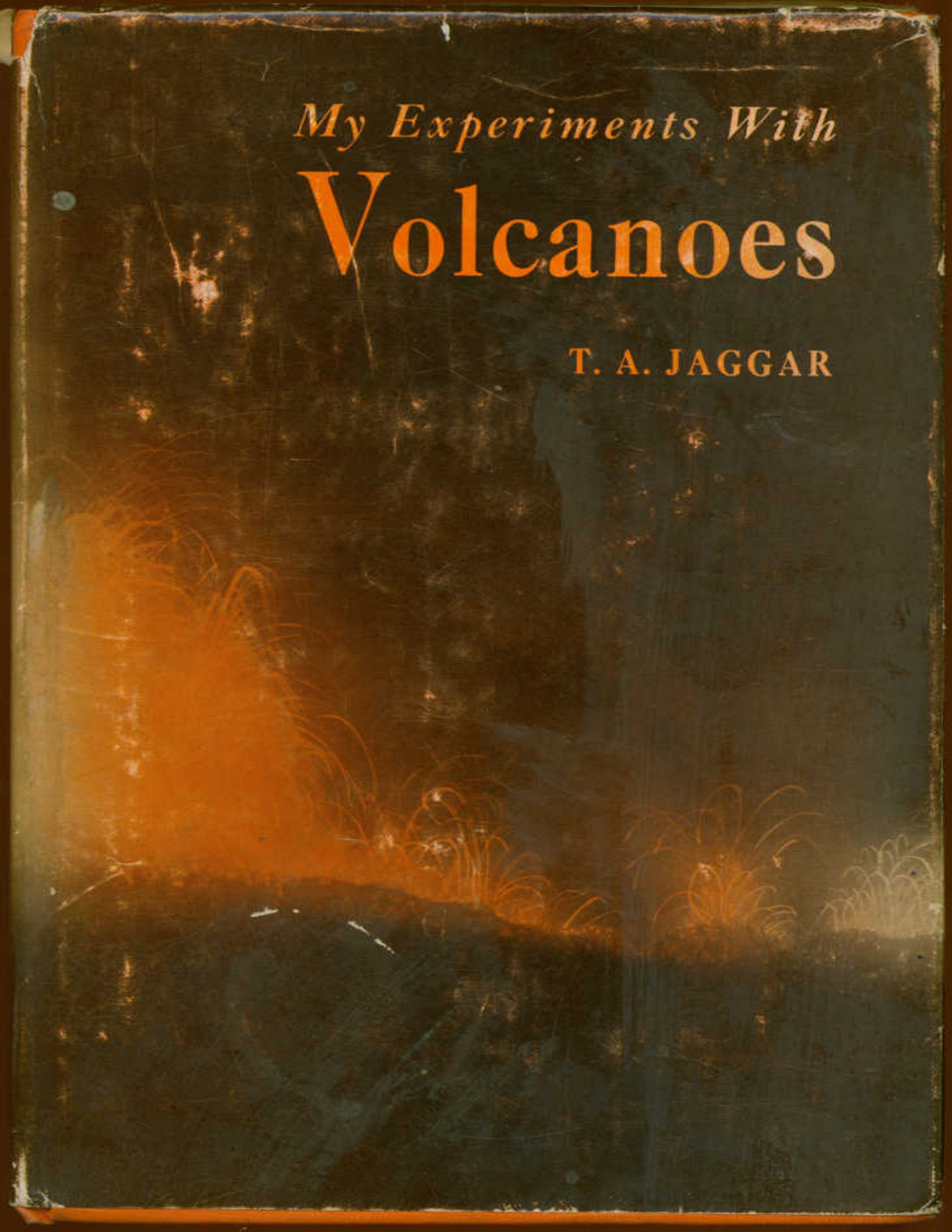


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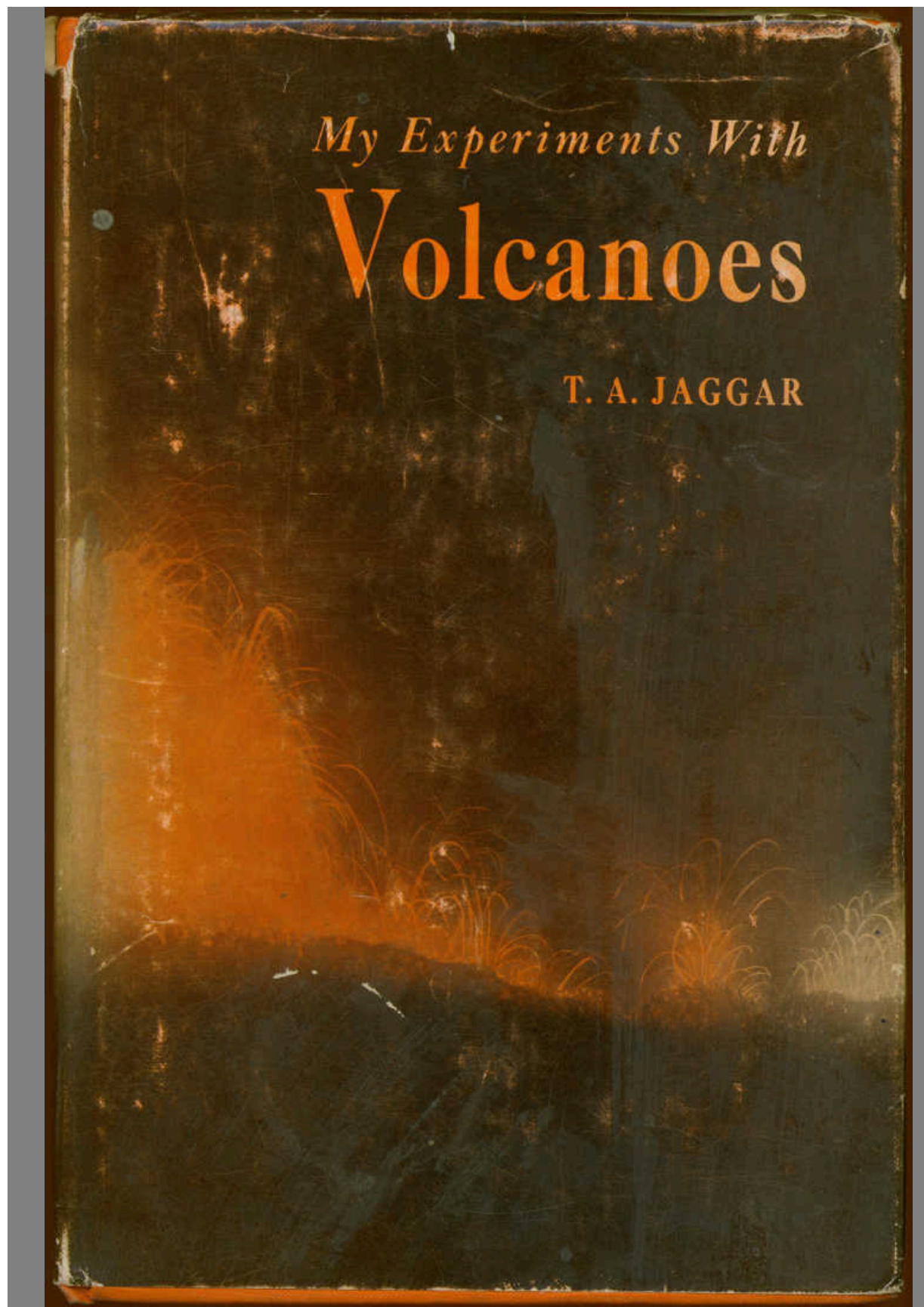
Volcanoes

T. A. JAGGAR



My Experiments With
Volcanoes

T. A. JAGGAR



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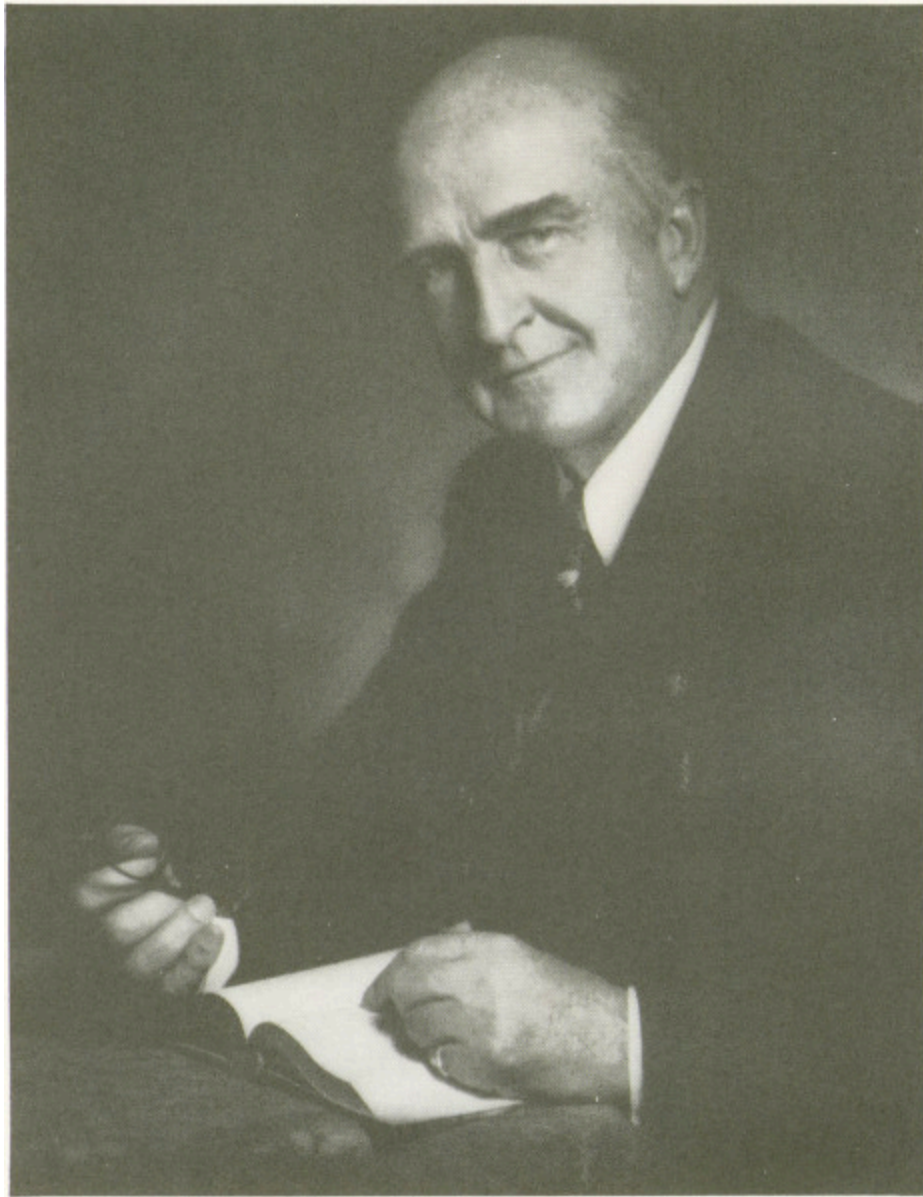
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EXPERIMENTS WITH VOLCANOES ***

My Experiments With
Volcanoes



THOMAS A. JAGGAR

January 24, 1871—January 17, 1953

My Experiments With Volcanoes

THOMAS A. JAGGAR

*“Through faith we understand
That the worlds were formed
By the word of God,
So that things which are seen
Were not made of things which do appear.”*

MCMLVI
HAWAIIAN VOLCANO RESEARCH ASSOCIATION
HONOLULU

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Thomas Augustus Jaggar, Jr.

January 24, 1871

January 17, 1953

It is the wish of the members of the Hawaiian Volcano Research Association to share with others the experiences they have enjoyed in their association with a truly great man.

On October 5, 1911, through the efforts of Thomas Augustus Jaggar, Jr., the Hawaiian Volcano Research Association was organized to assist in the support of the newly created Hawaiian Volcano Observatory at Kilauea, Hawaii. Accepting Dr. Jaggar's sincere belief that a systematic and continuous study of volcanoes would result in the protection of life and property, the motto the Hawaiian Volcano Research Association adopted was "Ne plus haustae aut obrutae urbes."

Dr. Jaggar arrived in Hawaii to take up his work at the Observatory on January 17, 1912—exactly forty-one years before the day of his death on January 17, 1953.

Dr. Jaggar spent the last years of his life writing the history of his sixty years of intensive, rugged, and hazardous scientific achievements. During many of these years, and up to the completion of his life's history, it has been well stated that one of his most valuable co-workers was his wife, Isabel, who shared with him the disappointments, the joys of discovery, and much of the physical work. It is the privilege of the officers, directors and members of the Hawaiian Volcano Research Association to present in book form this story of Dr. Jaggar's life.

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PREFACE

This, my latest book, is another experiment. After sixty years of volcanoes I have learned reversal of preconceived notions. Gradually I have learned a totally different approach.

Shaler of Harvard was my inspirer, worker in the wonders of swamp and ice and sea beaches. He set me to work and turned me loose; among books and storm waves and men; especially among men, young men, ever reaping something new. When I chose volcanoes for my field Shaler said, "You have certainly selected the hardest." It was a missionary field, for in it people were being killed. But the products of internal earth fluids, lava sea bottom, and vast Canadian ancient meltings, seemed to promise real natural history. Volcanoes squirt up the very ancient stuff of the solar system. Therein, I knew, must be something for future discovery. The investigation of it was a clear field, if action was the goal.

My field education in geology was by Hague, the friend of Archibald Geikie. By Emmons, skilled in ore deposits, and like Hague, trained by Clarence King. By Bailey Willis, son of a poet, a superlative draftsman and field man, and a brilliant experimenter. I went into the American West with these men.

But this story of a volcano experimenter's life would have reached nowhere without Frank Alvord Perret, whom I first met on the slope of Vesuvius in 1906. I knew at once that he was the world's greatest volcanologist. His skill was taking pictures. Mine was making experiments. We agreed that these two skills in action would accomplish what theories never could approach.

Perret was an inventor. He was an artist. He was a poet. He was a lover of little children, and a worshiper of the music of the stars. Always in delicate health, he circled the world. I was with him on Sakurajima, on Kilauea, and on Montserrat. We did not agree. He had a vast love of the romantic and bizarre. I was always a sceptic. But I thank Heaven that his posthumous and nobly illustrated book reached magnificent publication. His other books set a standard for all time for what the field science of volcanoes shall be.

Perret and his camera were my models. He gave me all of his pictures to use as I chose, and he and Tempest Anderson taught me volcano photography. The latter, a Yorkshireman, was a British geographer and we met on many volcanoes.

The purpose of this book is to tell what one man saw. I was actuated by the will to learn. I wanted to copy ripplemarks on the bottom of the sea, to understand what force pushed up Harney Peak as coarse granite in the Black Hills, and to imitate Yellowstone geysers spouting rhythmically. I wanted to know how cracks made the Cascade Mountains pile up in a line.

Finally, I studied the San Francisco earthquake rift, sliding open parallel to the shore for hundreds of miles. How thick was the crust of the globe? Then I was called to Hawaii, islands on a ridge 1,700 miles long with volcanoes at one end, coral atolls at the other. And I started a volcano experiment station at a very lucky time. Volcanoes proved surprisingly amenable to experiment.

Forty years of this lead far away from Lyell's geology—the geology of uniform processes past and present—and from brachiopods and trilobites. It lead to the ancestor of volcanoes. It lead to ancestral gas. It lead back 10 billion years. A lava splash might be a live souvenir of that age. More than anything else, this belief pointed our instruments down, to the inside of the globe.

Six decades of a man's life. Decades of geology, exploration, foundation, outspreading, prediction, and fruition. The fact of fruition makes the telling worth while. Geological education was unbelief. Fruition was belief, verified by growth of unified science. Culmination was not geology but science. Uniformity, evolution, and symmetry are in nature. Value and number are human. I have been called geologist and seismologist, volcanologist and geophysicist. I am none of these. I am interested in the evolution of what Hoyle calls "This quite incredible universe." I am just as interested in Bergson's "Creative evolution" as in Hoyle and Lyttleton's "New cosmology." And more interested in life than in either. The elements of fruition are a thick earth crust, a comparable pattern for earth and moon, and a mechanism for earth core. This is the story of sixty years of volcanoeering.

My Experiments With
Volcanoes

CHAPTER I

Young Scientist

“The gold of that land is good: there is bdellium and the onyx stone.”

It was the training of my youth under a father who loved God’s out-of-doors that led me to Audubon’s birds; to tramping miles over carries in Maine, Labrador, and Nova Scotia; and to fishing with another eight year old, named Willie Grant.

When I was fourteen my father the Reverend Thomas Augustus Jaggar, took our family to Europe, where botany and bird life were as much a part of my education as geography, French, and Italian. And it was during our visit in Italy that I made my first trip up Vesuvius. All of these early interests convinced me that I wanted to be a naturalist.

It was Nathaniel Shaler at Harvard who told me to go and study the beaches at Lynn and Nahant. So I walked and photographed, and measured ripplemarks. I found a headland and a longshore accumulation with scallops dwindling regularly along the high-tide level. I found swash marks a foot across forming as the tide went out. On the dunes were other sand waves beautifully regular.

Try it. Lie on your stomach and watch them. They are at right angles to the wind. Smooth them out and see what the wind does. It piles little flocculent heaps of coarse grains, each with an eddy downwind. The fine stuff migrates up the slopes forward with the wind, backward on the leeward side. The powder streams meet and lengthen the hills right and left.

I watched the swash marks. The swash of the surf full of sand rushed up the beach, cleared suddenly, and retreated, leaving a ridge along the beach. This elevation became the tide limit, and a new series started lower down. The swashes couldn’t climb over the ridge because the tide was going out. And so for hours ridge after ridge was built.

I watched high-tide scallops, six feet apart, forming heaps at the top of the beach. The swash waves ran into the bays between the heaps during the flood hours, making a rush up and a suck down. The rush up was muddy, the suck down was clear. Pebbles and sand were building up on the sides of the small promontories. Each heap was horseshoe-shaped, with the toe seaward. Forty or fifty crescents got smaller and more sandy toward the middle of the beach. Here was rhythmic force making repetition. The ripples and swash marks were repeated seaward. Clearly the headland of rock was making pebbles and sand, sending pulsations along the beach, instead of across it.

The ripplemarks were packed sand of the low-tide flat, formed totally under water parallel to the waves. The back-and-forth motion of waves made a pattern of sweep and eddy on the bottom. Were beaches, then, things of habit like birds? Here were four kinds of sand waves, all on one beach, all of them complicated by wind and water and tide; big and little; shapely and regular. The beach was alive. It was building from the end, it was rippling under wave action. It fed the wind as it dried, and the wind made an exquisite dune pattern of the grains. Perhaps beaches might be natural history, just as much as the birds that inspired my interest in nature when I was eight years old.

The mystery of the beaches drove me to a new discovery; to the university library, where I found French and English references to ripplemarks. I found experiments, soundings, fossil sandstone ripples. I learned that such great authors as the botanist De Candolle and Sir George Darwin had interested themselves profoundly in what happened to the sand grains. From the library I went to mud puddles in a tank and to experimentation. Thus I found my way from beach to books and from books to the making of baby beaches.

Later, at Harvard, zoology and botany were all cells and embryos and the microscope. The habits of animals scarcely entered into our studies. The natural history of Audubon and my boyhood had vanished. The new words were phylogeny and cytology, development of the individual, and cell development.

So in mineralogy the microscope and the tiny crystal governed; the molecules of the crystal, and the chemical atoms of the molecule. Science

was headed toward the infinitely little, though later, by way of the spectroscope, it was to leap to the infinitely big of the heavens. I never learned to think the universe finite.

Professor Shaler wrote in 1893, “In the next century there will be a state of science in which the unknown will be conceived as peopled with powers whose existence is justly and necessarily inferred from the knowledge which has been obtained from their manifestations. In other words, it seems to me that the naturalist is most likely to approach the position of the philosophical theologian by paths which at first seemed to lie far apart from his domain.” Just this has happened in the world of galaxies and electrons, producing Einstein and Planck, Jeans and Eddington, Hubble and Hoyle. And I suspect that sea bottoms and volcanoes are “peopled with powers” yet to be inferred.

Through Josiah Cooke and his wonders of projection apparatus; through Cook’s nephew Oliver Huntington and his mineral crystals; through John Eliot Wolff, whose assistant in optical microscopy I became; through Robert Jackson with his museum collection technique and the hexagon plates on fossil sea urchins; through all these I was introduced to the laboratory collections and instruments. I found a fascinating world.

The theater, too, furthered my education. Like many Harvard students, I “suped” for several great actors and actresses, among them Julia Marlowe and Sarah Bernhardt. And in one play I even had a speaking part: “My lord, Posthumus is without.” I also practiced legerdemain as amateur assistant to Kellar and Hermann, who called me out of the audience and pulled rabbits out of my coat and eggs out of my mouth. Thus I learned of the psychology of audiences, how to experiment in public, and how easily deluded is the average mind. Just so nature may delude, if the scientist doesn’t keep his wits about him. But I also learned the value of vivid demonstration before students. A great exponent of this method of teaching is Professor Hubert Alyea of Princeton. His chemical experimentation is marvellous. His chemistry textbook is modern physical chemistry at its best. He demonstrates that the art of the magician has come down to the twentieth century and that even mathematical science may pass over to the layman. I suspect that geophysics does not need to be buried under differential equations as it is today. Certainly experimental volcanology made exciting at the lecture table could work wonders in getting the globe explored.

At Harvard we were taught that geology was a detective history. Vaguely, the same fossils were the same age. Vaguely, man had come from a fish which climbed up on the land. It was much later that radio activity of rocks was accepted as setting ages in millions of years. King and Kelvin taught us that the age of the earth was 24 million years and the sun was dying. A half century later, 2,000 million years was the figure and the sun was heating up. Now cosmogonists talk easily of 10,000 million years as an item in star history. I have learned that one can have any theory he chooses, and that some new discovery will probably reverse it. A discovery is the uncovering of an appealing, bright idea.

The idea of geology as history based on Darwin's evolution never took root in my consciousness. Geology to me is the science of the globe. Science studies how things work, how things change, how they accomplish what they do, how they grow, and how they compare. It does not study the "why," or the necessity for an origin of anything. Originating is eternally in progress. Astronomy today is giving up origins. History based on a few relics seems futile. Relics, or specimens, must be compared with action.

Guessing that we must have come from a fish, with no evolution sequence in successive strata and no mammals whatever in very ancient strata and no preservation of soft creatures possible, seems a contradiction of Darwin's own testimony. He insisted on "the imperfection of the geological record." But he had no conception that the Cambrian was 500 million years B.C., nor that the fiery Keewatin of Lake Superior was 1,800 million years B.C. Darwin knew that the bivalve brachiopod *Lingula*, now alive in quiet seas, is exactly the same today as it was then.

Lingula is found fossilized in the intermediate geologic eras. We have no proof that intelligent beings in ships from unknown lands did not dredge him up in Cambrian time. Five hundred million years is so absurdly long that there may have been at least twenty different flowerings of intelligence on the earth, having no relation to us. Continents are places of catastrophe. Sea bottoms are places of constancy. Man lives on continents, and his fossilized bones are short-lived.

If each Adam preceded a new humankind of 100,000 years, the time since the Cambrian allows for 5,000 deluges, or eruptive conflagrations. Each one would exterminate that particular Adam's descendants. If glacial

periods are deluges, we know their scratched boulders back to 400 million years before *Lingula*. These older ice sheets were in Canada. But we know fiery floods of lava 1,300 million years before *Lingula*, on the north shore of Lake Superior.

We have not one particle of evidence that before the race was killed off primordial volcanologists, who were very queer looking chaps, might have studied those eruptions with expensive instruments. Certainly they had a lot of copper at their disposal. Perhaps the great lakes were a continental sea, and some ancestor of *Lingula* was scooped up for food by those doomed beings.

But geology at Harvard was not all history. When R. A. Daly and I were graduate students, we worked on Ascutney Mountain, studying ancient fire-made granites. The hills were lumps of the ancient pastes crystallized. The crystals were feldspars, mica, quartz, and iron oxides. Oldest prisms were lime phosphate, the mineral apatite containing imprisoned brown glass. How did the several kinds of red hot paste invade the altered sedimentary slates? Was brown glass the ancestor? Lava is brown glass. Some of the phosphate crystals contain gas bubbles and liquids. Daly, who published the work, found that ancient lava pushed up while deep in the claystones, and shattered a hole by heat and cracking. The pieces sank and the paste or gas foam was injected in successive lumps. Each new lump had more silica.

Apparently the fragments melted—some of the old sediments of Lower Silurian age were silica—and the invading magma was contaminated with more and more molten sand. So basalt turned into granite. Thus Ascutney Mountain in Vermont became a classic place for hot fluids squirting up and recrystallizing the under rock of New England. It made eventually, by erosion, the Connecticut River landscape.

Daly became a specialist on granites, I became a specialist on lavas. We became professors at Harvard and Massachusetts Institute of Technology.

Something new came into world geology when Wheeler, Hayden, King, Powell, Gilbert, and Dutton surveyed the Utah block fault mountains and the Rockies. They revealed the globe with a crust of gigantic cracked deep prisms, and an eroding surface. Davis of Harvard, the physical geographer, was at his zenith, and from Powell's and Gilbert's example came his classified river valleys. He devised systems of splendid topographic maps

and models, and demonstrations of glacier steam beds and deltas. He made surface wear and dumping debris a living thing, and the land forms a record of it.

Thus I was overjoyed when, in 1893, I received the summons to go with Arnold Hague to the land of geysers, colorful canyon, old volcanoes, and the source rivers of the Mississippi. My job was to take pictures with a huge camera, but I posed as microscope man, too. I climbed the highest peaks of the Absaroka Range, and I traveled with Hague and a mule packtrain back and forth across the range, collecting specimens. Hague had been with Clarence King during the 40th Parallel Survey for the Union Pacific railroads.

Hague's field method was to climb a peak, study the view, and ponder the visible strata, dikes, valleys, escarpments, and pinnacles for miles around, thus formulating each problem. Then we moved camp to a new place to solve the problem.

We sought the ancient craters. The volcanic tuffs and agglomerates covered thousands of square miles, dating from 30 million years ago and continuing outpourings until 2 million years ago, and there were lava flows, ropy or bouldery. Here were petrified trees; there could be found fossil leaves. The tree species told the formation ages of Tertiary time. Many peaks appeared but no volcano cones. The craters had been over what now were eroded dikes, or fissure fillings of lava, which stood out in crisscrossing walls. Where they clustered, ores were found: the Sunlight, Crandall Creek, and Stinking Water mining claims. These were the roots of lost volcanoes, lost by decay, tumble, rainfall, glaciers, and rivers. Underneath the mountainous lavas, appeared white marine limestone cliffs, and still lower appeared ancient granite gneiss.

The geology of ancient seabeds, fossils, eruptions, and glaciers was painted on a whole panorama of mountains and river basins. From a mountain top silently gazing through field glasses—which he was always losing and recovering—Hague would look around for hours. “That ledge is the Madison limestone, those are the Red Beds, those pink, rounded hills are Archean granites.”

After a day of packtrain travel I was free to fish or hunt. It was a privilege to hunt with Anderson, the old negro cook, whose gray beard and

bushy white wool belied his keen eyes. He had been a slave, later a soldier in General Custer's Big Horn expedition, and a pioneer and hunter. His father had been massacred by Indians, and Anderson swore he would kill any Indian on sight.

One of our hunting trips near Crandall Creek was especially memorable. "Mr. Jaggar, I smell sheep up on that shelf!", said Anderson. And he climbed up a pine tree growing at the bottom against the limestone cliff. He laid his Winchester rifle on top of the steep slide rock slope at the foot of the tree, muzzle upward, butt end downhill. "You mind my gun, I'll climb out on a limb against the cliff and get on the shelf, and yo' all hand the gun up to me." He reached the shelf, made of Cambrian limestone of trilobite fame, and sitting over on it immediately knocked down slabs of rock. They fell on the gun which started to slide down the slope. I grabbed for the muzzle pointed toward my throat, the stock wiggling right and left. The gun went off and I felt a nick in my ankle. Anderson had left a cartridge in the barrel with the hammer resting on it, but my nick was made by a pebble ploughed up by the bullet. So the trilobites took a shot at me. "Well, this is natural history," I murmured. Old Anderson was less philosophical. He cursed me for letting the rifle kick itself far down among the trees.

Elk, grouse, blacktail deer, antelope, rattlesnakes, prairie dogs, skunks, badgers, owls, whistling martens, wild sheep, and the grizzlies we never saw alive were all part of the great West. So were the bucking cayuses and kicking mules with which we lived, numerous ranchers, prospectors, soldiers, sportsmen, and guides. Once we were joined by a sheriff looking for an escaped desperado from Red Lodge Prison.

Just before I left the Yellowstone, I visited the hot springs and geysers. With more than 4,000 vents, the geyser basins are steaming areas in the forest. At Mammoth, the carbonate terraces show exquisite ripples and sculptured cups in steps. One hotter group of waters, through the igneous lavas and granites, becomes full of silica and deposits sinter. The other, through limestones, deposits travertine. The alkaline siliceous waters deposit such strong silica edifices as to hold the explosive steam boilers of the geysers. Both silica and lime deposits are led to gorgeous sculpturing and to brilliant colors at their borders caused by the blue-green algae, which live at temperatures up to 150° Fahrenheit.

The boiling waters have been superheated volcanically since Tertiary volcano times, when first dark magnesian, and afterwards siliceous, lavas were ejected. Here is the same order Daly and I found in Vermont; the dark rocks first, rifting through slate, the granites last, with quartz cutting the dark rocks. The cavities among the Yellowstone geysers show quartz.

The surprise to me was that the geyser basins were eternally breaking down, cracking, dissolving, making new geysers in the forest. Instead of being chiefly deposition, the hot spring action is chiefly erosion. It is a vast cycle of hot magma gases and rainwaters from Tertiary times to now; from 20 million years ago to now. A long time.

Remember that the last retreat of the glacier-period ice was only 20,000 years ago. That ice found the geyser basins in full swing. A thousand times farther back were the Yellowstone volcanoes in full activity, and they kept going while the continent lifted and pushed the Gulf of Mexico from the Great Plains to where it is now. And yet that 20 million years was only a twenty-fifth of the time back to the trilobites, and a Yellowstone seabottom bed of that age is under all the lavas. Our schoolbook history is pretty small.

In all directions the ground of Norris Geyser Basin is cracking and changing. The geysers are utterly unreliable, here today and mere hot springs or empty cracks tomorrow. Old Faithful intervals range from thirty-eight to eighty-one minutes, quite irregular. The New Crater was a squirting, scalding jet which killed the trees and vegetation all about. Its seemingly regular, twenty-five foot jets shot up at forty-five degrees inclination about every three minutes. Later, in 1922, I was to find this geyser totally different. Careful studies have shown that water of this elevation boils at 199° Fahrenheit; one geyser gave off 253° Fahrenheit, or fifty-four degrees of superheat, seventy-two feet down its shaft. This is the only place of superheated waters known on earth. The roaring steam of the Black Growler has eighty-one degrees of superheat. The quantity of carbon, sulfur, and chlorine in the waters is so excessive, though it is very small in the rock, that a source of heat from volcanic gas is certain.

The net result is thousands of boiling springs of rainwater, soaking a sponge of rhyolite rock over hundreds of square miles, erupting over a remnant volcanic furnace beneath, and eroding and dissolving out basins at the headwaters of the Mississippi.

Here is an object lesson in volcanic erosion. Here is a perpetual eruption of volcanic gases which has dwindled after millions of years of melting siliceous and carbonaceous rocks. It recrystallizes them as andesites, rhyolites, and obsidians, and mixes deep steam with rainwater to do the work of erosion and water solution and of deposits, over a vent at the heart of the Rocky Mountains. As usual, this vent has cluttered itself from age to age with the melt of the deep earth crust, namely basalt, which Yellowstone's lavas show repeatedly from bottom to top of its accumulations. And as usual, the vents themselves are hard to recognize, buried as they are under heapings.

In 1897 I returned to the Yellowstone, where I visited Death Gulch, a dismal solfataric gully with a trickle of cold, acid water near Cache Creek. Accompanied by Dr. F. P. King, I climbed up this gorge, where there was a bad smell and burning oppression of the lungs from hydrogen sulfide. It was a V-shaped trench 50 feet deep in volcanic puddingstones, whitened with alum and epsom salts. Bubbles rose through the water in many places.

The remains of eight big bears were found in the gorge, clustered in one place. The latest victim was a young grizzly with a clot of blood staining his nostrils from his last hemorrhage. Poison gas had killed him. Earlier visitors had found squirrels, hares, and butterflies and other insects killed by gas. Probably both sulfuretted hydrogen and carbonic acid gas do murder in still weather. However, we had the wind blowing up the gulch. We lit matches in hollows and carbon dioxide did not extinguish them. The same thing had happened when Mr. Weed in 1888 tested for carbon dioxide at Death Gulch.

Now, knowing the case of Mr. Clive, the Englishman, and his guide, Wylie, who were overwhelmed by hydrogen sulfide while photographing Boiling Lake on December 10, 1901, it looks to me as though the rotten-egg smell may play a large part in the killings at Death Gulch, as well as in some poison tragedies of Java. Boiling Lake is at the south end of Dominica Island north of Martinique. There are four solfataras and the scalding lake, the latter near the interior village of Laudat, at the head of a volcanic valley, and four miles on horseback from Roseau, a shore town southwest. When Mr. Clive, Wylie, and Matson—another native guide—looked down at the hot pool, Matson noticed it boiling without vapor, and called attention to the danger. However, they went on to the lake. Matson later reported, "I inhaled something offensive and felt as if I was dying. I ran, and lost consciousness.

I came to in a ravine and found Wylie lying where I had left him.” Clive, refusing to leave Wylie, sent Matson for help, but when rescue parties arrived, both men were dead.

At Boiling Lake there was no eruption, no vapor, only the very bad smell. All the symptoms indicated a sudden change in the pool from steam to excessive hydrogen sulfide. And five months later, at Pelée across the channel from Dominica, excessive hydrogen sulfide set off the great explosions.

In view of these phenomena it seems likely that Death Gulch in the Yellowstone also kills with sulfur gas, the odor of which is so strong there. Day and Allen associate hydrogen sulfide with the limited Yellowstone sulfate areas, of small water discharged, and such is Death Gulch. One part hydrogen sulfide in 200 parts of air is fatal to mammals, and it may come up in gushes. Carbonic acid asphyxiates, but it is not a poison and when it is free is so heavy as to mix with air very little. Death Gulch is not a place of lime deposition like Mammoth Hot Springs, where carbonated water decomposes underlying limestone.

Europe was to be the next step in my education. As assistant in petrography and graduate student at Harvard, I was encouraged by Wolff to plan for Heidelberg. There I was to find H. Rosenbusch, who had put system into the infinite series of minerals in rocks. But my journey to Heidelberg began with a geography congress in London and a geology congress in Zurich. These meetings were with such bigwigs as Lord Curzon, Henry M. Stanley, and famous arctic explorers, and I was surprised to find that all these VIP’s looked like ordinary men. Unfortunately for me, this realization came a little late.

Looking for a luncheon beer garden in Zurich, I picked up a small side-whiskered Englishman, and suggested we join a group of foreign geologists in a buffet. “Oh no,” he replied, “no beer. I only want a cup of tea and a biscuit.” So I left him and crudely and youthfully joined the younger men in the beer parlor for sauerkraut and wienies and Munich beer. Later at the opening meeting, the Geological Congress was addressed in French by the famous Sir Archibald Geikie, Director General of the Geological Survey of Great Britain and Ireland, and the author of “The textbook of geology,” the greatest of geology manuals. He was my pickup, whom I had deserted at

lunch time. I had lost the opportunity of a lifetime, for a tête-a-tête with the world's most famous geologist.

Before going to Munich, Harry Gummeré of Haverford and I trekked through Denmark in a third class carriage amid peasants smoking fearful-smelling tobacco in long china-bowl pipes. Then we crossed to Christiansand in Norway. We traversed the fjords north to Trondhjem by rowboat, in "stoolcars" with little girl drivers. Then we traveled on foot, and everywhere in rain. Waterfalls were so numerous we never wanted to hear of another one. We climbed up to Stalheim from Bergen, saw the Jordalsknut, a magnificent half dome in a vast granite canyon like Yosemite. We rowed around the Kaiser's yacht in the Nordfjord, and tried to pick him out on deck. We got soaked with days of rain in a backcountry village, and went to the inn, got into bed, and sent our clothing to dry in the kitchen.

The local Norwegian bank looked at our Brown Brothers letter of credit and said, "Nothing doing," which inspired us to compose a poem:

We're so happy we don't know what to do.
We haven't any clothes to wear,
We're wet all through and through.
We haven't any money and we ought to feel quite blue
But we don't, we feel so happy, we don't know what to do.

Fortunately, the innkeeper was amused by our poem and sympathetic toward our plight. He took our IOU's and told us we could have all the money we wanted and to send it back when we reached Trondhjem.

From Trondhjem we crossed Scandinavia by rail to Stockholm, like Venice a city of canals. Delightful maiden ladies kept the breakfast place and served us with many queer breads, goats'-milk cheese, and sublime cleanliness. The canal boat took us across Sweden to Göteborg. It was a little steamer, from the porthole of which we saw a cow comfortably grazing a few feet away. And we saw and were impressed by the superb landscaping of lawns, by tree horticulture, and by lock masonry. In both Norway and Sweden the people talked English, the national costumes were delightful, the girls were pretty, and everybody was clean and democratic.

The winter semester of 1894–1895 was spent in Munich, where Groth's mineral and crystal collections were the main attraction, and where I heard

the lectures of Sir Doktor Privy-Councillor Knight Karl A. von Zittel, author of six huge volumes on fossil shells, fossil horses, fossil dragons, and fossil trees, and a history of geology. We once saw him rigged out in gold braid and an admiral's fore-and-aft cocked hat for some imperial function.

He was a forceful lecturer. The assistant arranged diagrams on the rack, the students gathered, and then his majesty entered. Everyone rose and Zittel held forth with a rattan pointer: "Es gibt, meine Herren, ein ganze anzahl von ausgezeichnete beobachten über" and so forth. Then he whacked the drawings, and made graceful allusion to American investigators as he explained a giant stegosaurus.

In the "Heidelberger Geologischer Panoptikum," as an attic room on the Neckar was called, I afterwards posted a ditty based on "Ole Uncle Ned":

There was an Orthopod
Stegosaurus Marshii
Laid him down on his Jurassic bed.
He had a row of shovels down the middle of his back
But he didn't have a very big head.

Chorus:

Hammer, hammer, hammer on the stone
Chisel, chisel, chisel on the bone
There's no more rest for poor old Steg
For Zittel couldn't leave him alone.

Heidelberg days were memorable for the lectures of Rosenbusch, Goldschmidt, and Osann; for laboratory system; and for long collection trips. With specimen bag and hammer, we went to Saxony, Bohemia and the Vosges Mountains, the Black Forest, and the Oberwald. I had a large, pointed hammer named Umslopagaas, after Rider Haggard's hero who wielded such a weapon. When Palache and Brock and I were in a quarry and an unwieldy boulder had to be broken, the yell arose, "Umslopagaas come quick!" The collection of rock specimens at "classical" localities, meant the textbook rocks of Rosenbusch, or of Zirkel of Leipzig. Every student dreamed of having a private collection.

After the Ascutney experience, I was impressed by Schneeberg granite in Saxony. At the border of the granite are slates, baked in zones back from the

granite edge: hard horn rock, spotted rock, mica rock, then claystone. The colored geological map of Saxony was superb. This includes the mining district of the birthplace of geology in Europe, where in Freiberg, A. G. Werner had founded an arbitrary science in the eighteenth century, imagining granites to be crystallized from a world-wide ocean.

In one place I found a hand specimen with tiny granite tongues which had split their way, as liquid as alcohol, between the blackened folia of slate. The granite itself was all crystals, but here was proof of a fluid when the granite penetrated. What was it, how hot was it, a gas, a foam, a paste, or a liquid? The time this occurred was millions of years before Kaiser Wilhelm. I had found something similar in the Yellowstone, the little dikes of sylvan intrusives in Absaroka Mountains. The smallest tongues showed the most perfect granite in the microscope, of Tertiary intrusive stocks. It was as though in these siliceous invasions of basaltic agglomerate, nature made its best experimental granitization on a very small scale.

We soaked up the surprises of European scholarship. We pored over books in the bookshops, loaded ourselves with microscopes, goniometers, and four-volume textbooks. We found all the science of Europe in attractive unbound form and had it bound in half morocco. Mineral dealers were everywhere, offering beautifully labeled specimens. All things in Europe seemed inexpensive.

Rosenbusch, who had big brown eyes and a gray beard, came to look over my work on feldspar, in his laboratory. When I asked enthusiastically what make and model of German microscope I ought to buy, he turned me around and looked deep into my eyes: "Herr Jaggar," he said, "Es is nicht das Mikroskop, es ist der Mensch."

Another time he produced a dense black rock and said to Matteucci of Vesuvius, to Palache, and to me, "You are geologists. What for a rock is that?" We, of course, got it wrong, thinking it must be a lava. It turned out to be a black limestone, easily identified, had we scratched it instead of putting our lenses on it. He chuckled at the gullibility of geologists.

Osann gave a course on petrographic chemistry which met at 7 A.M.! We usually got there, but once or twice the teacher himself was late. We would gather around Osann, who was fat and genial, and say "Herr Professor, how

about some sausages and beer and a little breakfast?” He always replied “Why not? There is plenty of time,” and we sought the nearest cafe.

Some professors got up at two o’clock in the morning and wrote, taking advantage of the quiet hours. Rosenbusch had a high desk and wrote standing up. Their objectives were to produce enormous tomes listing all crystals and all rocks and all publications, in all languages. This is German science. Its password is “thoroughness.”

The net effect of German scholarship on me was a feeling of irksomeness and resentment, but what I learned of thoroughness and of mechanisms I value extremely. I honor the memory of those teachers, and I honor their pupils, who by specialism have penetrated deeper and deeper into the smaller and smaller things of matter. The ultimate is the background material between the galaxies of the universe and the unknown background particles of life. But for me, the middle field—the development of mountains, rivers and sea bottoms, continents and volcanoes, earthquakes and depressions of land, the sky, clouds, and waters—all the outside world, needed experimental engineers. Intermediate bigger things like the crust of the earth and moon, within the time that is measured in human years, seemed to be neglected by science, and yet to be accessible to the giant power of engineering.

Rosenbusch set me at one feldspar specimen for an entire summer. I wanted things moving, changing, and evolving. I wanted a narrative of that tabular feldspar crystallizing, or better, a dish wherein to watch it crystallize. To me it seemed that Faraday or Pasteur would have described the quality of a moving feldspar medium in pressure, heat, gas, liquid, or changing particles. The qualitative investigator would have a furnace and make many trials and produce synthetic feldspar, and he would write a narrative approximating what the under earth must do. He would make melt or froth conditions successful in imitating such rocks as basalt or granite, using hot gases.

The problem of basalt and granite began to be recognized in the eighteenth century. Werner guessed, and taught his pupils, that these rocks were sea bottom deposits. A few determined Europeans in the nineteenth century—Fouqué and Michel-Lévy, Doelter, and Morozewicz—melted mineral mixtures and made igneous rocks by cooling them. The motive was

approximation; the result was good and useful. No one reached melting by hot gases and absorption of hot gases. No one made granite. Volcanic rocks were imitated approximately as to crystals, but not as to gases. And the whole of volcanism was later proved to be gases, as is the whole of physics and astronomy and biology. Man is largely a puff of hydrogen.

These visions were what I brought back from Europe, along with much pondering of such experimenters as Daubrée, Lacroix, Stanislas Meunier, Reyer, and my teacher Goldschmidt, all brilliant imitators of the earth. Goldschmidt gave a course in blowpipe analysis which was completely original. His methods went far beyond those of his predecessors.

Meanwhile, W. M. Davis had written me to come home to Harvard and give the course in field geological surveying. This was in 1895–1896.

My teaching was devised to cut up the map of Boston. I pasted the pieces in notebooks and sent out students in pairs, equipped with map books. They were to keep pencils sharp, use a uniform system, and hammer off specimens from ledges. They were to examine the rock under a magnifying glass, then name it; but I cautioned them, “If you don’t know the rock, call it ‘FRDK, funny rock don’t know.’” Students marked the page opposite each map with symbols for the rocks on that map. Then they came together in seminar, and we made a colored map of the geology of Boston. Laurence La Forge, now professor at Tufts College, was my student and later my assistant. He published the results of our work many years after the study was made.

When teaching was extended into experimental geology and geology of the United States, laboratories were set up in the basement of Agassiz Museum and I was given carte blanche to furnish them. I equipped them with a water tank, a gas furnace for melting and recrystallizing minerals, pressure machines, an air compressor, and motors. Students were assigned experiments with wax, plaster, cement, sands, coal dust, and marble dust. They imitated strata, rivers, deltas, intrusions, and mountain folds, and familiarized themselves with the way solids break.

Each man took a special arbeit for his final thesis, and worked by himself with clock or metronome, thermometer or pressure gauge, spring balance or centimeter scale, and he reviewed the experiments of the past. Prominent among my students were Ralph Stone, afterwards state geologist

for Pennsylvania; Vernon Marsters of Indiana; Julius Eggleston of Riverside, California; and Ernest Howe of Yale.

In the course on United States geology were such students as Amadeus Grabau who became leading paleontologist of China; Stefansson the arctic explorer; Ellsworth Huntington, afterwards the distinguished Yale author and geographer; and Franklin Delano Roosevelt. With so many Roosevelts at Harvard, I quite forgot my famous student until his first visit to Hawaii, in 1934. Mr. Roosevelt had remembered his geology professor, though, and an aide phoned the Volcano headquarters to request that I be at Hilo when the President's ship arrived.

The United States geology course was the product of my two seasons in the Yellowstone and my interest in the great Hayden, King, and Powell surveys. The youthful geologists needed to know the continent and its details.

The big Washington monographs and folios have made a gallery of underground pictures of one of the greatest continents, and these are supplemented by the work of the Canadian geologists. America shows Appalachian folds and thrusts, fault blocks of the Utah plateaus, and eruptives of the Rockies. It contains the amazing metamorphism of very recently upheaved sea beds along the Pacific shore. It records the remnant sea bottoms and dust-storm deposits of the vast plains, bearing beside the obvious buffalo skulls, the old bones of whales, reptiles, and rhinos.

Superposed on all this is so-called physiography, the science of falling materials and water, the rotting of the lands, and the accumulation of debris. A net of rivers over ground and under ground is what stands out, and the living river pattern has changed incessantly through the ages. But through and over it is a moving process of the ages, kinetic, alive with glaciers, hot springs, underground heat or surface cold, soaking rains and rushing storms, earthquake and uplift, fault motions and sinkings. Everything is in motion to one who senses slow motion, occasionally breaking down resistance and charging ahead. And geology is a sense of slow motion and its jumps for 5 million years, with this human year, here and now, of great importance. Geology, like humanity, is not just history.

Under all are gas and heat; Saratoga Springs, Yellowstone, the Comstock Lode, and Mount Shasta. The series gets hotter from New York to

California. And out at sea the refuse of the continent is dumping all day long. And science is anxiously waiting to learn how hot sea bottom is.

In addition to laboratory work, I wanted to conduct cross-country hikes for such subjects as botany, geology, and zoology in the forests and swamps and hills of Massachusetts. And it was in connection with these plans that I learned a lesson in simplicity. I went to President Eliot, remembering the high sounding “Pierian Sodality” name for the college orchestra, to get a classical calendar name for my cross-country tramps. He said, “What, in brief, is your idea?” I replied, “In ordinary language they will be natural history walks.” He took a pen and said, “Why not this for a name?” On the paper was written “Natural History Walks.”

An important part of our curriculum was the Tuesday evening geological conference, during which any graduate worker could give a paper. To these conferences came, at different times, Brooks, Spurr, Schrader, Goodrich, Mendenhall, P. S. Smith, Mansfield, Matthes, Lane, Crosby, Barton, Douglas Johnson, Daly, and all the Harvard staff. The men got confidence in public speaking and exhibiting, and the professors commented in kindly fashion. Topics ranged from summer work in the far west and current studies in meteorology under Ward to petrographic or experimental work with projection apparatus under Wolff and me. Jackson and Hyatt brought in fossils, and the Geological Survey was always in evidence as a goal for young men, or a subject for review. Shaler’s comments were accompanied by a string of good stories. The conferences taught students how to teach by making them speak in public. It was one of Shaler’s most productive inventions, and has been copied far and wide.

Walcott in the Survey looked to Harvard to produce field mappers of rocks. Graduate students had the choice between process and history, geography linked to school teaching, microscopical petrography and crystallography linked to the minerals and rock collections, or evolution linked to museums and fossils. Beecher of Yale had found hairs on the legs of fossil trilobites. Someone else had found fossil bacteria. A group of petrographers got together and founded an artificial classification of fire-made rocks based on chemistry—no use at all to the field man with a rock specimen. Agassiz had built a magnificent museum. The research motive was based on collections; the public exhibit motive was based on evolution and big, rare things. The publication motive imitated Europe; “be as

technical as possible, detest reporters and newspapers, and never be popular.”

In 1897 Harvard University gave me a Ph.D. degree, after a double thesis and an oral examination. I passed the examination very awkwardly, as my capacity for remembering text book information is nil. My theses were (1) on an invention, a mineral hardness instrument; and (2) on the included fragments found in Boston dikes.

The microsclerometer, as the instrument was called (that is, a microscope scratcher), was designed to diamond drill a mineral to a fixed depth. The hardness was measured by the time consumed, on the theory that the energy required for the standard hole varied with the time, and the time with the hardness. The number of rotations with a constant speed motor is a measure of the time.

The paper was published in America and Germany, and elaborately reviewed by a microscopical society in England. The instrument was borrowed by H. C. Boynton, a graduate student in metallurgy, and he got good results on the microscopic crystals that constitute steel. The inventing and constructing with the aid of Sven Nelson, a Swedish mechanic of ability, were to me an education in themselves. For one thing, I learned how enthusiastically science feeds on ultra-little things.

My petrography of included quartz fragments in basalt dikes was partly published, but made no hit at all. It was outdoor work, it concerned the granite problem, it revealed the “fluid” of granite minerals as “waters or vapors” having no effect on augite, the green fusible mineral of basalt. But the same fluid was revealed as corroding quartz inclusions, harder and supposedly more infusible.

If temperature had anything to do with it, the granite fluid could melt holes in quartz inclusions, but the mantle of augite dark crystals which the basalt had plastered on the outside of the quartz fragments remained unmelted. This was my first adventure with the ancient problem of fusion, or melting. I became convinced that granite fluids, like the makers of gold quartz veins, are low temperature vapors or gasses. This agrees with what is now well known, that silica has a low melting point. But melting and temperature are not the whole story.

To me, the spreading of one's fame by scientific papers was commercialization. "You must make your name known" and "what have you published?" rang through the scientific halls of learning. No suggestion of art, literature, drama, beauty, or philosophy ever came to me from my scientific colleagues. Some literary friends, like William Garrott Brown and my classmate William Vaughn Moody, thought readability important. Brown warned me against the dullness of small papers in scientific writing. Agassiz warned me against exactly the opposite, namely, against popularizing or being interesting. This antithesis between science journals and art probably never comes into the field of vision of many young scientific writers. They see only "Write for your scientific peers and for no one else, that is your world." All my life I have been plagued by "be as technical as possible" versus "tell the public what it all means."

I suspect that our system is producing diagrams and statistics in geology (and perhaps in science generally) and no longer produces works of art. I know few geologists who are fine draftsmen. They accept photography instead. I know none who is a literary stylist. They write for ultra conciseness and tabulations. The nineteenth century taught classical English and drawing.

Geology is a science of the dreamland of the earth's interior and of millennia of the ages and of the overwhelming expanse of rich, productive, unknown ores under ocean bottoms. It is a field for men of letters, and for new Magellans, Humboldts, and Darwins bursting with imagination and the will to explore.

This seeming digression is really germane to the purport of this book. It is one man's review of a half century of evolving discovery. Also a half century of evolving error and departure from the ways of the leaders. The leaders, from William Smith's thoroughness with strata in England, to Clarence King's summary of a thousand miles across the Cordillera, explored upward and outward. It persuaded governments. Persuasion before the court of public opinion no longer uses and employs explorer men of letters. The United Nations is not employing Clarence Kings on the world geology of the remaining three quarters of the earth.

The confusion, the secrecy, and the loss of art are occasioned by vulgarization. In 1875 real men of distinction explored the earth. Now that

is left to incorporated establishments, teaching trusts, and calculating machines. Clarence King was a linguist and was the son of a trader in China. His Yale training under Dana and Brush gave him real culture. His founding of the United States Geological Survey was the evolution of a genius who disliked politics and whose friends rejoiced with him in great prose, good pictures, and fine sculpture. Then he was wrecked by a false ambition and the decadence of the very thing which made him great, the simplicity of high thinking, noble writing, and cultivated friends. Lacking today are cultivated boys with an ambition to explore the globe, both under the sea and in the wilderness.

Geology in 1897 was a jigsaw puzzle, with a choice between the museum and the field, between the easy thing of collections, fine microscopes, and the scientific societies, and the hard thing of exploring the globe. Collections and instruments were an overpowering attraction, particularly when photography and experiment were involved. But roughing it in the wilderness has made some of the finest characters I ever knew.

Geological surveys of the west continued to occupy me during the summers. I worked in the Black Hills of South Dakota under Samuel Franklin Emmons, and my associates included John Mason Boutwell, John Duer Irving, Philip Sidney Smith, Bailey Willis, and N. H. Darton. Boutwell was to become a copper geologist and copper magnate in the mines of Utah; Irving, Professor of Economic Geology at Lehigh and Yale; and Smith, head of the Alaskan branch of the U. S. Geological Survey.

Being with Emmons, Willis, and Darton in the Black Hills field was to learn variously how geologists work in the field and how their minds work. Emmons was of the Boston Brahmins, a Harvard man, mining geology his specialty, with the Clarence King tradition of the Great West, the 40th Parallel Survey.

Bailey Willis as Chief Geologist spent a week with us in camp, and I saw his genius for drawing in line, and he explained the four-step pacing method. Willis mapped distances by pacing across mountains, counting in his head, while talking at the same time. He compiled in color a geologic map of the United States. His marvellous experiments on mountain folding, his explorations in all the continents and his poetic faith in hydrogen and crystallization as internal forces made his name immortal.

N. H. Darton mapped the Great Plains; and his genius was for hard work, long field hours, color photography at its very beginning, and an extraordinary eye for detail in the field.

Darton showed me how to find the Chadron Formation on the divides, white clays easily overlooked. Darton's many years, traversing the entire West, and publishing superb monographs of artesian waters and of immense fossil sea bottoms, summarizing the geology of whole states from Texas to Canada, ranks him among the great geologists. I learned from him detail of infinite discovery possible in every rock ledge. He found tiny fossil shells everyone else had missed. Powell and King had painted impressionistic geology. Darton followed and painted thousands of miniatures, but also combined these into large books.

Charles Doolittle Walcott was Director of the U. S. Geological Survey at that time, and no greater geologist ever lived. His Cambrian fossils, those of the first great fossil-making "Mediterranean Sea" of North America, lay buried in the United States from shore to shore. Unswervingly he followed every inland sea of 531 million years ago and thereafter, through advances three times across the continent. Lands were of moderate relief and climates were mild. Marine animals and sea weeds, large and small, were abundant for 80 million years. And remember that a million years is a thousand times the interval since William the Conqueror.

The continent Walcott mapped of that ancient time was the North America of today, with sags that let in shallow sea strips and pools where the Cambrian shales and limestones now lie. He wrote a description of that vast history, and all his later summers were spent in the Canadian Rockies, where fossil-bearing strata make the most startling mountain peaks on earth.

My Black Hills surveys of 1898 and 1899 were near Deadwood and Spearfish and Mato Tepee, the Devil's Tower National Monument. In those badlands with weird desert gorges, appear the bones of ancient rhinoceroses and many grotesque animals, huge and tiny, of 40 to 60 million years ago. We found little bones in white earth on the divides still preserved against erosion.

Our big job was to map the laccoliths near Deadwood. A laccolith, or rock cistern, is a lava body which in very ancient times squirted into the cracks of the strata. The lava had penetrated between the strata of the

northern cover of the Black Hills, swelled to lenses between the strata; and, particularly, it selected and penetrated the soft shale beds which grow thicker and more numerous upward among the formations. Thus after erosion of the present landscape, both large and small lava lenses were revealed as resistant hills, the largest toward the bottom of the pile of strata and the smallest and steepest toward the top in thick, black ancient mud deposits.

Mostly, the laccoliths were injections of volcanic fluid up a crack, which met a hard bed and bent to squeeze the paste or lava into a soft layer. The result was an underground lava flow which ruptured the beds. Apparently the first rush brought up fragments of the rocks below. This fragmentary stuff of mud and gravel was overridden by the lava, until the latter penetrated horizontally a mile or two between strata, arched the layers above, and solidified at the Devil's Tower with vertical columns like the Giant's Causeway in Ireland.

This group of subterranean volcanic eruptions between strata probably came under sea bottom at the same time that the Yellowstone upland began its open-air outpourings farther west. But in the Black Hills there is no sign the laccolith lavas ever broke up to the top country.

The Black Hills, like the Rocky Mountains, were a long time rising in waves of action, whereas the lava intrusion was a relatively short episode of one of the latest of these spasms. However, that episode entails a long story of numerous injections. It takes us down into crust and along through the millennia.

Always think in millions of years. It is wise also to think in millions of miles and to remember that the sun and the Milky Way are parts of the same system as the earth. And remember that a ledge or a boulder doesn't worry about living 20 million or 100 million years. A skull is a boulder. That old brontotherium rhinoceros with a forked horn, standing eight feet high and fifteen feet long, lived in the upper Oligocene, when clay and volcanic ash were being deposited in the Bad Lands of South Dakota. Probably vast flood plains of rivers were his habitat, swamp reeds and leaves were his food, and floods washed his bones and buried his skull where we find them today. The country of open glades was probably like the safari land of central Africa.

Brontotherium's skull in Chicago Natural History Museum dates from about 30 million years ago. The bones are scattered, and few complete skeletons have been found. Man's ancestor may have started 10 million years ago, but the nearest approach to an ape who lived in the trees of old Bronto's forests was an opossum. Furthermore, nothing like flint tools have been found in the rhino strata. The apes started in Europe and Asia in the next geologic period, and some fossilized monkeys have been found in South America. But men and monkeys are too soft. They don't make good fossils.

The bones we found were of turtles, in clays upheaved on the top of the Black Hills uplift. These clays were afterwards eroded into the present valleys, and probably were contemporaneous with the riverbed silts, where the rhinoceros skulls were found. So our turtles and rhinos were no doubt neighbors in 29,998,000 B.C.

Our sojourn in the Black Hills was not without adventure. One evening when Boutwell and I were riding home to Deadwood, I dismounted and jumped into the shrubs of a gully to knock a rock specimen off a ledge. From beneath my feet came a buz-z-z like a swarm of bees. I had jumped right on a rattlesnake and could feel his coils against my ankle, and no leggings that day. Boutwell called out, "Oh let me see him! I've never seen a rattlesnake." I made a suitable reply and, somehow, leapt clear before the snake had a chance to strike.

Another adventure concerned my gold watch, a gift from my dad on my twenty-first birthday. I lost it from a chain which broke against the saddle pommel at some dismounting point. I advertised for it by placards at railway stations and, amazingly, it was returned. A Salvation Army man found the watch, badly trampled by my horse, at a back country place, brought it to me in Deadwood, and received the reward. I took it to the maker in Waltham, where it was restored; and I am wearing it fifty-four years later, converted from a hunting case to a stemwinder.

John Irving of Yale, whose father had been a mining geologist in the Great Lakes district, was one of the most lovable companions I ever camped and tramped with. We were together in the Black Hills, where we hired a wagon outfit to cross the Hills to the Devil's Tower. The personnel was a masterpiece of improvisation. The cook was a fat boy who told

marvellous tales of adventures. Among other things, he had been a human ostrich in the circus, and he assured us that chewing up glass and swallowing it did no harm if you knew how. So elaborate was his cooking that again and again we ran out of grub. Furthermore, meals were generally late, but we knew better than to hurry the supper and his finishing touches. When finally a meal was ready, he advanced to our tent, bowed, and called out, "Gentlemen, you will now proceed to sagastuate."

Johnston the teamster was an ambitious South Dakota high school graduate and farm boy who wanted to learn all he could from geological surveyors. A few years ago, in the nineteen forties, I received a letter from him in southwest Africa saying that he had been successful in placer mining for gold and diamonds and that he was writing a book about it.

Arizona was my fourth field of fire-made irruptions; after New England, the Black Hills, and the Yellowstone (old, middle-aged, and young). To the Bradshaw Mountains between Prescott and Phoenix and lying south of the Grand Canyon, I was sent with Palache to make the Bradshaw Mountains folio.

At Prescott we had the rare privilege of talks with Clarence King. An aged bachelor dying of tuberculosis, he was living in a cottage with an old negro servant. King was a fascinating talker and writer. He had been the first director of the Geological Survey and was the author of "Mountaineering in the Sierra Nevada." His great summary volume of the 40th Parallel, the survey along the Union Pacific, is one of the classics in literature and in geology. His model, unhappily for him, was Alexander Agassiz, who made a great fortune out of Calumet and Hecla copper. When King went into mining to make a fortune he contracted tuberculosis. He died soon after we saw him.

The problem of what makes granite was never better illustrated than in the Bradshaws. One formation, in upright bands for miles across country, showed dark schist, diorite, granite, diabase, granite, light schist, quartzite, granite, gabbro, and schist again, like a succession of dikes, slabs, and veins side by side. A mountain spur, like a bookshelf with colored books on edge, is called Crooks Complex, and was named after Crooks Canyon. The trend was with the pinched strata but the stuff was mostly igneous.

It was as though a mechanism of melting-up was mixed with intrusion of fluid, but what fluid? A glass? or a gas? There was no smearing, but clean-cut dikes and schist slabs on edge. In the big granite hills there were contact breakups with fragments of schist imprisoned in granite, but not smeared or streaked. The impression was of millions of years and thousands of episodes, all dike-making and guided by the upright lamination or vertical structure of the ancient altered tightly folded clay and sand strata, squeezed together by horizontal pressure.

Since learning of the million-year periods taught by radioactivity, and of the many million years within a single era of geology, I have begun to wonder whether these very old formations may represent hundreds of millennia, with granitization happening over and over again, in each geological revolution of upheaval and mountain building above.

Granitization, then, is a process of heat pressure, gases, melting, and crystal making, of which the ancient words magma or emulsion or paste give no conception. And volcanism, up through the deep crust, is the mystery devil. May it not be nucleonics and melting of deep crust, rather than chemistry? And is not the mystery devil always hydrogen gas?

At the beginning of the twentieth century I visited two places which are close together and related to the Bradshaw Mountains. One was Searchlight in the southern tip of Nevada, the other was the Grand Canyon of the Colorado River.

I shall never forget my arrival in Searchlight. A strike of miners was going on, and Stanford geology students had been sent in as strike breakers. Big Bill, the sheriff, brought the boys across the desert from the railway. His buckboard was in front and the Stanforders followed in a wagon. The strikers lined the road out from Searchlight, intent on loosing the horses. But when they saw Bill's star and his notched six-shooter, they dropped their hands to their sides and stood like a row of tin soldiers, while Big Bill led the way through at a gallop, cursing them roundly.

When I got off the train at Ivanpah, a small place with only a few houses, I spoke to a young station agent where the ancient Wells Fargo sign hung. He told me that the Quartette Mine team would meet me soon, and shortly a cloud of dust on the desert proclaimed the vehicle which came dashing up, a phaeton rig with two big horses. The five men inside were armed, with

rifles and pump shotguns protruding. One man pulled out a heavy leathern pouch, and another stood over it with his rifle. "Come on, Jack, lets go over to Wolf Saloon." "No," said Jack, "not till I get my receipt." The mild station man yanked out a receipt book, filled the blank acknowledging \$20,000 in bullion from the mine, threw the pouch into an open safe, and Jack with his receipt departed, leaving the gold brick to the mystic protection of that sign, "Wells Fargo and Co." Two able-bodied bandits could easily have held up the whole rail terminus.

When I started for the mine, accompanied by detectives and guards, we all carried pistols in holsters strapped under our arms. En route, we spent Christmas amid the smell of sagebrush and the glorious sunset lights of a purple desert. Once more I murmured, "So this is natural history."

I was employed to examine the Quartette Gold Mine, and the geologic mystery of the origin of a million dollars in dirt between a level 200 feet down and another at a depth of 500 feet. The million dollars was along a crushed, slipped, so-called vein, where a fault followed the upright bedding of just such gneisses, granite dikes, and schists as had made Crooks Complex in the Bradshaws. Where gold was richest, minerals were richest—beautiful orange-colored wulfenite, green chrysocolla, blue azurite, onyx, quartz, and calcite. Everywhere were quantities of gouge, or crushed clays, from grinding walls. Native gold particles were distributed through all this.

The schists were filled with lava fissure fillings, and the mine was where this pattern of bands was interrupted by a very ancient greenstone or basalt body. Hot fluids of the volcanic period, deep underground, had accompanied fault slipping or fracture where the ore was, the vertical fault parallel to the upright layers and across the greenstone contact.

Ore and gold particles were directly related to fracture, to the fault slipping on an upright crack of one mountain block against another, to the hot vapors depositing the mineral collection, and to renewed crushing and sliding on the mountain blocks. This was during or following some part of the volcanic period when all the cracks were injected with andesite lavas, or what the miners call porphyry. The origin of the minerals was in lead and copper sulfides which lie deeper down.

A hundred miles to the northeast is the Grand Canyon, and all around are granite mountains, just as in Arizona. These Searchlight schists are the same

Algonkian ancient strata, recrystallized and granitized, that make the inner gorge of the canyon, and are traversed up cracks by volcano-making lavas, such as dot the north bank of the canyon with crater cones. Above in the canyon are the horizontal strata from Cambrian up to the Coal Measures and beyond. The vast maze of castles and turrets is a net of branch valleys of the Colorado, trenching through these old seabed deposits.

Including Searchlight ore, the whole history going backward is top country desert, deep trench, strata piled in rivers and sea bottoms for 500 million years, and lastly faulting and cracking that squirted steam and made gold minerals over and over again during the last 100 million years. There were at least a dozen revolutions that lifted and lowered mountain ranges and continents for 2,000 million years, and the remains of iron-eating bacteria and of seaweeds and other living things that go back for 1,500 million years. Through it all are granite injections as a process, as a mystery, going over the whole range of years in different ages, and meaning what?

One of the puzzles of Grand Canyon, Bradshaw Mountains, and Searchlight—if not also of New England, the Black Hills, and the Yellowstone—is faulting. A fault is what a geologist means by a crack down deep where the country rock has dropped down on one side so as to make a discordance across country. Earthquake faults make a visible bank or step or sidewise slip, changing the surface after an earthquake.

The northwestern states are partly mapped as fault block mountains. The island of Hawaii has a series of fault step blocks southeast, slipping toward the ocean. The steep east face of the Sierra Nevada is a fault fracture.

Professor Shaler once stopped me on the street and said of my field work, “Jaggar, you don’t teach faulting enough.” Faults were shown along straight lines on the color maps of formation in the old Boston books, and were located by guesswork if glacier deposits covered up the ledges. It seemed to me that faults ought to be proved or else omitted from the maps. Probably I too was wrong, for faults or cracks completely concealed by soil and strata are tremendous unknown lines on the globe.

The Searchlight ore body is certainly a fault fracture, and so are those of Tonopah and hundreds of mines. It was digging that proved it. The cracking

and slipping and steaming and mud-making on the fissure are what brought up the minerals.

A question arises as to how much the Grand Canyon itself and its tributaries are guided by fault fractures under valleys. My impression was in 1901, and it still is, that “Jaggar ought to teach faulting” more than he then did.

The primitive ocean blocks of earth crust sank, while continents remained high, leaving the earth crust a mosaic of blocks large and small, high and low. Between the blocks spout the volcanoes. I have never agreed with C. E. Dutton that volcanic heat energy could come from shallow pockets under those fault blocks. Even he acknowledged the weakness of the argument. If the earth crust broke up and the blocks variously sank in the core matter, leaving continents as a complex of high blocks, then the blocks are deep and are still moving. The movements are in years, year-thousands and year-millions. Volcanism up the cracks releases core energy. So does much of fault movement, namely earthquakes. And these facts geologists do not appreciate.

So we get faulted river courses and fault cracks up which came fluids that transformed sediments of rivers, lakes, deserts, and seas into granite, felsite, and greenstone. These are the ancient names. There are hundreds of other, geology names. But geology produced no Faraday.

I disliked geology in 1902. And I disliked mining because of its secrecy and its devotion to profits. Geology failed to tell businessmen the mystery of granite, of felsite, and of greenstone. Astronomers told the same men of mysteries, and they were fascinated. Physiology led them to cells, plants, animals, and chemicals in the blood, solving mystery after mystery. Men, money, inventions, engineers, buildings, and staffs grew by leaps and bounds in those sciences. The best geology could do was guesswork—a mastodon, a big reptile skeleton, a guesswork color map—while seventy percent of the earth was seabottom rock, unmapped, and twenty percent more consisted of fractures covered with soil.

Seeing the Carnegie and Rockefeller laboratories and observatories, I grieved for field geology. The public did not even know that granite, the mystery, is the commonest rock and that quartz, the gold maker, is the commonest mineral. Nor did they know that both are almost absent from

the whole Pacific. Nor that geology is almost ignorant of their origin and injection, if it is injection. Here was the globe, the end product of astronomy, the most fascinating research in the whole range of science. The source of all raw materials of commerce, yet its fire-made rocks and its seabottom rocks remained a mystery.

Before leaving the Grand Canyon, let me record my impressions of the erosion. It is a gorge a mile deep usually described as “cut” by the Colorado River. As I shall show in discussion of experiments with the Grand Canyon model, it is possible, in stratified layers yielding grit to flowing rainwater, to cut a deep canyon by surface runoff. It is possible for underground water and tributaries from side rainfalls to increase the volume of such a stream greatly in a hundred miles. But Dutton’s showing of upheaved and downdropped big blocks of broken mountains, and such obvious breaks as the Tonto and Bright Angel faults shown to tourists as traced out by Bright Angel Canyon, prove that the earth crust is broken. And Searchlight showed a fault to be a water supply.

The enormous canyon appeared to me to be a million-year break system of earth-crust rotting. The water is a giant modern grinding mill of rainfall, underground accumulation, and transport. But with five great erosion surfaces shown in the discordances, from 2,000 million years ago to the present day; and with upheaval of the high plateaus in block faults, and bent strata age after age; and farther north with recent volcanoes that spouted up the cracks, it seems more vivid to think the valleys at least partly water-filled cracks and chasms. Volcanoes cannot be shallow. The canyons and the great bend are different from the Green River source, because of upward push in waves. The up-push of the Uinta Mountains is well known to have been slow. It kept pace with the ruptures followed by the river. Going back to Daubrée, rivers follow cracks much more than do the textbooks.

In 1899 two things happened which affected the rest of my life. First, Director Walcott asked me to furnish estimates for a Hawaii geologic survey, a request which eventually led me to Hawaii. Second, the Yakutat Bay earthquake snapped on an astonished world, though most of the world didn’t know it.

The Yakutat Bay earthquakes in Alaska, in September 1899, were accompanied by the pushing up of the bedrock shoreline by forty-seven

feet. Lowered beneath the sea were whole forests, on glacial deposits pulled down by submarine landslips. It was an uninhabited region at the foot of Mount St. Elias, along a fjord penetrating far into the mountains. It came in line with the Aleutian trench, under the Pacific, 4,000 fathoms deep. The earthquakes lasted two weeks.

This colossal movement of blocks of the earth's crust hundreds of miles across gave one the impression that we knew little of what was going on. Remembering that seventy-two percent of the earth's surface is covered by oceans and that less than ten percent is really inhabited, I awoke to how much there was to learn. If whole forests and their roots could float away into the Pacific currents, with all their plants and animals and seeds and bacteria, what might not have occurred in past ages, when such jostling of crust blocks was common.

But before I was to experiment with live volcanoes came a decade of laboratory experiment.

CHAPTER II

Imitating Ripplemarks

“The Constitution is an experiment, as all life is an experiment.”

At the end of the century my experiments with the sclerometer, and with the class in experimental geology, steered me for years into laboratory experiment. Europe was headed toward geophysics and geochemistry, meaning chiefly mathematical and statistical analysis. My vision was nonmathematical, though I used pressures, temperatures, clocks, and yardsticks to measure erosion, sediment, warping of strata, and melting.

This took me away from petrography, for the polarizing microscope was dealing with infinite series of minerals and molecules. I could see nothing but infinite penetration into the smaller and smaller. Clarence King and Frank Perret had been on the way to infinite journeying outward to the bigger and bigger.

The guiding formula was “erosion, sedimentation, deformation, and eruption.” Measure these on the globe, imitate them with mud pies in the laboratory. Compare the global examples with the mud pies. Try to get the mud pies to illuminate the gigantic stream systems, flood plains, sea bottoms, folded mountains, intrusions, and lavas of the earth. Then try to measure in the field those processes with observatories. So, to me, came the transition from collections to experiments.

The machinery of nature, whether with sand heaps or sand grains or coral pebbles, is the same. It is impelled by currents flowing over loose materials which make eddies in the lee of lumps. The eddies are either billows or cyclones. At the middle they are billows; at the ends they are cyclones. The billow eddies obstruct the heaping. The cyclone eddies lengthen the heaps right and left of the current direction. George Darwin studied the eddies by means of a drop of thick ink in a glass tank on top of a ripple ridgelet. The ink migrated to form underwater billows and cyclones, or vortexes. He used a dropper to place the ink globule, and then watched the vortexes form as he oscillated the tank.

Low parts travel fastest, namely the points. High parts build on the upstream side, and travel slowest, and the stuff tumbles over the crest line and is corniced by the eddy. Snow does it, pebbles under sea do it, and marine life adapts itself to it, wherever the food supply is best.

In the study of ripplemarks, Harry Gummeré, a graduate in astronomy, was my collaborator. Ripplemarks are made by back-and-forth eddies on the bottom, while big waves oscillate the water. We moved the bottom instead of making waves in the water. A glass plate sprinkled with sand under water in a tank was oscillated back and forth horizontally. It was clamped under a carriage which oscillated on wire tracks stretched across the tank. A string pulled the carriage against an elastic on the other side. A wooden wheel and crank, set upright edgewise, had holes and pegs to pull on the string, and the crank turns were timed with a metronome. The holes in the flat wheel were a centimeter apart, so that a revolution of the wheel pulled the string for every two centimeters of travel of the carriage. Thus the sand-covered plate was jerked back and forth under water two centimeters, four centimeters, six centimeters, and so forth; once a second, or two seconds, or three seconds, and so forth, by beats of the metronome.

The result was beautiful ripplemarks on a glass which could be lifted out of the water, dried, and placed over blueprint paper to preserve the record. The sizes of ridge to ridge ripples were from a fraction of an inch to two or more inches. The little ones diminished to zero when the jerking was small, the big ones washed out when the jerking was too big.

The blueprints showed that both length and speed of strokes (amplitude and acceleration of motion) made the ripples increase in size, and somewhere between the largest and smallest sand ripples was the optimum perfection of ripple form. The blueprints look like mackerel skies. And mackerel-sky clouds are billows of condensation between an upper cold stream of air and a lower moist one. In between are the same back-and-forth billows of vortex as in our sand.

At a geological conference at Harvard I showed blueprints made directly from glass plates covered with artificial ripplemarks. At the same time I exhibited rock slabs of fossil ripplemarks and photographs of others shaped like horseshoes. These were variants of the rippledraft process seen on sandy beds of running streams. I also showed photographs of swash marks

running along the upper steep slopes of beaches. And of the wind-formed rippled drift of dry sand dunes. From the deserts of Peru come photographs of *medaños*, or crescent dunes, hills of sand tapering to curved points at both ends. The points are downwind, the high horseshoe toe of the hill is upwind, and like a coral atoll the edifice is current-formed.

Ripplemarks can form in hundreds of fathoms of ocean water if the storm waves on the surface of the sea are big enough. A particle of water on the crest of a wave is lifted up and down in a long vertical ellipse. A particle deep down under the wave is lifted fore and aft in a long horizontal ellipse. Under a three-hundred-foot length of wave in the English Channel in deep water the bottom particles of water are shoving sand back and forth, and making packed ripplemarks.

A big sand grain becomes a lump for small sand grains to bump against. They make a heap which piles up and lengthens out. The heaps merge and we get a tightly packed and ridged sandy bottom. Each ridge has an eddy first on one side, then on the other, as the water particles reverse in direction. Oscillation builds first flocculence, then alignment, then even spacing. The opposite sides of a ridge have equal slopes.

Rippled drift is made by a current in one direction. It is usually not so regular or in such straight ridges as ripplemarks. If a stream of water is jetted over sand round and round in a ring-shaped tank, ridges will migrate along the bottom, but they are smeary. The regular ripples in dry sand on dunes have flatter slopes upwind, steep scarps downwind. They are regular, probably because wind blowing is intermittent and back currents occur. So they become more like ripplemarks.

On the bottoms of water streams, the horseshoe rippled drift requires a nice adjustment of lumps and side points migrating downstream. All rippling requires a sand of mixed sizes of grains. If they were all alike they would not ripple, for the larger grains have to obstruct the smaller ones in order to produce the ripple pattern. Rippled drifting as a whole is a building mechanism. Mixed with wave currents which move beaches along, including beach pebbles, it can be compounded into building oceanic islands. The crescent dunes of the desert are dependent on the prevailing winds being loaded with a sand supply at a windward erosion source.

Oceanic currents depend on the winds, like the trades in the tropics, and an obstructing bank or shoal adds surf action to the streaming. If corallines and *Tridacna* clams and crabs add organic cements, a horseshoe hill is built on the sea bottom. Big eddies will do the same kind of work as little eddies. This phenomenon extends all the way from the galaxies of stars with their beautiful spirals, to the spiral eddies in molten lava rushing down a pit crater, or to the streaming of protoplasm in a plant.

De Candolle, the great botanist, studied rippledraft in order to try to solve the most abstruse problem in all biology, the unsolved mystery of cell division. At some critical point a budding cell decides to form a partition and divide in two. Why or how? De Candolle thought that the protoplasm granules circulating around the cell walls might start regular lumps on those walls, and so build rippledrafts and make eddies.

Thus a current and an eddy and mathematics might start many of the doubles, triples, hexagons, and stars of the world of shells and living tissues. And the cells could pile up in symmetry in the submicroscopic world.

The erosion of the earth's surface reveals symmetries. River maps look like trees with branches and with rivulets as twigs. Other symmetry is in the horizontal plane of the ocean, where headland furnishes pebbles and the sand sweeps into pure curves of beach and bar and cusp. So a delta builds into a lake of leaf shape and annual layers are added as the flood seasons come.

Some of the fingerlike drainage of erosion cuts into plowed lands during a rainy spell. This suggests what might be done with a spray, a mud bank, and a tank, to see how the finger valleys form. This erosion of the runoff of water was imitated in the Harvard laboratory.

A beautiful river pattern on a slope, like the trickle of raindrops on a windshield, was made by tipping up a rectangular glass plate covered with very liquid clay. A portion clung to the glass, and exquisite fernlike streams formed on the upper half of the plate, with a bank of distributaries of V-shape on the lower slope.

This glass plate was used for a surface of stamp mill slimes, of thicker beds, and was eroded with an atomizer and water by means of a barbershop

air compressor. The slimes are very fine pounded sands with angular fragments. To get a stream pattern, this is necessary, so as to have fine grit to cut down the rivulets between the coarse grit remnants. This resembles the requirements for ripples.

The spray was kept going for hours. Meanwhile the river pattern at the steep sides of the sloping plate ate into the bank of sediment, robbing the streams of the main slope, because the side streams were oblique cascades. They dug deep, took off the water, and left the main slope streams without their headwater drainage. The pattern of the main slope became the headwater branches of the side streams, the streams which in plan drained over the edge of the uplifted plate right and left. This was somewhat like stream robbery.

For example, the Lewis River at the south end of Yellowstone Park once drained Yellowstone Lake, including the Lamar River, which is now the headwaters of Yellowstone River. The Yellowstone plateau formerly drained south into the Snake River and the Pacific Ocean. The Yellowstone River headwaters suddenly tapped the system, thanks to geyser erosion and acid corrosion, and the Yellowstone Canyon cut down rapidly, reversing to the north the outlet of Yellowstone Lake. Thereafter the lake flowed into the Mississippi and the Gulf of Mexico. At some critical time about the glacial period the continental divide made a leap of thirty miles from the present head of the Canyon to the neighborhood of Lewis Lake, or from one end of Yellowstone Lake to the other. This is stream robbery.

Spray and runoff and rainfall and wash did not alone cut down the Yellowstone Canyon. The essentials were the rotting of rock and the pull of gravitation on the fragments. The Yellowstone rotted away on the north side, but it was hard granite and mountain-built quartzites on the south, toward the Tetons. Hot spring rotting, geyser erosion, acid waters, and sulfur decomposed the north country. The underground water head followed the easiest channels, and the canyon was the result. The canyon line encircles Mount Washburn, the old volcano, and conceivably is over an old crack concentric to the dome.

Water is a transporter, and cracking opens ways to the rotting agents. Only in rivulets and floods does water actually corrade, or grind, the bottoms of streams. In our spray and fern patterns there is analogy to

rainfall springs on flat strata, but nine-tenths of the elements of erosion are left out: jointing, weathering, ice, faulting, gravitation, rotting down, quaking, solution, sliding, and last, spring water.

Erosion by sliding continues by wind action in desert mountains, and on volcanic cones under bombardment, and by rocks snapping under chill and sunshine on the moon. Creep of loose stuff is the greatest eroder on earth. Rainfall cloudbursts certainly help, especially where soil is not held together by a mat of roots.

The process of erosion is supposedly slow, as all geological processes are slow, if we neglect the possibility of such submarine landslips or supramarine upheavals as occurred at Yakutat in 1899. But even New England has floods, hurricanes, landslides, forest fires, and cloudbursts which are exclamation points on an otherwise sleepy history. And in the past it has had ice sheets, and subsidences beneath the sea.

In other words, the making of valleys and stream patterns for the map is accented occasionally, and the occasions may come in climatal waves unknown to us. The stream patterns in the Bad Lands, Tennessee, Pennsylvania, the Grand Canyon, and New England make very different maps. The rotting of the rock, limestone caverns, rainfall, faults, and sloping underground strata bearing spring water all influence these maps. What is erosion and what index is written on the land to say the Grand Canyon and tributaries are being carved downward faster than the Mystic River in Boston?

Ralph Stone tackled the Mystic River, and marked ledges and set stakes opposite the flood plain meanders. The idea was that ledges split by winter freezes, and that the meanders of a stream build on one side and cut on the other. Maps were made repeatedly, and the ledge cracks were measured in millimeters. Some movement was found, but a college year was not enough time. If we could combine as a motion picture, photographs from the air taken once a year for many years, doubtless the film would show that the stream meander pattern is migrating toward the sea like a wiggling snake.

Stone next made a model three inches thick in a tank of water, by sedimenting sixty-one very thin layers of marble dust, coal dust, clay, red lead, and sand. He tipped it up as an island and sprayed it in periods which lasted one to ninety-two hours, up to a total of 719 hours. A forking stream

and its delta were formed in the lagoon of the tank. The stream cut a canyon with waterfalls, treelike branches, esplanades, and a flood plain. There were three principal hard white multiple strata layers in the model, separated by sand. The white layers made waterfalls and were eaten back to form the canyons.

When the cross section of the delta was sliced with a knife, it showed three white layers foreset at thirty degrees under the tank pool and separated by more sandy strata. The bottommost of these was the sediment of the top thick marble dust layer of the model as first eroded by the spray, and the top frontal layer of the leaf-shaped delta was the product of the erosion of the canyon bottom on the lowest of the white layers. This must happen in nature where one formation in reverse order is derived by river erosion undermining a stratified older pile of sediments.

We called this the Grand Canyon model, and it showed many features similar to those of South Dakota Bad Lands and the Colorado River drainage. It was strictly rainfall erosion and stratification soakage and seepage. The model surface sloped ten degrees, the high divide at the top had a backslope of forty-five degrees, and everything was sprayed for two months with special hose nozzles, making during part of each day a mistlike rainfall.

The steep backslope did not trench itself at all despite its steepness. This slope, on the contrary, absorbed moisture and carried the rainfall underground down the dip of the strata to add spring water to the main streams. The backslope was a “steep escarpment,” supposed in physical geography to migrate by trenching backward, but the rills never gained volume enough to cut into it. All the water volume acquired its grit for cutting from the large surfaces, which were gradually tilted in the direction of the rivers.

When the complete series of experiments on erosion and sediment was published, it showed that the treelike branching of rivers is dependent on underground water surfaces; that meanders on a flood plain are partly a bubbling-up process of flood-plain soakage; that when side tributaries form by undermining, the upstream branches cut off underground water from the downstream branches; and that when a country is tilted in one direction, there is a tendency to parallel streams, separated by intervals controlled by

underground water areas reached by the undermining tracery of headwater springs.

This arborescence in a spray model is a regular and delicate adjustment, where a bunch of tributaries is not mere catchment of rainfall, but is the product of sheet flood in belts of underground water related to the tilt of the country. Arborescence of river drainage on a surface of flat strata, like the coastal plain of the southeastern United States, is a rhythmical pattern of exquisite design capable of reproduction and study in the laboratory. It is a mathematical forking and headward development dependent on volume of water, undermining impermeable strata along permeable ones. And after the "tree" map is formed, the bulb of branches and twigs and underground leaves of spring water holds all the downslope country in its "shadow," so that no new rivers can form there. This is what makes our great maps of river systems. It is not haphazard. It is a vast ocean of underground water, with mountains of water and valleys of water.

A great lake marks an underground soakage water level. A riverbed marks an underground seepage topography. The sea of water inside a continent is just as much a map of hills and dales of water as the land is a map of the hills and valleys of geography. The water is dynamic, it is flowing. The land surface is dynamic and rain fed; it is creeping soils. Together, groundwater and rivers are melting down the landscape as a living thing. Man dams the water and uses the power of the erosion melting down the land.

When we went to Haystack Basin north of the Yellowstone Park, we found that all of the mountains surrounding it were audibly crumbling. Ultimately, the continent is all one thing: a falling body of rotten rock, ice, water, sand, boulders, and soils, self carved into valleys and mountains, always tumbling. And down below are the fault blocks, prisms of earth shell over the white hot core. And that also is eternally in motion, irrupting, earthquaking, lifting, falling, scraping, heating, cooling in waves through the ages. Man is very tiny, but if he listens he can hear the earth's heartbeats.

At hot springs the water mantle meets the hot earth shell. So the geyser basins of Yellowstone, California, New Zealand, and Iceland are a hot part

of the great erosion system of groundwater. This brings us to the next group of experiments, the making of artificial geysers.

Geysers as eroders show that the under earth is hot and is invaded by rainwater. In exceptional volcanic places the water is boiling hot. The Firehole River of the Yellowstone is carving down basins of solution faster than the regular geysers are building up siliceous sinter. Here is boiling-spring erosion by solution. It may be called the extreme thermal aspect of ordinary spring-water erosion. How does spring water erode? By bubbling up under the beds of rivers. The bubbling out of springs starts rivers, and flood rainfall starts soil gullies; land sculpture is the result.

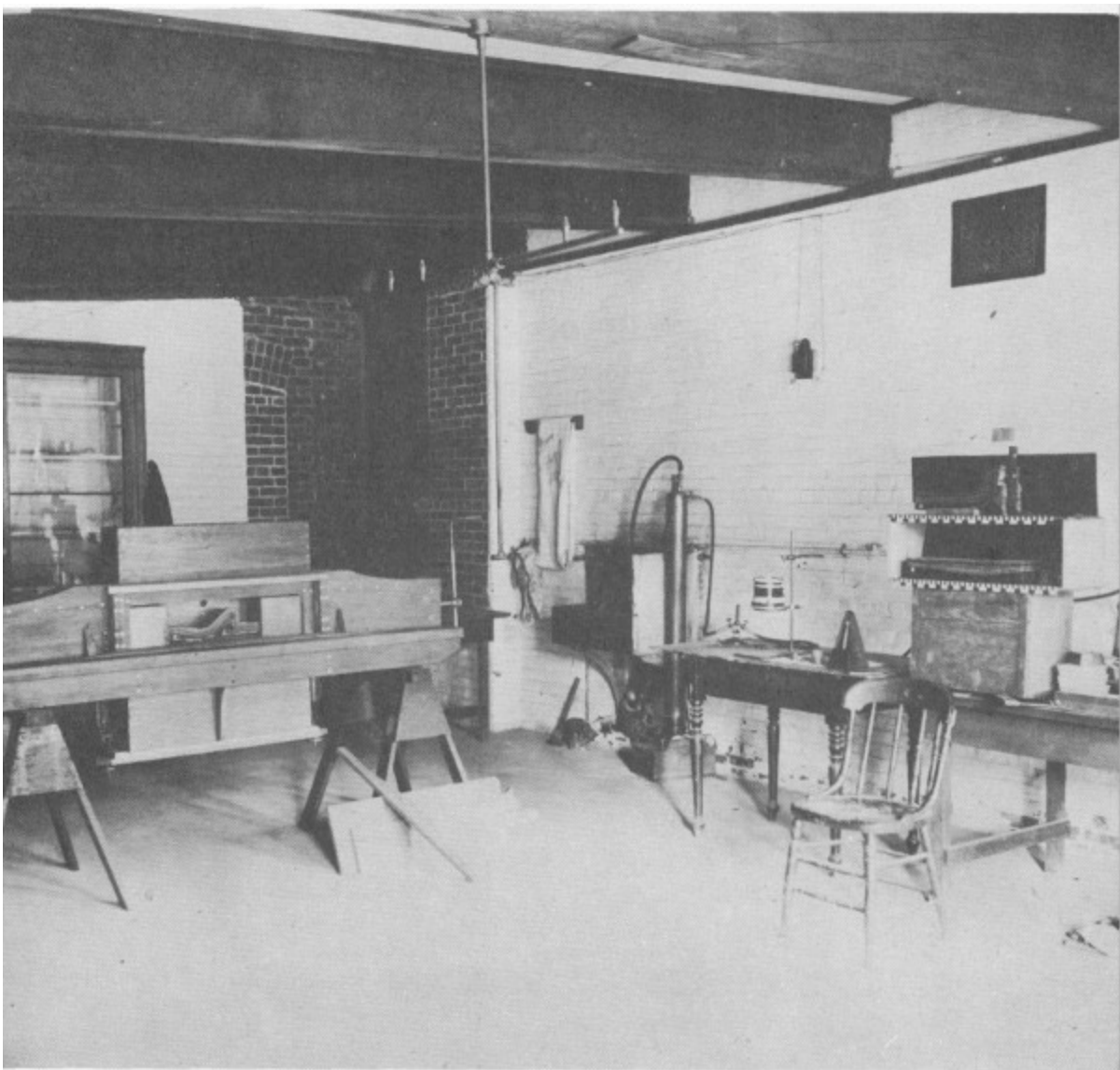
We introduce geyser experiments here because boiling springs make drama out of ordinary springs, just as active volcanoes make drama out of buried volcanoes. Ordinary springs and buried lavas intruding invisibly are much more important and extensive than geysers and volcanoes. Most people never think of a spring as one of millions bubbling up the beds of brooks and rivers and sea bottoms.

Most people never think of volcanoes erupting—properly speaking, irrupting or inrupting—under Kansas or Brazil. Nobody denies those places are hot underground, but it all seems remote. Yet every spring is thermal if there is heat escaping through the rocks around it.

Geyser basins lower the country around them and leave hills in relief. The proportions of basins and hills depend upon the runoff of rotting and dissolving rock. The shape of a hill standing high, what Davis called a monadnock in New England, depends on its whole history, not on its hardness. Ascutney Mountain stands high as a lump because surrounding slates have rotted down. Mount Monadnock may stand high because the springs under the river pattern of cracks neglected it in the rotting and crunching of a continent.

Dynamic weight eternally falling makes low places. Hardness against weathering makes a mountain high only as a relic or residual. It is a node in the gigantic process of gravitation rotting and the spring squirting of groundwater. The water heats, rises, dissolves, siphons, springs up, and transports dirt. Underneath is a definitely heated earth crust.

Accordance of summit levels of mountains and hills as one looks across country does not have to represent an upraised plane surface. There is more undermining where the spring squirting is most voluminous. When spring squirting is equal, the opposed slopes of a valley adjust themselves. The tree line, the snow line, the rain line, and the wind line are definite levels of erosion. Under it all the rotting rock is falling toward the earth's center, slowly, creakingly. The everlasting hills are not everlasting, they are everfalling; rocks, boulders, slopes, waters, gravels, sands, and muds. And adjustment to the atmosphere and groundwater surface is irresistible.



1. *Experimental Geology Laboratory, Harvard University, 1900*



2. Fountain at edge of lava lake, May 17, 1917

The notion of erosion pulling down hills to a flat plane near sea level is fascinating to geometry-minded people, but not to the mechanically minded. A flat plane near sea level in the Mississippi delta is where the river has swung right and left against valley walls, over its own flood plain. A flat plain, secured by ice sheets or planed off by encroaching wave action as land sinks is mechanically probable. In these circumstances we look for river or ice or wave-beach deposits. But an “almost plane” occasioned by the multiple action called erosion down to base level is to me the delightful

dream of map students. If a landscape has been planed off, a machine router or planer did it. The great rivers of China have had a long time to bang back and forth against their confining boxes of rock and on top of their own mud.

To return to geyser-spring experiments, I built a simple quart flask surmounted by a four-foot glass tube. At the top the tube rose through a cork in the bottom of a two-foot pan. In the side of the cork of the flask was a second tube with a hose leading up to a reservoir bottle of water. The reservoir bottle could be raised or lowered. If the water in it was level with the pan, there was hydrostatic equilibrium: the pan a pool, the bottle a source, the flask and tube full. When we applied heat to the bottom of the flask, the water boiled, the pan overflowed, and some cold water from the bottle chilled the flask. The pan had become a boiling spring.

Next we lowered the reservoir bottle. The reduced head of water permitted no overflow at the pan, and steam bubbles accumulated in the four-foot upright tube. The boiling point was controlled by four feet of water pressure. If the bubble lift reduced this to three feet, there was a lower boiling point, the pressure was reduced by overflow above, and the whole flaskful boiled. The geyser tube became a regular geyser at intervals of a minute and a half, with eruptions enduring twenty seconds.

This was a miniature of Old Faithful in the Yellowstone. Old Faithful is bigger, its intervals average sixty-five minutes, and they range from thirty-one to eighty-one minutes. It jets up 150 feet for a period of four minutes. It throws out 3,000 barrels of water at each eruption. Our little machine threw up about a pint to a height of four feet.

We hear much about soaping geysers as an artificial stimulus. The apparatus in our laboratory showed the effect of soap right away. When some soap was put in the pan, the intervals of a minute and a half shortened to one minute. Soapsuds accumulated in the tube and depressed the water to the neck of the flask. The multiple bubbles, film against film, made the water system viscous. The myriads of tiny steam bubbles formed so fast that they shortened the lifting time for the column. If the height of the reservoir bottle was so adjusted that the geyser didn't quite know whether it was a geyser or a boiling spring, the soap made the decision, and the thing went off with a bang.

This simple group of experiments makes springs very real. The Yellowstone explosive springs differ from other springs in having superheated steam from live lavas to heat them. The rock is cracked and the water is doing a job of solution and deposition. It deposits stout tough silica around some openings and builds them up against the head of groundwater, and they become geysers. It deposits lime dissolved off underlying limestone at Mammoth, and this makes sculptured terraces but not explosive springs because the temperatures are not so hot. In both lime and silica regions, blue-green algae, which love hot water, decoratively sculpture the pools.

Like a magician I exhibited the artificial geysers before New York and Boston science academies, and gave the summaries of the results of our geyser experiments, as follows: (1) Boiling springs are like other springs, controlled by the head or pressure of underground water in the hills. (2) Upstreaming of heated water and building up of silica (convection is the scientific jargon) may push the vent of a boiling spring even higher than its source (reversed head). (3) In this delicate condition, even rainfall or sinter building up or outburst at a lower level or clogging of a pipe may change spring to geyser or geyser to spring. There are many more boiling springs than there are geysers, and many more hot springs than there are boiling springs, and the word cold means nothing at all. There may be boiling springs under New York City if you go deep enough. That is why the riot of geyser apparatus is worth thinking about. (4) Irregular geysers overflow continually, regular geysers discharge their waters only during eruptions. Both are methods of feeding rivers, just like any other springs. But there is a lot of volcanic heat underground.

This brings up the question of how much a volcanic eruption is like a geyser. Geologists apply a glib word, phreatic, to Japan's Bandai Volcano, which blew steam and rocks out of the side of a mountain and dammed a river. Hawaiian volcanoes squirt liquid basalt up a crack with flames and red fume and sulfur gas, and almost no steam at all. The answer seems to be that the Palisades of the Hudson may once have been Hawaiian lava eruptions and, further, that lava is still erupting there if you go down deep enough. New York doesn't know about it, but it sensed it in 1886, when it felt the Charleston earthquake.

All that Catskill water supply of the great city is in cracks above the level of the deep lava, and extends out under Long Island Sound. If the Hudson fault fissure wiggled a little more than usual, and if the deep lava lowered and pulled down some of the Atlantic water, an eruption like Bandai is not impossible in the Watchung Ridge of New Jersey. This is not likely; but the globe has been through revolutions and cataclysms, and the Watchung explosions might start a new geyser basin. Something like that happened in northwest Wyoming in the Pliocene age, during 11 million years, next preceding the ice ages that began 2 million years ago. And the Yellowstone was the result. We shall see more volcano geysers.

Next, the making of deltas became a hobby in our laboratory, in connection with the old leaf deltas scattered on the New England landscape, partly covered with trees within the grounds of the country villas about Boston.

Delta deposits extend upstream, within the mould of the cavern within ice of the glacial period. Thus the map shows a snake-like ridge of gravel, ending in a maple-leaf flat, with lobate frontal slopes. These slopes were much steeper where the dump of the stream on the delta fell over the beach line at the lagoon or lake level in which the delta was built. This was like the delta shown in Stone's erosion model.

Stone prospected the idea of torrential deltas in a tank, while E. W. Dorsey and I started a tank imitation of the glacial sand delta. In the glacier, the ice tunnel had been supplied with water by melting through the ice crevasses, just like tunnels seen in Switzerland, floored with sand ground up by the ice. There was thus a torrent pouring along inside an arched tunnel, the mouth of which emerged on a delta in a pool, with water surface either at the tunnel level or above it against the rounded front of the ice mass.

In imitation of a rounded bank of ice with a pool of water in front and with a subglacial meandering cave fed with sands and a torrent, an apparatus was built and supplied by a hose. A sheet of lead was bent in the form of the glacier surface, with an arched opening, and set in our tank. This fitted over a tunnel of sheet iron, soldered so as to meander in plan, and fitted at its upper end with pipe and hose connection. A sheet-iron funnel rose from the upper end of this artificial cavern, wherewith to supply

different colored sands to the model subglacial river, represented by the hose jet and iron tunnel. The iron tunnel ended flush with the leaden arch.

The object of the experiments was, first, to set the leaden glacier in a pool of water in the laboratory tank. Next, to jet water through the tunnel, supply sediment in successive colors through the funnel, and let that accumulate on the bottom of the tunnel and in a delta in front of the artificial leaden glacier. The deltas and their sliced cross sections in different experiments represented the noted difference of kinds of sand supply or difference in water level of the pool. In one case the water level was below the ceiling of the tunnel where it emerged from the arch entrance. In another, it was above the cavern mouth, so that water of the cavern stream, debouching from the submerged cavern mouth in the lagoon, spurted up with its mud and made a half crater against the glacier front.

These experiments illuminate the gravel-quarry sections of Massachusetts. In those cuts in eskers (serpent ridges) and sand plains (glacial delta fans) were seen topset beds, or flood wash, or foreset beds at forty-five degrees which are the sublagoon frontal wash, and occasionally backset beds where cavern wash gushed upward.

So our cross sections, cut with a knife in the delta, and the winding cake extending upstream in the cavern showed topset, foreset, and backset strata after draining the tank and lifting out the apparatus. From the embryo delta the flood-plain beds overlap the earlier frontal, or foreset, beds. The frontal beds are always under the lagoon. The flood-plain beds (topset) were made by a meandering river course under the air. Always this plain is built at beach level as a wash fan shaped like a leaf, with the cavern stream bottom as the stem of the leaf.

New England has been covered with mountainous ice, miles high. Subglacial streams and subglacial clear ice caverns are abundantly found at the lower ends of all glaciers in the world. They merely represent the melting snow and ice in pulses of sunshine, snow at the source, ice in the course, crevasses and gravitation making water seep through. This water shapes a channel for itself and erodes a sewer system of scouring along the bottom of the subglacial valley. This grinds and melts the bottom ice into arched caverns; and the sediment builds up on the stream bottoms,

eventually carving the roofs of the caverns into high arches or arcades. The subglacial caverns are self constructed drainage pipes.

The glacial stream is really a river flood cutting its valley. The ice river grinds and scrapes, and the water under the ice pipes and drains the melting. The ice carries chisels of broken rock. The enormous weight, in gliding plane layers of ice, flows in accordance with the crystal laws of snowflakes and ice crystals. The moraines, or debris fields, at the sides and on top and underneath the eroding ice jumble yield mud and sand and boulders. The torrent underneath removes the rubbish.

The delta in front follows laws of sedimentation. If there is no lake in front, the delta is a flat wash fan, or valley flood plain. All these things become clear to the student who makes a baby glacier out of tinware, sand, a tank, a hose, and a faucet.

I have spoken of cataclysms, or what early geologists called catastrophe, happening occasionally in the world of erosion and subterranean geysers. Such were the Yakutat crash and the Bandaisan explosion. But each glacier-period field, like an ice mountain over Europe and America, constituted a cataclysm lasting 500,000 years, and this happened four times even in the centuries of early man. The Mediterranean and the Great Lakes are offspring of such cataclysms. But Lyell carried the doctrine of uniformity to extremes; he thought that what man sees is what always happens. I do not believe Lyell ever realized that earth or sun might conceivably explode in a month of our time. Again, this is not likely.

The opposite of uniformitarianism is occasional catastrophic trigger. The process of erosion pulls the trigger for sudden deformation. Slow deformation pulls a trigger for eruption. Eruption triggers internal intrusions. The Frank Landslip; the Charleston, San Francisco, and Napier earthquakes; the Pelée eruption; and the Yakutat upheaval all created terrific surprises for geologists.

The gigantic intrusions through millions of years from the core of the earth, made of white hot star matter, percolating to surface volcano belts up 1,800 miles of permanent, primitive cracks, are mostly balanced by the crustal weight. This is the adjusting globe. But the intrusive mechanism, under tides in the rock and in the oceans, always in motion, pulls the trigger for the big geologic revolutions.

The very deep broken earth blocks shift, volcanism between them heats the surface, floods the surface with gas foam, and lifts areas of surface by heat swellings; and on the surface, what was a glacial period gives place to a volcanic period. The last of these was the Miocene Tertiary, with large-scale volcanic eruptions all over the world.

Comparing Boston with the Black Hills showed underground eruptions in the latter, for which a warping uplift pulled the trigger. These were the rock cisterns or lenses of porphyry injected among the strata. The time of this was Miocene or Eocene Tertiary, probably later than most of the volcanoes of the Yellowstone, farther west.

Boston, on the other hand, was making black basaltic dikes, probably identical with the volcanoes of the Berkshire Hills and New Haven, of the age of the big reptiles, 150 million years before the Black Hills injections. The trigger which pulled off the Boston eruptions was the Appalachian warping. That which fired off the Black Hills was the Laramie revolution that pushed up the Rocky Mountains.

The injection of lava lenses in the Black Hills was a form of deformation of strata which we experimented with in the laboratory. The layers of sandstone, limestone, and old ocean muds covering over the arch of these hills were injected by dikes or fissure fillings from below. How would injections behave?

With Ernest Howe as my associate, I arranged a square tank for sedimenting sand, plaster powder, coal dust, or marble dust in layers under water. Under it was an iron cylinder in which wax could be melted. A screw piston pushed the molten wax up to inclined or upright slots in the middle of the tank box. The water was drained off, and the hot wax was injected up into the strata. The tank sides were taken down, and hardened lenses of wax were sliced vertically with a hot knife to show what had happened to the strata by the process of wax intrusion.

In some of the experiments 300 pounds of shot were piled over a cloth layer on top of the strata to imitate the weight of natural sediments. This was before injection of the hot wax, and the result was a neat dome of deformed layers at the surface, a domical hill over a lens of wax inside. This hill was eroded with a spray of water to show what kind of radial valleys

would form. Such radial streams were found in South Dakota, with infacing escarpments, around some of the dome hills made by laccoliths.

From the beginning it appeared that a lens of injection would form, that the strata would arch over a dome of wax. The arched strata stretched on the crest and the breaks gaped upward, while the side bends cracked gaping downward. It was there that the wax could break its way upward and make a volcano. Some nice little experimental volcanoes of wax-built cones and craters formed on top of the model.

As with all folded strata arched downward under weight, the cracks on the bend of a downward arch, or syncline, admit lava from below, whereas the cracks on the upward arch, or anticline, are held tight, closed by the weight of strata above. Thus an intrusive dome will not erupt through its crest, but through its sides.

The results of all these tests showed that rigid beds carried the arching force and that soft beds were most invaded and pushed aside by the wax. The steepness of curvature of arch varied with the load. An inclined pipe formed an irregular lens thickest away from the incline. In a hard bed ruptured on a downbend, concentric fractures around a dome let the lava up to higher strata.

On the crest of a hard bed the fractures are like the spokes of a wheel, but they do not make dikes; they yawn open upward. Liquid wax tended to spread as a thin sheet in soft layers of strata, stiffer wax tended to arch up in a steeper dome. Rapid injection made a higher and smaller dome than slow injection.

Compared with the arching up the whole long mountain oval of the entire Black Hills dome, with granite on the crest, this intrusion of wax only imitates the small domes of lava intrusion or injection, where the injection carries the energy or stress. Indeed, in nature, even the lava lenses are influenced by the buckling that is going on in the strata under stresses of crust warping. For the warping crust of the earth is always pulling the trigger and straining the strata. The lava rising from below seeks out the weak places and assists the buckling, as well as following the most incoherent mud or shale beds.

When it comes to the big oval of the whole Black Hills uplift—swollen up like the Rocky Mountains during millions of years and within which the lava injections were only an item—we are dealing with a push from below or an expansion that swelled up the pre-Cambrian ancient rocks as well as the later granites. Such swellings were doubtless made again and again in Massachusetts. There, also, we find lavas and granites and Red Beds and glacial boulders, older than the Appalachian Mountains, as well as younger. The younger Triassic lavas are definitely erupted between fault blocks.

All that our experiments showed was what melted stuff will do in strata under weight, when the force of melted stuff overcomes that pressure to find a place for itself, although the weight may be more or less lifted by big arching that is taking place on a big scale. The arching is bigger than the hydraulic or gas pressure squirting.

There is another possibility besides buckling. This is faulting, or movement of deep crust blocks the boundaries of which do not appear. The deep crust is a movable mosaic above the core, and this movement renews itself, now here, now there. Dutton shows that we may think of the Rocky Mountains this way all the way out to the Pacific coast. We may have the core fluids sucking down the blocks, the volcanic fluids pushing up the local strata. And the volcanic fluids in cracks are the degenerate gassy top remnants of the core fluids which man has never seen and which are 1,800 miles down.

The boundaries of the crust blocks do not appear because the whole first shell of the globe is buried under lavas and intrusions and crystals and mud, meaning by mud, countless dumpings of lakes and rivers and seas through 3,000 million years. Such is the kind of thinking started by making wax injections.

It will be seen from the experiments that whether we are imitating underground heat with a Bunsen burner to start a geyser, or overground cold with delta apparatus to simulate a glacier, we are dealing with erosion of the earth's surface. Erosion started with the first attack on lava by the atmosphere or by sea water. Never was the pristine lava anything like the magma inside the globe; it snapped and chilled and oxidized. Whether we call it basalt or obsidian, it degenerated. Moreover, it degenerated in the outer crust when it loosed its gases, heated itself and the rock wall, found

groundwater and free air, and started oxidation new to it. Thermal action is just as much concerned with erosion as is rainfall or snow. Therefore, whether injecting wax and swelling strata or imitating geysers and ripplemarks, we were experimenting with volcanoes, for the crust of the earth is fundamentally volcanic. For the purposes of this book these facts demand reiteration.

In what are called geosynclines, or earth sags, the great beds of strata are accumulated. They are the dirt washed from highlands into midland seas. They are strata of sandstone, mudstone, or limestone; thin films in comparison with the fire-made earth crust. It was the wrinkling of basin fills by expansion or end push that built Himalaya and Appalachia. The mountains are etched out of foldings and overthrusts and faults by rotting and water transport. Pressing the strata endways to wrinkle them is called mountain building, much better named strata wrinkling. The thickest of them reached twelve miles vertically, but what is that to the earth's crust of 1,800 miles? The crust lifts and lowers fault blocks. The little strata basins expand with heat on their bottoms and get pulled and pushed by underground lava intrusions. Also they get squeezed by global contraction between crust blocks, and shoved up and down by the agelong wobbles. The biggest wobble was the downdrop of the great oceans over fault blocks when the crust first cracked and settled over the core. Those oceans have shifted and adjusted in waves of global action ever since. The crust has kept the earth a sphere while lavas erupted and weighted down the blocks. This block wobble extends into the innermost continents. Eruptions up the cracks migrated from the continental seas to the shores of the present oceans. They changed composition as they did so, because they changed from under-air eruptions to under-sea eruptions, fifteen pounds pressure to 600 atmospheres pressure. From erosion eruptions with enormous heat, to deep sea eruptions with enormous chilling and pressure. And the latter are the volcanoes of the present day, mostly concealed except for the islands and sea borders.

Meantime, the crust blocks continue to wobble up and down, and quakes continue to creak under the rock tides of sun and moon pulls. The creaks and wobbles are our big earthquakes, tidal waves, and eruptions. Such big accumulations of eruptions as the Cordillera or the Hawaiian Ridge is a terrific weight in a few million years. Both heaps have been at it since

Miocene time, or for about 18 million years, banging down through the crust blocks on top of the core. Whether such balancing of heavy weights on top of the crust blocks is due to change of lava weights or sediment basins, six to twelve miles of rock vertically, the down squeeze and underflow is called by the Greek word isostasy. It means standing level and is a poor word because the earth's crust never stands still. The blocks are eternally adjusting and creaking over a fluid core, the globe is whirling, the sun and moon are pulling, the volcanoes are erupting, and the solar system is shooting through space. Terra firma is never static. And our little atmospheric lives on top of it never stand still. We are hot, and we ourselves do a great deal of eroding.

This oration is introduction to the next series of Harvard experiments, which dealt with squeezing and wrinkling strata in imitation of the folds and faults of the Appalachian Mountains. Bailey Willis, at the Geological Survey, made a press of wax models of strata. A heavy oak piston was advanced by a screw crank. The models were waxes mixed with plaster for hard strata and waxes mixed with Venice turpentine for soft strata. They were cast to imitate actual successions of hard, thick limestones; less hard sandstones; soft mudstones; or slates. The piston advanced at a measured rate against one end of the model, the other end being a fixed box, the strata lying horizontally. The elongate Appalachian basin had a continent (the piston) to the east; a wide flat fill of limestones or sea bottom to the west (the box); and the deepest trough of pebbles, sands, and muds on the east, toward the rivers of the eroding continent of that ancient time. The heavy limestone tapered from the west into these thinner beds and made a stiff rib in their midst. The final result of their wrinkling was linear folds with axes north and south parallel to the trough, and close set at the east. The folds overturned toward the west, the overturns developing into overthrust fractures westward when the beds ruptured. Also, the folds became bigger, flatter, and wider apart westward under the deeper sea, the famous one being the Cincinnati arch.

The evidence in the middle eastern states is that the trough bottom sank as the heavy shore sediments were dumped by rivers into the sea. The west-central states received a wide flat of limestone. Uplift of the continent shallowed the ocean and pushed it, narrower, over to the great plains. So there were left a deep trough of weak beds, a massive limestone, and an

overlap of continental wash across the uplifted later continent of the present time. The problems to be studied in Willis' models were how folding would affect such a pile, what transmitted the wrinkling force, what started a single fold, and how soft and hard strata behaved under horizontal pressure.

He found that hard, thick layers of limestone transmitted the push farthest. That soft beds piled up on each other near the piston. That these beds showed beautiful overthrust faults inclined away from the piston. And that the start of individual folds was favored by very small initial bends in a transmitting layer. These downbends away from the continent would be made as the trough bottom sank through the ages. The nature of this sinking in upright slices of the bottom rock is probably downfaulting. Each vertical slice would make a step-bend as it sank.

The bottom of Willis' box did not admit of down motion by underflow, nor did the piston pressure create an opposed horizontal force that might have come from the ocean area. In restraining up motion over the folds that formed, Willis piled bags of shot on top of the model to represent downweighting. The folding in the Appalachians was down at the bottom of the heap where things were hot and compressed, and heat could extend individual strata.

In our pressure chest we extended the Willis conception. We made two pistons at opposite ends of an oaken box, with thick plate glass panes at one side, so as to watch the folding. The two pistons would distribute the end pressure better and admit the possibility that all the pressure did not come from the continent. The bottom under the model was an inner box that could move down, hung on heavy spring balances. These could be screwed up to a pressure upward to compensate the load of shot. Thus the first fold could arch downward as well as upward. This imitated a possible lowered trough bottom. The piston rate of advance was controlled by metronome, one man at each screw.

For examples, models E, F, and G had four white and four black layers, all alike in substance, at fast, medium, and slow rates. The quickest was shortened one inch in five minutes. The slowest was one inch in an hour and three-quarters. The quick-squeeze model flexed smoothly, all folds seemed to flow, and the model held together compactly. The slow-squeeze model shortened the same amount, cracked in many places, was brittle, and

did not hold together compactly. This appeared to prove that slow motion will fracture where quicker motion will hold strata intact, under otherwise identical conditions of substance, of folding and shortening, and of vertical confinement.

We verified Willis' conclusions that stiff and thick beds transmit the pressure farthest and that overthrust tends to form in soft beds, which thicken near a piston. In one model we got overthrusts in opposite directions on opposite sides of the model along a single-fold axis, with a twist in between. While an experiment was in progress, the chest creaked occasionally, the equivalent of an earthquake. One model was cast to represent overlap of strata near shore, like a coastal plain. When squeezed, it made a group of overthrusts away from the piston acting as shore rock.

In burial of strata there is a possibility whereby they wrinkle, and wrinkle most in one direction, which piston pressure does not imitate. That is the heating by burial and expansion or lengthening of controlling layers. In a long basin like the Appalachians, the wrinkling under expansion across the greatest length is easiest, because the axis of stiffness is parallel to the long trough. Transitions off the coastal line from one sediment to the next—sand to mud, mud to lime—will be weaknesses to start bends when expansion pressure takes place under burial along the layers separately heated. These bends develop into wrinkles and the wrinkles, into propagated folds, with the axis parallel to the initial change of weaknesses. Expansion lengthwise on folds, once begun, may make long flat arches pitching in one direction. This heating by burial distributes the folding better and farther than pushing abutments, and makes initial bends. All bottom strata heat and expand in all directions. The direction of easiest yielding to a folding impulse is across the weak transition belts. After that the motion is taken up by linear folds and fractures in one direction.

The models, after continuous or intermittent squeezing, were removed from the chest and sliced with a hot wire for sectioning and photographing. In one, brittle, broken series of folds in a hard layer, the model was taken apart on that layer and the surface photographed. The crest of the folds showed jointing or regular cracks. One set paralleled the fold axes as would be expected; the other set crossed the slopes diagonally and in curves. These last indicated the strains of a twisting nature on a single layer between a downfold and an upfold.

What makes the end thrust, or piston push, in nature? According to the old idea, it was contraction of the inner earth by loss of heat. Willis wrote that the basin sank, isostasy or deep flow was at right angles to the length of the basin, and general contraction took effect by reason of the deep flow. The deep flow was toward the lighter continent, from which the sands were originally lost.

The recent notion that radioactivity heat is in the outer shell denies contraction of the inner earth. Furthermore, I do not believe in a shallow underlayer of lava fifty or less miles down and capable of flowing horizontally under shifting weight. I do believe in a deep underlayer of fluid 1,800 miles down, under a block-faulted crust. This fluid core adjusted itself to the ocean-bottom blocks originally, making the upright slices moving-down controllers of the Appalachian basin. There is no proof that sediment weight did it. It is more likely that igneous, or fire-made, lava, as the thick outer armor plate of the globe erupted in acts of intrusion, lubricated the vertical slices. Intrusions are under every sedimentary mountain range on earth. It is more likely that an agelong up of ridge fault blocks and a down of the basin fault blocks decided where the central continental basin should be, all of it well within the permanent side ridges of North America. For this was a continental mediterranean sea, and the warping of its highland of Philadelphia and its basin of Cincinnati was a mere episode in the 2,000 million year history of Atlantic and Pacific borders of the continent. The sinking of the intracontinental sea, relative to the staying up of the highlands, was a wave in the history of globe and core. Erosion and deposition were results, not causes. They were results of the volcanic history of the ever moving active mosaic of the globe. The permanent North America remained high, relative to Atlantic and Pacific deeps.

The folding of the sediments merges into intrusions of magma in the southern Appalachians. Here arose the granite problem on a tremendous scale, which is repeated in our Ascutney Mountain in Vermont. What it was doing under the bottom of those vast fields of limestone from Ohio to Illinois we have no idea. No more do we know what is doing under the vast fields of lime and red ooze at the present bottoms of the deep oceans. But we do know that fire-made rock squirts up under all sea-laid sediments which anyone has ever studied on islands or continents. This fire-made

rock, solidified, has thickness and a bottom. We do not know its thickness nor its bottom. We do know that under it are big cracks 2,000 miles long rupturing it into volcano systems. The conclusion is that the globe is mantled by a layer of igneous matter which has spouted up cracks since more than 3,000 million years ago. How did this matter migrate by new intrusions, to pull, push, heat, and wrinkle through 500 million years the dirt accumulated in shallow Appalachian trenches from Alabama to Indiana? We do not know.

The last of the Harvard experiments that I took part in concerned melting up powders of basaltic minerals and rocks, letting them cool down gradually, and then sectioning them for the polarizing microscope to see how they resembled lavas. V. F. Marsters of the University of Indiana helped me. Based on the European work of Doelter, Fouqué, Michel-Lévy, and others, we used a French furnace with gas flame blast and small crucibles of diatomaceous earth mixed with clay. The specimen powders of crushed natural basalts, or mixtures of pyroxene, feldspar and olivine, were kept glowing for forty to 150 hours, and cooled either rapidly or slowly. The belief in those days was that slow cooling was the main control of coarse crystallization. Quick or slow cooling certainly does produce these effects in lava flows.

From quick cooling, we generally got radial bunches of crystals or spherulites, in a glassy groundmass. From slow cooling, we got diabase structure or coarser crystallization, with some openwork hollow crystals. And there were little grains of magnetite and spinel. Much time was wasted on furnace safety and methods, and on fire-punctured crucibles of platinum, carbon, and graphite.

Nothing had been learned in 1900 about stirring, nor about gas as an ingredient in basalt. It was not until years later, at the Hawaiian Volcano Observatory, that Emerson proved that aa lava was made by stirring a crucible. Aa is crystalline. Emerson got glassy lava by quiet melting. No one has yet subjected lava to hydrogen blasts like those of a Bessemer furnace, nor to other gases. There is a big field here for imitating Mauna Loa and Etna fountains, and for critical petrography of artificial basalts. Modern work has been concerned with physical chemistry of limited mineral systems. So far as I know, no one has mathematically synthesized

natural rocks as an object in natural history since the work of Carl Barus for the U. S. Geological Survey in the nineties.

CHAPTER III

Expedition Decade

*“The voice of thy thunder
was in the whirlwind.”*

Whereas small scale experiments in the laboratory helped me to think about the details of nature’s experiments, there remained the need to measure nature itself. The deep lavas of South Dakota, squeezing among shale beds, posed many questions. What penetrating of strata goes on under Vesuvius? Does lava inrush tilt or lift the ground? Does this measure up to eruptions in or from craters? Cannot experiments with craters themselves be made by dwelling there? Certainly the progress of lavas can be measured as they flow forth.

The decade following my mud-pie experiments saw me assistant professor at Harvard and head professor of the geological department at Massachusetts Institute of Technology. These appointments were under Presidents Eliot, Pritchett, and Maclaurin. From 1901 to 1910 I continued to serve the Geological Survey, writing up back reports. Then nature took a hand. Along came earthquakes and eruptions in Guatemala, a terrific disaster in the West Indies, expeditions to the Caribbees, Italy, the Aleutian Islands, Japan, Hawaii, and Central America, another in north Japan, and disastrous earthquakes at San Francisco, Valparaiso, Messina, and Costa Rica. The destruction of St. Pierre in Martinique set the stage for field work on volcanoes and earthquakes, work which I was to continue for a half century.

When the evening papers of May 8, 1902, announced the sudden annihilation of 26,000 people that morning at 8 o’clock at St. Pierre, Martinique, I went immediately to President Eliot. Knowing that I had been urging field study of volcanoes, he agreed that I ought to go to St. Pierre and wired Secretary of the Navy, William H. Moody, to arrange for transportation. Immediate financial support came to me from Alexander Agassiz, the National Geographic Society, and numerous friends; and my Harvard colleagues agreed to give my lectures.

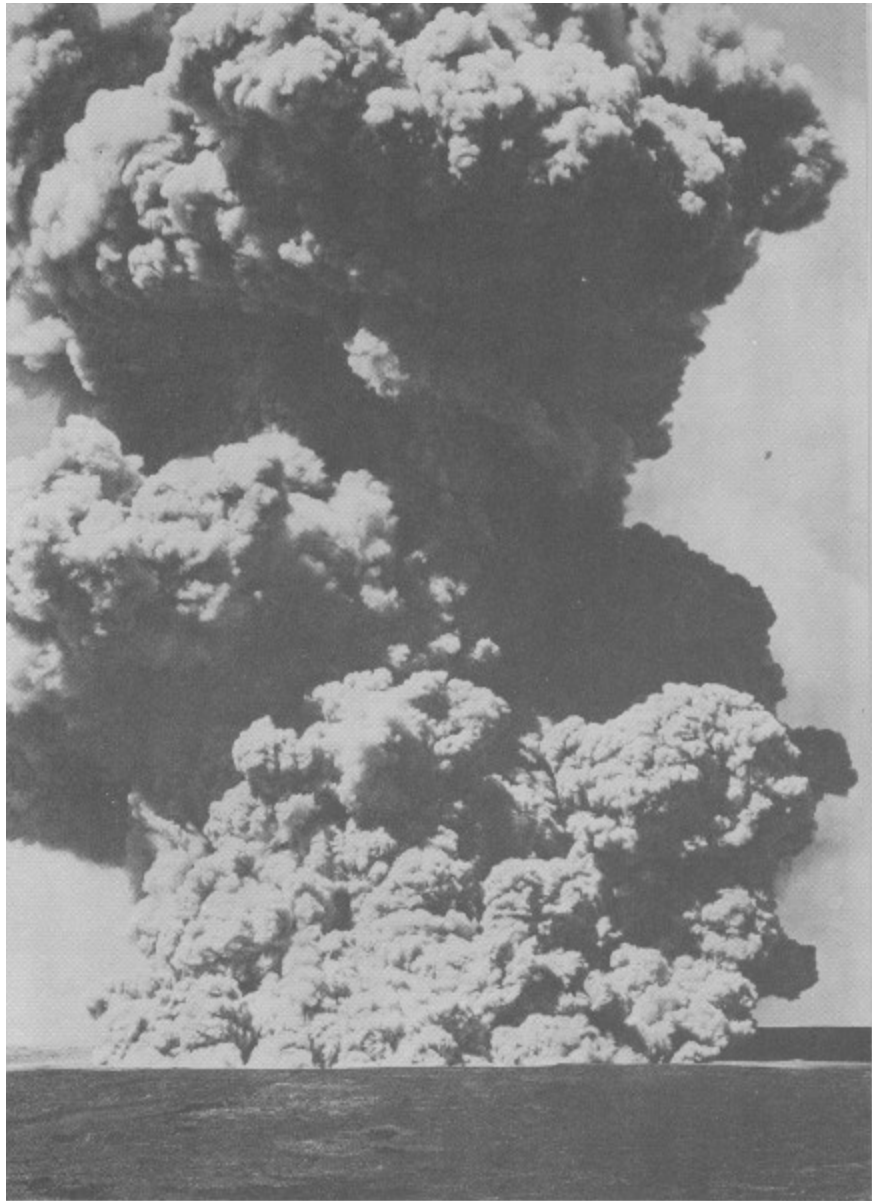
I reported to the training ship *Dixie* in Brooklyn, where I found Captain Robert Berry, a stalwart Virginian, in command of a cadet crew. On board were I. C. Russell of Michigan, author of “Volcanoes of North America”; E. O. Hovey of the American Museum; Curtis, the maker of topographic models; R. T. Hill of the Geological Survey, and expert on Caribbean lands; and numerous other scientists, and newspaper correspondents.

The voyage to the West Indies was unique. On the navy cruiser were stores of food, tents, clothing, and medical supplies for the refugees and an oddly assorted passenger list; all assembled because of warfare against mankind by two utterly unknown volcanoes, Soufrière on the British island of St. Vincent, and Pelée at the north end of the French colony of Martinique. Geologists gave lectures to the crew on deck; and in turn, we learned about naval discipline and efficiency.

When we arrived at Fort de France, thirteen days after the terrific disaster, we were transported at once to St. Pierre on the naval tug *Potomac*. We landed and walked through the ruined sugar city, the streets puddled with molasses and rum. Thousands of dead were buried underfoot amid the rubble, for the day before our visit, there had been a second blast from Pelée, the 4,000 foot volcano smoking four miles away. This had thrown down what roofs remained after the first explosion.

We arrived opposite St. Pierre May 21, 1902, and saw a smoking, dusty line of ruins along the shore. Before we landed we were warned that if the tug’s whistle should blow we were to make for the boats. The dusty hill lay on our left like a gray snow landscape, not at all like a cone. The crater was a gorge in an ordinary mountain under clouds.

We wandered through the dreary ruin and found masonry completely destroyed and no visible large volcanic fragments. The streets were full of rubble, and everything was coated with green-gray powder. Roofs were gone, an occasional timber was burning, and bodies were still numerous in the shells of houses. We saw a baby in an iron cradle, a man face down in a tank, and a big man on his back in a deep baker’s oven. His flesh was shriveled and drawn away from his joints by heat. Elsewhere eight or ten bodies were crowded at the foot of a cliff.



*3. Explosion cloud rising from Halemaumau during explosive eruption,
May 13, 1924*



4. Crag in lava lake, January 23, 1918

The end of the town toward the volcano, all backed by cliffs, was deeply buried under gravel, but the southern end had a covering of only a foot or two of sand. The second explosion was greater than the first one, demolishing third storeys and the second belfry of the cathedral. The beautiful bells “whose soft liquid notes used to ring across the bay with touching cadence at the Angelus hour” lay tumbled in rubbish, splinters, and steaming vapors; their ancient embossed inscriptions half buried in dust.

The bodies were mostly shriveled to a crisp from the second eruption, for earlier the bodies had not been much altered. The odor was a haunting one that returned in dreams—of foundry, steam, sulfur matches, and burnt stuff, and every now and then a whiff of roast, decayed flesh that was horrible. It was impossible to realize that this Pompeii had been a thriving

French town two weeks before. Not a roof was left, and scarcely a timber; steam came through little holes in the wet brown sand, and a sickening whiff showed whence it came.

It was hard to distinguish where streets had been. Everything was buried under fallen walls of cobblestone and pink plaster and tiles, including 20,000 bodies. A New England town would have blown away as white ashes before the giant blowpipe acting on the flame of burning rum.

I looked toward the gray old volcano, with shrouded summit. The landscape was dusty, like old statuary. Mountain slope and cliff were denuded of trees. An overturned factory boiler had holes punctured by flying stones. A circular marble fountain basin was chipped away on the volcano side by bombardment. Old cannon used as mooring posts at the quay had been uprooted violently. The green landscape ended abruptly at the city along a sharp line, with coconut palms half green, half brown. There was no motion except steam jets on Pelée's slopes.

Suddenly I wondered what those steam vents were doing. At first there had been one or two along the sea front; but now there were eight, ten, twenty, spurting high and scattered all over the volcano. A physician, Dr. Church, was standing near me, and we agreed that we disliked the outlook. Now there were forty jets, like so many ghostly locomotives run out from the Pelée roundhouse. Meanwhile, white-coated officers and scientists were scattered about in groups under the cliffs, some out of sight of Mount Pelée.

We looked toward the USS *Potomac*; she had seen the steam, and her own white steam presaged quick, repeated toots of her fog horn. Pellmell the passengers came tumbling to the landing. The sailors had no sooner started the boats than two more white-coated figures appeared, and we had to put back for them. The mountain looked as though it were rifting in a hundred places preparatory to an outburst, and there were many stories of new craters forming. What we saw was actually the product of a smart rain shower, falling on red hot dry gravel; but we were to learn later about rain rill explosion. Wherever a stream rill runs down to such contact, a jet of steam forms at once.

The main water gorge of the Pelée crater was blown clear of clouds as we steamed past, and we saw a cup under the summit amphitheater where a lake had been, with a pile of scaly looking hot boulders in its midst

steaming violently. This crater extended into a deep gulch to the ocean, whence had come a disastrous mud flood on May 5 which buried a sugar mill. This had happened three days before the destruction of St. Pierre. Water preceded steam. The cracks under the gulch undoubtedly dipped away from the city, and from an unknown chasm athwart the gulch line ejected water and superheated steam toward the city, like a jet from a hose. This happened on May 8. The ejected material had been in dry steam, and red hot, accounting for early reports of lava at night.

I saw molten rock five weeks after the *Potomac* trip, when the crater cone was above the rim of the gorge, apparently large fragments of brown angular material resting on finer gravel. Cauliflower clouds of reddish dust spurted up the bed of the gulch below every half hour, and migrated down the gulch. This was followed by a low growl, perhaps from avalanches. The basin widened during the month, and the dome gained in height and breadth. A bright incandescent crack at night was seen to cross the heap obliquely. A sudden increase of glow was followed by a rumbling, as though the dome were heaving. Breadcrust bombs of andesite, cracked on their surface in deep gashes, and picked up on the mountain at both Pelée and Soufrière were pieces of the internal lava.

A chance clearing of the whole dome came two months after the obliteration of St. Pierre. This we photographed, when brown dust was rising, and steam jets appeared southeast on the dome and in the gulch. On top was an extraordinary spine, shaped like a shark fin, with steep escarpment to the east, curved and smooth and scraped to the west, pushed up and out of a central rupture of the dome. It was like paste from a tube, a hard central pencil of lava that had been shoved up by the expansive force within. Jagged surfaces of breaking showed on the vertical east cliff and long, smooth, arched striations of scrape appeared on the rounded west profile of the protuberance. Other hornlike projections showed on the dome. The summit spine was 200 feet above the surface of the heap.

On July 6, 1902, came the first report of the famous Pelée spine. It crumbled in August, and a year later a new spine, facing in the opposite direction, reached a height of 1,000 feet. It was a central tongue of the semisolid lava of the dome, sufficiently plastic to be urged out by forces within. Otherwise the dome was a nearly solid extrusion covered with fallen bombs. This was the magma, or lava, of the Pelée-Soufrière eruptions. Dike

ribs extended radially from the spine athwart the dome. I published an erroneous explanation that the dome of boulders consisted of old fragments melted by a superblast and was not true lava. I was so far right, however, as to anticipate the gas-heat theory and melting of all volcanism.

The direct crisis of these Carib islands in 1902 was introduced by Soufrière Volcano on St. Vincent, 100 miles south of Martinique, at 1 p.m. on May 7, nineteen hours before the St. Pierre disaster. Soufrière exploded, as the common saying is, through a crater lake pit southwest of its 4,000-foot summit, the crater edge being 3,500 feet high. It is notable how many volcanoes are 4,000 feet high, and how many have crater pits, not at the top, but along a rift below the peak. Just this was the case of Pelée, just this characterizes the calderas of Kilauea and Mauna Loa. A dozen other volcanoes could be named where the vents are through the flank of the heap.

Hovey, Curtis, and I were taken by the *Dixie* to St. Vincent, where the hospitable English colonists provided us with houses at the base of Soufrière, and with servants and horses; and the Government supply steamer took us around the island. We made the ascent of Soufrière to the edge of the great crater and looked down at boiling waters far below, green and muddy, and sending up a column of steam on one wall.

We three Americans guided by T. M. MacDonald, a Scottish planter, made the first ascent after the fearful eruptions of May 7 and 18. Leaving our quarters at Chateau Belair, we climbed on foot from the southwest base, with six stalwart negroes carrying instruments, water, and food. In the ruins of Wallibu sugar mill we encountered a wild-eyed East Indian coolie and his helpers looting sugar.

The Wallibu River received the brunt of the heavy, dry, red hot, gravel of the eruptions, drifted like snow and crusted with wet mud. Water supplied by the river broke its way into the eighty feet of incandescent fill of the valley. Instantly a steam explosion was hurled up in white volutes, and the river dammed its own channel with the stone shower from upblasts. This forced its own waters into fresh hot cinder and so maintained explosive action. One such exploding river sent up a column three quarters of a mile high, indescribably majestic, causing the natives to report new craters. A shower of mud and sand fell on our party.

The old road crossing Soufrière mountain was destroyed, the river flats were deeply trenched, and difficult ridges and hollows were encountered at every step. The gulches were deepened into gorges, the slopes above furrowed with a feathery rill drainage pattern. Each spur between gulches was like a very steep roof, with a smooth pathway uphill along the watershed. This made progress easier. Big tree stumps of *Ficus* jutted ragged through the hardened mud, the branches charred and sharpened by sand blast.

A whirl of volcanic sand made an unpleasant stinging shower of dust, and sulfuretted hydrogen smelled of rotten eggs. But near the summit the air was fresh and the sunshine bright. A rain would have made the mud slippery and perilous, for the gulch slopes were practically cliffs. Finally we did come to mud clots, resembling a cattle wallow, knee deep and sticky. Large blocks of rock two feet across lay on the surface, flung-out pieces of the old crater walls; and there were some bombs of new lava.

After three hours we assembled at the rim of the old crater, which before the outbreak had been full of a high crater lake. Suddenly we came to an immense chasm almost circular, then the profile of a black precipice falling away 2,000 feet; and up its face we saw a silent steam column purling away in billows. The bottom was a green pool of boiling water, muddied by springs from the wall; and a hundred tails of white steam joined the column on the wall.

The inner walls showed horizontal bands of old lava, and intrusions both in lens shape and as dikes. There were red brown puddingstones made up of fragments. A funnel-shaped intrusion looked like the diagram cross section of a volcano, making a perfect T of gray lava, like a mushroom. A large fissure, filling west, rose from bottom to top. A northern rocky horseshoe rim, or somma, at the top made the peak of St. Vincent. The crater lip was a mile wide and the interior a half mile deep; and the green puddle at the bottom was 1,200 feet across. The base of the wall column sputtered fiercely and sent up spurts of black mud and rock fragments. The lake level was 1,100 feet above the ocean, 800 feet lower than before the eruption; and the pool was shallow, with mud flats and islets. We operated cameras, compass, and sketch books; paced off a base line; and noted that the northwest corner of the crater had been blown away to leave a big notch.

When we returned to Chateau Belair, the negro peasant women brought out their children to gaze at us, the godlike men who had dared the crater. Mr. MacDonald had to steer us through the crowd, and we felt like the twelve apostles after a miracle.

The Soufrière eruption during the first week of May was more voluminous and violent than that of Pelée, for Pelée was concentrated on one target. Soufrière wrought havoc east and west, whereas Pelée was in a sector southwest of the mountain. They were equally devastating, however, and both made downblasts of superheated steam and gravels. Scalding dust killed people, but so did water waves, conflagration, steam, stones, drowning, and burial.

Soufrière's dust fall was reported all the way to Trinidad and Barbados; and from ships east and southeast, directly against the trade winds, from 100 to 900 miles away. The dust column penetrated the antitrades of the upper atmosphere. Sounds were loud 150 miles away, but not heard close to the mountains. In the red hot gravel were innumerable landslides, river waters rushed into the gravel and made false eruptions, and shore cliffs collapsed.

No lava, except as fragments, appeared in St. Vincent, whereas it rose as a crateral heap in Pelée. Floods of rivers radial to the volcanoes appeared both before and after the first eruptions, and scientists erroneously attributed them to cloudburst rains. Later, exact descriptions by natives showed that the sources were hot waters gushing out in places where there was no rain.

A succession of eruptions at increasing intervals from May to December actuated both volcanoes. In succeeding years, explosions dwindled; but over Pelée's crater rose a mighty dome and spine of stiff quartz-basalt lava, like ointment from a tube.

There was, on Pelée, a splitting of the bottom of the long crater gulch. Cauliflower steam volutes charged with dust gushed up the cracks, hard-edged in profile down near the shore, soft and diffuse near the crater. Scalding waters in the gulch bottom carried mud. The mountain was cracking open along radial gulches, and squirting up steam and geysers, but this all concealed itself with sediment. Nobody ever saw the cracks open.

The migrating steam clouds charged with gravel were called glow clouds and were believed to “flow” as gas fluids from the crater.

An elucidation of all this mystery came many years later, after a thorough study of all reports. The glow clouds, which were at first confused with the gigantic blasts that had destroyed the city, were gradually explained. It became apparent that radial cracks are ancient characters of lava domes, and that lava domes lie under heaps of agglomerate. Pelée and Soufrière are heaps of agglomerate. Kilauea and Mauna Loa are lava domes. Vesuvius is an intermediate type of volcano.

I remained in the field from May to July, returned to Mount Pelée, cruised through the northern Caribbee Islands, and went to the bottom of the deep crater of Mount Misery, on St. Kitts. My guides on St. Kitts were two colored men, Johnny Eddy and Samuel Jim. In the crater we found steam and sulfur and a rotten-egg smell, on the bank of a cold crater lake. We descended by seemingly vertical cliffs covered with roots. This was a typical fumarole, or solfatara, one of the unsatisfactory characteristics of craters. We collected specimens and took snapshots, wondered how often such places change suddenly, and knew hydrogen sulfide gas only by the smell. It all jibed with what I was later to discover in Hawaii; that the only way to know a crater is to live with it, and that gases can melt lava.

As I look back on the Martinique expedition, I know what a crucial point in my life it was and that it was the human contacts, not field adventures, which inspired me. Gradually I realized that the killing of thousands of persons by subterranean machinery totally unknown to geologists and then unexplainable was worthy of a life work.

The story of Rita Stokes made a tremendous impression on me. In Barbados hospital I talked with this young white girl and her colored nurse, Clara King, who had been passengers on the SS *Roraima* which was at St. Pierre when the city was destroyed. When I saw them they were swathed in bandages. Clara's burns were severe on knee, arm, and hand. Rita's were on her head, hands, and arms, and one seriously disfigured ear. Both were somewhat injured for life. Mrs. Stokes, a boy, and a baby girl in the cabin with them had been killed. All saw the adjacent mountain sending up puffs, as the ship lay at anchor off the St. Pierre waterfront on the morning of May 8, but they were reassured by the ship's officers.

Suddenly the steward rushed by shouting, "Close the cabin door, the volcano is coming!" Mrs. Stokes slammed the door just before a terrific explosion came which nearly burst the ear drums. The vessel was lifted high and sank down, and all were thrown off their feet by the shock, and huddled crouching in one corner of the little cabin. Scalding moist ashes poured in through a broken skylight in inky darkness. Next came suffocation, relieved by the door bursting open and air rushing in.

When a little daylight came back, Mrs. Stokes and the little boy were plastered black with hot mud, the baby girl was dying, and the nurse and Rita were in great agony. A heap of scorching mud had collected on one corner of the floor, and as the young girl put her hand down to raise herself, her arm plunged to the elbow in scalding sand. They were all taken out to the deck where mother, boy, and baby died. The ship was on fire, and the nearby city was a mass of roaring flames. More ashes fell and scalded the victims. Curiously, third degree burns were left on flesh, through underclothing not burned at all.

Clara said that the mountain appeared gray with smoke rolling west, that the weather was very calm, and that the dust smelled like gunpowder. She saw no flames during the blast and did not know what set fire to the steamer. The fires probably came from the city. Ashes came in sputtering splashes like "moist marl." No rocks fell and the grit in cabin and on burns was wet sand. Before the blast there had been falling dust but, according to Clara, no difficulty in breathing. The sun was brownish red.

The bow of the ship was pointed seaward, and the vessel heeled over left, then right. The stern, toward the conflagration, caught fire first, the bow later. There was no rumbling, only shock and rattling thunder all at once, no noise before or after. The only people Clara King saw toward the shore were some men on a raft.

I wrote President Eliot and the American Relief Committee about the case of Rita Stokes, half American and the only white woman saved in St. Pierre. And I rejoiced to learn from her guardian and uncle, J. E. Crony of Barbados, that she was provided for. The sum of \$450 was sent to the committee, and \$6,000 in trust was set aside for her. She was never separated from her devoted nurse, Clara King.

Apart from the experiences of the wounded, I found much to contemplate in the findings of numerous geologists; in the accounts of doctors, sailors, naval officers, resident government men, the local newspapers, and photographers; in the specimens we collected; and in the work of great newspaper and magazine correspondents.

The facts and photographs we collected were baffling. They did not correspond with the text books. Two volcanoes a hundred miles apart suddenly spouted death downward. Obviously they were connected along the island chain, with ocean to the east and ocean to the west. Telegraph cables were broken. Why? That which lay under the ocean was totally unknown, both events and topography. The biggest part of these volcanoes was submarine.

Earthquakes at Pelée were relatively small but often continuous. Tidal waves were local and accompanied by downblasts of steam. The downblasts were at first supposed to be due to fallen avalanches from the upblasts. Then it appeared they were really sloping jets from concealed holes or cracks in the gulches, with inclined orifices amid the blocks of a cracked-up mountain. For at Pelée the blast that destroyed St. Pierre shot from the crater gulch in cascades of water and steam, while observers on high ground saw the horizon, or clear sky, over the crater.

The speed of the blast was six miles in two minutes, or 180 miles per hour. This was different from the glow clouds in the later months, migrating slowly along cracks in the gulch bottom.

Man's perception of speed relative to himself has nothing to do with actual speeds. It may be argued that a miniature volcano erupts faster than a big volcanic system, but not if the whole terrestrial plexus of systems is taken into account. An eruption of Mauna Loa is a very slow affair, in comparison with the 10,000 underground squirts of lava in cracks totally unperceived, except as tremors on seismograph.

Pelée's eruption was like turning on a hose. A structural valve or orifice, suddenly opened by underground heaving of the mountain block and letting out steam and mud, appears to be the only reasonable explanation of what happened. And the only agents possible were glowing stiff lava heating boiling water underground. Both of these were later identified.

Grove Karl Gilbert of the U.S. Geological Survey, who had criticized favorably my manuscript on the Black Hills intrusive lavas, wrote me not to drop the enigma of Mount Pelée, because he found the published reports unsatisfying. In 1949, forty-seven years after the disaster, I published "Steam blast eruptions," dealing with Pelée. In the interim I studied many volcanoes.

Alexander Agassiz, who had been urging me to do a memoir on volcanoes, financed a trip to Vesuvius when it exploded and poured out lava in 1906. Ottajano northeast of Vesuvius was demolished by jets of gravel and stones; and Boscotrecase at the south was invaded by black streams of heavy, sprouting, bouldery slag. Here was a change of habit, from heaping up lavas for thirty-four years, to collapse, internal avalanching, and pure steam explosion accompanied by remnants of stirred lava flow.

Why thirty-four years? A third of a century? Three times the sunspot interval? The previous steamblast explosion of Vesuvius before 1906 had been in 1872. In the case of Mount Pelée and Soufrière the intervals since past explosions had been fifty-one years and ninety years. But it should be pointed out that the Carib volcanoes had two years of terrifying rumblings, odors, and quakes just before 1902. Groundwater exists in large volume under all three volcanoes. Soufrière, Pelée and Vesuvius all began the steamblasts with collapsing craters, that is, with internal lava going down into the bowels of the earth. The lava usually showed in Vesuvius, whereas at Pelée and Soufrière it merely made fumaroles, or gas vents. Man, a mere microbe, could make nothing of hot sulfurous cracks.

On April 25 the electric train slowly pushed us up as far as the observatory station, beyond which all was destroyed. Outside Naples the fields were covered with two inches of gray-green dust, and pines and palms were loaded with a two or three foot drift of sand. Near the observatory a heavy six-inch mantle of sand and dust buried the lava fields. The Vesuvian cone was covered with straight sand slides, whitish gray, which occasionally slipped downward. The landscape was shrouded in drifts of white ashes revealing obscurely the slaggy contortions of lava beneath. Pure white steam boiled up from the cavity in the peak, surrounded by an older rain cloud, like a hat on the volcano's crown.

My companions—Dr. Tempest Anderson and Messrs. Yeld and Brigg—were all from Yorkshire. We started the ascent of the twenty-nine degree slope in a strong west wind. The steam settled down on the summit, than alternated with clear spells. We followed the west profile of the cone straight up, noting how the funicular rails were twisted by landslides. Everything was covered with pebbles, sand, and dust, with here and there large fragments up to five feet across. We found solid footing on the radial elevations of either scoured old lava or packed fragments. The gullies were filled with deep sand.

The rim we could see ahead was the edge of the crater itself. The abruptness of the fall off, when we finally came to it, was startling in the extreme. The wind was pelting our necks with stinging sand grains which, incidentally, were ruinous to my new Kodak. Only occasionally did sunshine sift through the mixture of sand, steam, and cloud. We could make out an inward slope of thirty-five degrees, terminated 100 feet below by a jutting, fuming precipice. The circular curvature of the crater was embayed. The only noise was the howling wind. We could not see the opposite side of the collapsed cauldron a half mile across. The summit was 4,000 feet above sea level by aneroid measure, 350 feet lower than before the eruption. There was a great notch northeast toward Ottajano where thousands of tons of gravel were hurled clear over the top of Monte Somma, the encircling old ridge. The east-west diameter was left much greater than that of the north-south. The radial ridges and gullies were like a corrugated roof, and sand made a flattened angle of scree at the base of the scoured cone. The corrugations were not rain erosion, but were made by backfallen debris sliding. I got some photographs and Mr. Perret gave me others.

The big thing was the line of mountain blocks of earth crust. In Italy it is made up of Ischia, Pozzuoli, Vesuvius, Lipari, and Etna, whereas the Caribbee line is made up of Mount Misery, Montserrat, Guadeloupe, Dominica, Martinique, and St. Vincent. Such a line of broken earth blocks is a volcanic system. Hundreds of miles long, it is never quiet. A single place seems quiet because superficially we are totally unconscious of the other places. A microbe on the scalp knows nothing of the skin of the toes. Men are mere microbes on the skin of shore, sea, and island. And they are remote from any consciousness of sea bottom.

Vast distances and long intervals are writing records, but man does not measure them. He measures civilization, wars, and dynasties, not the adventures of the ground he dwells upon. Ground he considers static. Actually it is intensely dynamic. Occasionally it explodes and man is destroyed. Earth history and volcanic systems make wars look very small.

The tremendous accumulations of broken rocks over lava beds on the cone of Vesuvius, and on all the Caribbee Islands, recall the breccias, or volcanic conglomerates, of the Yellowstone and of the High Plateaus of Utah. Floods of basalt alternate with vast falls or outwashes of volcanic gravel. Avalanches, landslides, torrents, floods—call them what you will—cover immense areas of the Cordillera. Vesuvius and Pelée pile up cones, but the Caribbees and Italy are also heaped with agglomerates. Erosion destroys cones, but erosion makes agglomerations or valley fills of rocks and mud. This is the history of every volcanic system on the globe. Stübel discovered smooth basalt domes like Mauna Loa under every volcanic system.

In 1904 Vesuvius had vented a lava flow which stopped in September, and its cone was sharp, with only a little crater and inner conelet on top. In 1905 lava had flowed from a northwest split. On April 4, 1906, a splendid black cauliflower cloud arose. The northwest flow stopped and a southern radial rift made lava mouths progress 500, 1,800, and 2,400 feet below the top, more than halfway down the mountain. From the lower mouth came glassy pahoehoe, or smooth destructive streams intensely incandescent and liquid, quickly cooling to aa, or sprouting rough fudge, black crusts, and clinker. The molten porridge flowed as a snaky avalanche into the masonry village of Boscotrecase.

On April 7, at the crater, a column of boulder-laden steam shot up four miles, snapping with lightning. New lava mouths sent forking snakes crushing and swallowing parts of the village. A graveyard was neatly filled within its masonry wall, showing that internally the rocky torrent was a liquid.

Meantime trajectories like those of a hose sent falls of gravel for miles, to Ottajano on the opposite side of the mountain. These also came from the central crater. On the west flank, at the observatory, the house was rocking, and heavy stones forced its occupants to retreat. Matteucci and his staff

went halfway down the cone, to return next day. Explosions dwindled during the next fortnight, though one day an adverse wind from the crater carried carbon dioxide and hydrogen sulfide almost asphyxiating some persons. Thereafter cauliflower clouds of white steam arose and the noise of big avalanches was heard.

The clinker field that invaded Boscatrecase was 16 feet thick, and houses were cut in two by a slaggy torrent. In Ottajano, on the opposite side of the mountain, flat tile roofs collapsed, buried under three feet of heavy gravel, some of it the size of an apple. Nearer the crater, boulders five feet in diameter were thrown a mile. The volcano was probably blocked inside by welling lava on the Boscotrecase side, which caused it to vomit steam and earthy avalanche material obliquely outward on the opposite, Ottajano, side.

The Italians have a word, *sprofondimento*, which means to make profound by insucking, that expresses what happened. This plexus of uprush of slag and inrush of avalanche, against a water-steam geyser, both happening at once, was very different from the quiet outpouring of lava during the preceding years. It definitely meant rupture of earth blocks, deep escape of that lava probably at the underocean part of the radial cracks, and deep entrance of spring water into incandescent vacated chambers. It meant a rupture crisis, collapsing the peak, and a new geyser quite unrecognized. The eruption ended when the slag pressure was relieved, the mountain blocks had settled, and the frozen slag had shut off groundwater. The remaining lava entered into decades of deep accumulation and gas bubbling, the solfataric phase. That which ended the thirty-year upbuilding was probably downward pressure due to weight of surface heaping of the cone. Cracking released water inward.

The next thirty-eight years were to culminate in a similar crisis for Vesuvius which lasted ten days, and again its peak collapsed. This was in March 1944, when our American troops entered Naples. It is interesting that these culminations have been from a third to a half century apart, but the meaning of intervals can only be really understood when volcanoes like Etna, Stromboli, and Vesuvius are grouped together. The same thing is true of Kilauea and Mauna Loa, and of Pelée and St. Vincent. Ponte reports the eruptions of Etna as ten years apart, similar to the sunspot interval; and Perret notes a ten-year interval for the smaller eruptions of Vesuvius. We measured an eleven-year interval for Hawaii, with culminations close to the

minimum of sunspots. A culmination is when lava goes down and keeps quiet, or when sunspot numbers go down and remain few. No one knows why, or of any connecting cause.

Three eleven-year culminations make a third of a century, when at Kilauea and Vesuvius, something bigger happens. Sunspots have numbered a suspiciously similar curve at corresponding dates.

Photographs of Vesuvius taken just before the 1944 collapse showed the 1906 crater hole completely filled and overflowing. There was an inner flat floor, a conelet standing in the middle. The 1944 eruption collapsed the conelet, split the big outer cone, and sent flows to destroy San Sebastiano and several villages. The torrents of ash killed people and the electric station of the funicular railroad was destroyed, as usual. The mountain split in several directions.

Just as in 1906, the stages of the 1944 outbreak were lava flows, mixed lava gushing intensely liquid, crateral caving in, tremendous gas emission, black ash changing to vapor and white ash as the emission increased, and ultimately white steam. The black ash was the contemporaneous lava with dark augite; the snowlike white ash was ground up old lavas, containing the white crystals, leucite.

The liquid phase took an unusual fountain form, resembling that of Mauna Loa in Hawaii, and nine spells of bright incandescent explosive fountaining occurred. The collapse began on March 13; the fountaining occurred during March 20 to 22, with jets of bright liquid lava and flames, 1,000 to 3,000 feet high; and the crater became a lava lake. The flames were occasioned by hydrogen within the lava itself, and perhaps some carbon gases. This liquid fountaining phase was the culmination of explosions, making pumice, with water vapor the gaseous product. Ash fell four feet deep three miles away, and some fell on the Adriatic coast. Both white steam clouds and black ash clouds arose with the fountains, white and black side by side.

The net effect was to leave a bowl 1,500 feet in diameter and 800 feet deep, floored with avalanche gravel. This reconstructed the funnel of 1906, and as in 1906, the height of the rim was 4,100 feet after eruption. In other words, the thirty-eight years had filled the vast crater, only to have 1944

engulf and eject the contents, and strew them down the slopes, adding an immense weight to the outer shell of the cone.

A hundred million cubic yards of lava was poured out, and 50 million cubic yards of ash now lie on the volcano. Three times as much was carried far away, and the volume of gases was ten times as great. The rock fragments, probably 200 times as great, were engulfed by avalanches.

The big achievement of an eruption is to wedge open a mountain, let the internal lava effervesce and go down, admit ground water, and make spectacular fireworks of burning gas and meltings. Release of pressure by splitting open the crust permits a great show of fiery foaming, but no geologist sees the profound accomplishment of lava sinking and flowing away by underground channels. It may flow out along the Mediterranean Sea bottom. At Vesuvius, it may slip through deep cracks in the direction of Sicily.

Certainly a periodic adjustment of the big system (Vesuvius-Stromboli-Etna) has taken place deep down in the earth, and the thirty-eight years of accumulation mean a stress by weighting down. The pressure of 100 million tons of stored lava inside a weak cone mountain and ready to effervesce with heat and give up its hydrogen is what science too often forgets.

The continental crack system between crust blocks and full of rain water is waiting to assist the crisis, while the blocks are poised over uprising gases of the ages. The gases of the ages, reaching to the core of the globe, are eternally melting the walls with white-hot core matter, walls of siliceous rock blocks 1,800 miles deep. In this system, Vesuvius is a tiny pimple. Incidentally, the 1944 earthquakes were recorded in largest number during the period when the liquid pumice fountains were in action in the nine different spells between March 20 and March 23. This means that the maxima of engulfing crater, seething slag, outrushing gas, crunching mountain weight, and avalanching inner walls were all happening together. The clogging of vents forced the ground water steam into pulsations. This could not last; the mountain blocks settled and resumed pressure, deep lava drained off, heat dwindled, and gas was relieved. The bigger volcanic system asserted its downward weight of the adjusted globe.

By making much of pulsations and thirty-three year intervals, we are dreaming of an ideal volcano such as might be constructed as was our

geyser apparatus. But there is no question of the reality of tides in rock, as well as in ocean; of day and night; cold and sunshine; year and century. Continent and ocean are positive, globe and solar system are positive. The ideal volcano is part of a tidal system and is limited in size. Therefore science has a right to inquire how it happens that through centuries most volcanoes stay 4,000 feet high. It has a right to look for averages and periodicities, just as a doctor looks for respiration, temperature, and heartbeats.

Like men, volcanoes are not all alike, but both men and volcanoes are orderly organisms. The object of volcanology is to find order and relate the small orderliness to the big regularity of globe and solar tides.

My 1906 visit at the end of the Vesuvian eruption crystallized my lifework idea, begun at Pelée; but my accomplishment was dwarfed to triviality by that of Perret, whom I first met while he was assisting the Italian volcano observatory. He was a photographer and observer of rare merit. He had been living in Naples and photographing all the Italian volcanoes, and he had worked out a solar control diagram for predicting volcano tides. Italy had made a volcanologist out of a physicist-engineer. Discovery of Perret meant to me much more than any phenomenon of geology.

Frank Alvord Perret was an electrical engineer from Brooklyn, and a genius with an ordinary Kodak. He took at Vesuvius, by sheer daring, the most remarkable photographs ever made of an active volcano. His knowledge of astronomy, meteorology, and physics made him see in a volcano something to study close at hand, as Benjamin Franklin studied a thunderstorm. He developed and printed his photographs himself, and colored his lantern slides. He helped Matteucci, the observatory director on Vesuvius, and was decorated as Chevalier by the King of Italy. He tramped close to lava vents and explosion clouds, and took hundreds of pictures.

Perret and I had exactly the same conception of a volcano. We thought of it as a living organism to record, just as rainfall is recorded by the weather man. For our recording, we had to invent volcano instruments. Though the camera was Perret's supreme instrument, he had been an electrical inventor all his life. Businessmen in Springfield, Massachusetts, financed his work in

Italy; and I went to Springfield to lecture and encourage their research association, the predecessor of our Hawaiian association.

Perret photographed Etna, Stromboli, Teneriffe, Sakurajima, Kilauea, the Carib cones and other volcanoes, and performed heroic work at the Messina earthquake of 1908. When, in 1929, Pelée entered into another of its periods of exploding and heaving it was studied critically by Perret who had established a museum and observatory at Martinique. He finally settled down at his museum in St. Pierre, and was of great service at the Montserrat earthquake crisis of 1933 and thereafter. He was not physically strong and the volcanic dust gave him pneumonia, but several times he recovered from attacks. He died in New York, having been forced north by the second World War.

I also met the Yorkshire oculist, geologist, and photographer, Dr. Tempest Anderson, on Vesuvius in 1906. This was another happy meeting. He too was a skilled volcano photographer, and had taken pictures in New Zealand and Iceland with his privately built cameras, using methods of extreme originality. He afterwards made for me a camera with small glass plates, dark chamber, arm sleeves, no plate-holder, alpenstock tripod, bottle strip-testing developer, self-drying metal case, and great perfection of rigidity and focus. We were to meet again and again in different parts of the world. He became one of the British experts sent to Soufrière by the Royal Society. He died of typhoid on a volcano voyage to the Philippines.

Shortly after my Vesuvius expedition I moved from Harvard to become head of geology at Massachusetts Tech. My teaching overlapped that of Professors W. Niles and W. O. Crosby at Tech and Wellesley, while for a time I continued my Harvard work. It was at this time that I began to think of possible ways of financing an expedition to the Aleutian Islands and their forty active volcanoes. The year 1906–1907 was a time of financial boom, so I went to Calumet and Hecla, the great copper company of which Agassiz was president. To my astonishment they subscribed \$1,000 to start the Technology Expedition. State Street and Wall Street raised this to \$13,000 in ten days, and I learned much about the availability of money during a boom of the stock market. President Pritchett of Harvard approved the expedition, and I organized it for a sailing schooner from Seattle, with nine in the crew and seven scientists.



5. *Scientists of Technical Expedition to Aleutians, 1907; left to right: Jaggar, Gummere, Vandyke, Eakle, Sweeney, and Myers*



6. Captain George Seeley of the *Lydia*, *Technical Expedition to Aleutians*, 1907

We set sail in the spring of 1907 and spent four months in that ocean of gales, fogs, rain, and cold between Dutch Harbor and Atka—the eastern half of the Aleutians. One man, Colby, was a bear hunter who explored the Alaskan Peninsula and reported on coal and gold. The scientists were two geologists, two mining students, a physician who was also botanist and entomologist, and an astronomer. They were Eakle, Myers, Sweeny, Vandyke, and Gummeré. The sailing master and mate were uncle and

nephew, both Nova Scotians named Seeley. The following poem by the master tells the story better than I could.

AN ALASKAN IDYLL

An Eastern College of renown
Had purchased in Seattle town
The schooner Lydia of ill fame
And Seeley was the Captain's name.
The Scientist they numbered seven
Their subjects ranged from H——I to Heaven
One on volcano's one on stars
Botany, bugs, short cuts to Mars.

Like knights of old were they prepared to shoot
The mighty whale, ferocious malamoot.
Good fellows all. I hope they'll lenient be
To him who writes this verse upon the sea

Prof. Jaggar, man of earthquake lore.
To climb mountain peaks and them explore
By delving mid their bowels, Pray dont scoff.
Could tell you how it was, the cussed thing went off
At other times on many a foreign shore
Had studied deep in seismologic lore
By looking down their throats and note the smell.
Could tell exactly just how far we were from H——I.

Prof. Gummere of the Drexel Institute
On mighty Mount Makushin burnt his boot
The crater sure was hot but when we did inquire
Found it was done while drying them to near the fire
Angles and dips of the magnetic kind
Dry bulbs and wet were ever on his mind
Strong in debate on theories scientific
Passed many a weary hour on the Pacific

Dr. Vandyke the foothills oft would skirt
Oe'r turning stones and delving in the dirt
Beetles and bugs, all things that fly and crawl
Were his delight, and well he knew them all
If one were ill, or hovering near the Brink
He'd bring you back with ointment made of Zinc
Fauna and Flora that is bugs and flowers
Were his delight. On them he'd talk for hours
Of stature slight by nature energetic
The way he'd chase those bugs was quite pathetic

Colby and Cody hunters of renown.
Whose specialty was bears, white, black, or brown
The Aleutian Is'ds yield but fox and rat
But little did these Nimrods care for that
Blood was their hobby they but lived for gore
And Colby's stomach ever called for more.
They left us early much to our dismay
To hunt the grizzly down in Bristol Bay.
With grape nuts, flour, bacon in galore
They chase the caribou, what could they ask for more

Next is little Dr. Eakle with the twinkle in his eye
Who could cook a flapjack, pound up rocks.
Or climb the mountains high
I'll bet when he gets home again on California's shore
He will never travel the Bering Sea in a Sch'r any more
He left us at Dutch Harbor and took another way
To Berkeley's Alma Mater on San Francisco's Bay.

Messrs. Sweeney next and Myers, young men of good repute
The latter on his bugle would oft delight to toot
At any hour at any time either by night or day
Reveille, Mess call, any old thing
He'd lug her out and play.
He left with Eakle much to our regret
And in my dreams I hear reveille yet.

Now I've roasted and I've toasted these fellows good and true
Just incline your ear and listen
While I whisper unto you
With a better lot of shipmates have I never yet set sail
Mid the light Pacific breezes or the wild Aleutian gale
I'll remember each and all of them
And I hope they'll think of me
And the trip they made in the Lydia bold.
To the darned old Bering Sea.

George Seeley
Sailing Master of the Technology
Expedition 1907

[Uncorrected from the original manuscript]

We collected specimens and made notes on geology, magnetism, topography, weather, photography, ethnology, plants, insects, birds, ores, shipping, volcanoes and navigation—materials for years of laboratory study. The journal of the expedition, thirty-seven pages long with photographs, was published by the *Technology Review*.

Like every such volcano expedition, we were hampered by the necessity of using a sailing vessel, by bad weather, by rain which interfered with photography, by long spells on the open sea in fog, and by inaccessible craters amid the ice of mountain tops. From the administrative viewpoint, two things stood out: the need for an amphibian boat, independent of harbors, and the need for a land station more or less permanent, wherefrom an amphibian boat could operate to reach and land on determinate beaches. A permanent station could work on specimens in bad weather. These discoveries determined the policy that was to eventuate in the Hawaiian Volcano Observatory, to the building of amphibian boats, and to five other Aleutian journeys by 1932.

I might describe sliding down the slippery grass of Unalaska, on the steep slopes peculiar to the Aleutians; exploring ice craters on top of Makushin in Unalaska; or getting storm bound for five days trying to reach Atka's Korovin'ski Volcano on foot. But these tales have been published elsewhere.

The most exciting of the Aleutian volcanoes is Bogoslof, a peak submerged north of Umnak, with its crater, a line of erupting crags, just at sea level. We had good luck with weather and landed on Bogoslof in the forenoon of August 7, 1907. Hundreds of sea lions, bellowing close to the dories, would pop up and stare at us and then plunge frantically beneath the waves. On the beach we found one bull asleep, but he awoke and awkwardly floundered to the sea. The islet was then four peaks with sand flats between, the central one a steaming mass of lava protuberances shaped like potatoes. Next to it was a half cone broken in two, with a horned spine like a shark fin; Pelée all over again. It was also similar to New Zealand's White Island. At the two ends of the island were older, peaked lava rocks. The active heap was 450 feet high with bright yellow coatings, and a ring pool of hot salt water around it, yellow with iron-stained mud. The rocky cliffs were covered with thousands of murre, their chicks, and eggs; and the birds darkened the sky in flight. The stench from offal and rotten eggs was intense.

The sea was full of fish, the beaches were full of sea lions, the hot lava and air were full of birds. Thus life and deadly volcanism lived together. The active rock was refractory basalt, semisolid, crusting and breaking into blocks as it rose from a submerged crater.

On September 1 after we left, the crater exploded, throwing sand and dust a distance of 100 miles to the east. The middle heap was engulfed, leaving only a lagoon; and the remaining peaks were shrouded in a heavy mantle of debris. Such a history of building and bursting and spreading out as a shoal has gone on for more than 111 years. Bogoslof is the peak of a submarine Pelée, several thousand feet above sea bottom. It is always active, the index volcano of the Aleutians.

It was about this time that the need for observatories began to be recognized. Something new and of grave menace had come into geology, terrible steam blasts capable of shooting out horizontally and explosively. And even as I write in 1952 these have been taking human lives at Mount Lamington in Papua and Mount Hibokhibok on Camiguin Island of the Philippines.

At Vesuvius, under Palmieri, an observatory had been established about 1859. The director was interested in meteorology as affected by Vesuvius, and annual reports were published irregularly. Successive directors became interested in making instruments for volcano science and Mercalli, the director in 1907, published a book in Italian on the active volcanoes of the world. When I went to Mount Pelée I was mindful of the venture at Vesuvius; and Professor Lacroix of Paris established artillery officers near St. Pierre ruins after the disaster, to watch and report as a volcano observatory. They furnished details and photographs of the many eruptions and the growth of the lava dome and spine. Doctors Hovey, Flett, Anderson, Lacroix, and Heilprin returned to Mount Pelée and added much to the observational and photographic record, and Dr. Stübel published a special book inspired by critical study of the Caribbees, in comparison with Andean volcanoes.

Hovey and I put through a resolution in 1907 at the meeting of the Geological Society of America, "strongly recommending the establishment of volcano and earthquake observatories." Perret and I were both inventors of instruments, both experimenters, and both convinced that the expedition method alone would never solve the volcano problem. The brothers Friedlaender of Zurich were establishing a "Zeitschrift für Vulkanologie," in Naples, and a laboratory with German, Swiss, and Italian assistants. The Carnegie Institution established in Washington a geophysical laboratory devoted to high temperature physical chemistry. We others were influenced

by field ambition, and since 1899 I had fought for a Hawaii geological survey, for I was convinced that Kilauea Volcano there must have an American volcano observatory.

My experiments on erosion, sedimentation, deformation, and eruption convinced me that a field experimental science was bound to grow up in each of those parts of dynamical geology. All of these needed field observatories to determine index of erosion, index of sedimentation, index of ground movement and earthquake, index of volcanism; these indices to be quantitative just as the thermometer and barometer and wind gauge made climatology a quantitative science of the air. I found almost nothing being accomplished in these new field sciences. No one dreamt of attacking the Mississippi as a field of pure science of erosionology, compared to the Amazon. It was felt that these things could be left to commerce and the engineers.

By index of eruption I mean the geographical peculiarity of Vesuvius, for example, as an eruption center. Perret tried to reduce this to diagram form. I published, in Washington, a plea for geophysical observatories.

An earthquake in 1908, predicted and photographed by Perret, had killed 125,000 people in Italy at Messina, near Mount Etna. Hence I felt more strongly than ever that something must be done. So it was that in 1909, at my own expense, I made a journey to Hawaii and Japan with my family. Everything within me converged on making a life work of the results of my Pacific journey.

In Honolulu I was invited to show my colored lantern slides of the Mount Pelée disaster and to describe Massachusetts Tech's plan for a seismograph station on Blue Hill near Boston. When the Honorable L. A. Thurston of the *Pacific Commercial Advertiser* interviewed me after the lecture, and asked whether Kilauea Volcano on the island of Hawaii would not be better than Blue Hill, I replied that it certainly would have many more earthquakes and, in addition, would offer volcano lavas to observe in action. Thurston asked, "Is it then a question of money?" I replied that it was, largely, but that it also entailed persuading Tech authorities that I was right.

After visiting Kilauea, where I stayed at the Volcano House and saw Halemaumau lava pit in action, I went on to Japan. There I visited the

seismograph stations of Professor Omori and traveled to active Tarumai Volcano in Hokkaido. Tarumai, which was undergoing an interesting eruption at that time, is a 4,000 foot cone in pine forests on the north island of Japan. (Notice the usual 4,000 feet.) It had broken out explosively, sent up a great spiral of cauliflower clouds of steam and ash thousands of feet, and followed this by piling up a lava dome in its summit crater, the dome lifting the crater floor and protruding above the top of the mountain.

This was an extrusion of andesite, more refractory and giving hotter steam than Kilauea vents, as measured with an electric thermometer. We got 450° Centigrade with Bristol thermocouple in sulfur-covered cracks hissing on the actual face of the lava dome. Kilauea had given 300° Centigrade in the famous “postal card crack” where visitors browned their cards.

The stiff rising lava dome of Tarumai was a duplicate of the lavas of Bogoslof and Pelée, but Bogoslof was a crater at sea level, and Pelée’s big dome and spine above the mountain top developed in the second year of eruptions. I found further inspiration in a visit to Asama volcano in central Japan. Here, just as at Tarumai, the hard lava lay in a rigid swirl, hissing and steaming at the bottom of the summit crater after the crater had announced eruption by “cauliflower” uprushes.

It was evident that hard lava push-ups from the bottom of craters were characteristic of the Pacific and Carib shores, in contrast to Hawaiian and Italian flow-downs. The pressure upward breaks a mountain, the slag and boiling groundwater inside churns up avalanche gravel and dust, columns of dust-laden steam rush out, the break-up lets up lava, and according to its frothing gas and heat and the air temperature, it is capable physically of either foaming out liquid through radial cracks or pushing up semisolid and piling as an aa heap.

The net effect is flat lava shields for Hawaii, with flows into and under the ocean, and shapely high cones for the Andes and Japan, with Italy somewhere in between. The difference in the lavas is a matter of internal meltability, due to chemistry and gases.

In the first decade of the twentieth century this was new to me as a geologist, for the books did not explain internal gas in lava. Geography understood nothing of the relation of a volcano to lines of cracking earth crust and depth of crust, and gigantic explosions dominated history as

exceptions. Refractory slags were then believed to be stiff by reason of chemical fusibility, and gas in solution in a melt is not understood even today. The Japan journey explained the textbook contrast between oceanic Hawaii and continental Ecuador, both volcanic, and the further contrast with Yellowstone agglomerates, and intrusions of the Black Hills of South Dakota. Clearly Hawaii must be studied, and experimental geology extended to the globe as a laboratory.

On my return to Honolulu, Professor Ralph Hosmer, forester, met me and reported that Honolulu money was available, if Massachusetts Tech would send me to Hawaii to found a volcano experiment station. Then and there the Hawaiian Volcano Research Association formed by business leaders in Honolulu became a reality, to crystallize later into an educational corporation.

In 1910, while I was still a professor at Massachusetts Tech, the United Fruit Company invited me to go in one of their ships to study the earthquake destruction of Cartago, Costa Rica. I saw an opportunity to study seismology in the field, as I had studied volcanology in Martinique. The United Fruit Company owned the railroad and much of the national debt of Costa Rica. F. R. Hart, treasurer of M. I. T. and director of the fruit company told me to make my own plans and the company would pay all expenses. Knowing that engineering is of first importance in earthquake disaster, I invited Professor Charles Spofford, head of our Civil Engineering Department, to go with me, and he promptly accepted.

Our journey was from New Orleans, in one of the splendid snow-white steamers of the fruit company. This ship, going by Belize in British Honduras, took us to Limon on the Caribbean side of Costa Rica, a place of banana plantations and Jamaica-negro labor. From Limon we took a mountain-climbing, narrow-gauge railroad, to the high and healthful capital, San Jose. We passed the ruins of the city of Cartago, with its earthquake tumbled churches and wrecked lower buildings, all covered with heavy roofs of red tiles. Don Anastasio Alfaro, government scientist, showed us seismographs and maps, and we called on President Jimenez, who owned a dairy farm on the high slopes of Irazu Volcano directly above Cartago. I arranged with the President to have the government make an official inquiry all over the Republic, suggesting a study of ten grades of earthquake damage, adapted to Central American habits. These grades, from mere

alarm up to wrecked churches, were to apply to what had happened in each place. According to the answers, we would make for each place a numerical value of intensity and plot these on the map.

We visited the wreckage of Cartago, where the quake had come like the crack of a whip on May 4, 1910, just at the supper hour. An American railway conductor and his family were seated at table and with the first jarrings, they all pitched forward under the dining room table. When the low adobe house fell on top of them, the table saved their lives. A pathetic object was the hollow square of the Carnegie Palace, designed by a Costa Rican architect to promote Central American peace. It was improperly braced, and everything came down, including the ornate stone wall around the grounds; and a cracked gate post held a melancholy buzzard in the hideous ruin. This and several of the big churches, cracked and disrupted, gave Spofford food for his architectural notes.

The President's farm on Irazu was a lovely place of green glades, fat cattle, and attractive Spanish dairymaids, at an altitude of more than 9,000 feet. The crater of Irazu at 10,300 feet was a tumbled depression on the top of the mountain with a steaming solfatara on one side, and a lot of circular holes inside, within a rim more or less circular.

Poas crater was very different, with a crater lake of boiling water surrounded by bright-colored horizontal layers of ash. We found buried bombs from a recent eruption which had punctured the soil with holes one or two feet across. There was wild adventure for me in being given a horse at 4 A.M., equipped with a rotten saddle, which slipped when I mounted him. The horse resented me in the early morning darkness, having just left his grain, and immediately bucked off both me and the saddle. More adventure followed. On the ride up the mountain and in the midst of the forest we encountered a jaguar trap which had recently caught two big cats. It was a pen, roofed with logs baited with a fowl, and disguised with brush; a shutter fell and closed the opening when the bait was touched. On the way down we had a terrific tropical thunder storm, with sheets of cold rain, and I got chilled to the bone and was sick with dysentery for two or three days.

There are a dozen volcanoes like these two on the backbone of the Costa Rica rocky mountains. They trend in a ragged line from the Panama boundary on the southeast, to Nicaragua on the northwest. All have records

of explosive activity, but lava flows are rare. Beginning at Nicaragua the line of the Cordillera, capped with volcanoes, continues through Salvador, Honduras, and Guatemala; and some of the lower ones have lava flows. Cosequina is famous among them; and conspicuous as a frequently active volcano is Santa Ana in Salvador, one peak of which is Izalco, the index volcano of Central America, erupting frequently. Other index volcanoes are Kilauea for Hawaii, Stromboli for Italy, and Bogoslof for the Aleutians. The next line of volcanoes, also trending northwest, extends from Guatemala into southern Mexico. The Costa Rica line overlaps the northeast side of the Nicaragua-Salvador line, and this in turn overlaps the Guatemala line, and so on. The chains of volcanoes are over an echelon of cracks, surmounted by heaped-up lava peaks on the continental divide.

From the point of view of experimenting with volcanoes, the exploration of the Cartago earthquake and Poas and Irazu craters and a study of their relations typified the unsatisfactory combination of upheaved mountains of strata and of volcanic eruptions and underground friction. This extends all the way along the Cordillera from Patagonia to Alaska. I say unsatisfactory because from the science standpoint, the action of eruption or earthquake is far scattered in time and place, and only local observatory geophysics and traveling scientists will do the work. Cartago is directly at the foot of Irazu Volcano, but the volcano did not erupt simultaneously with the earthquake. In the same way Messina is at the foot of Etna, and Tokyo is at the foot of Fujiyama; and the great earthquakes do not accord with eruptions. Sakurajima in 1914 was an exception, it had a quake after outbreak.

The direct outcome of my study, on the map of Costa Rica, of lines of equal earthquake effects, showed the maximum of the 1910 quake on the continental backbone, and the lines were crowded together along the western mountains. However, they spread out wider and wider along the Caribbean coastal plain, which is an elevated sea bottom on the northeast side of the country. In other words the terrific jolt was a deep slipping or scraping under the volcano line, and the elastic waves of like strong effects were close together in the mountains on the Pacific side, opposed by hard rock. On the other hand these waves, much feebler, widened out their lines in going through flat, soft strata on the Caribbean side. The answer seems to be that along the jagged rupture which underlies the volcanoes there is continuous upward pressure of lava, which occasionally is accelerated into

a big bump or slip, now here, now there, as the whole great mountain range volcanically heaves through the ages.

Our next journey was from Barrios across to Guatemala City, where we had distant views of such volcanoes as the pure cone of Agua and the sharp peak of Santa Maria, which in October of 1902 had blown out its flank and left a vast hole. The Guatemalan plateau of rich soil and abundant market products rises gradually from the wet banana lands on the Caribbean side to a height of 4,870 feet at Guatemala City. This is on the line of volcano cracks. Then the land plunges abruptly in a precipitous down-faulted slope, to a low flat shelf along the Pacific Ocean. This shelf is covered with the merging of many deltas formed by the streams and torrents which drain the well-watered plateau. Along this line at the top of the precipice is the chain of volcanoes, with rich coffee lands at their feet on the upper slopes. Coffee plantations were destroyed by steam, mud flood, and ash blasts in 1902, and similar destruction was destined to begin again in 1923.

A large model of Central America has been built in a park in the open air in Guatemala City, showing magnificently the upland plateau and its mountains, the flat slope to the east, and the long straight steep plunge to the Pacific coastal shelf. This is one of the best illustrations of the block faulting of a continent, lifted like a huge flat slab along a crack, and tilted away from the Pacific. The Pacific block dropped down.

The same structure is true, on a larger scale, of the line of the Andes, lifted as a volcano-covered slab, down-faulted along the Chilean coastal plain. The upland slopes away to the basin of the Amazon. In these studies we are experimenting with volcanoes on the scale of geography, but the principles involved apply to Mexico and to the Cascade Range in Oregon. They probably apply also to the Aleutian, the Kamchatkan, and the western Pacific arcs, considered as upheaved and eroded ridges. They are arcs because they are ancient calderas.

We traveled by steamer along the Pacific coast to Panama, where the canal was being finished. We were impressed by General Goethals and his associate engineers, and with the marvellous organization of big engineering as the United States could administer it. Yellow fever had been conquered, ships constantly brought dairy products from New York to canal employees, houses were screened and unglazed, and the jungle was cut

back to limits of safety from the mosquitoes. We found lively young American college graduates, both men and women, playing tennis in the deep tropics, where earlier hundreds had died of fever. We arrived just at the time when sides of the Culebra Cut were continuously sliding inward like a glacier, to close up the ditch. The ground under a village at the top of the bank was cracking in long crevasses, and habitations had to be abandoned. The only answer was to dig away the hill with hundreds of dump cars, until the slope was flat enough to stop sliding.

An amusing episode occurred at the Pacific end of the canal, where giant monitors, or hose nozzles, were being used to cut away the banks. Engineer Williamson had conceived the idea of mounting these monitors on concrete barges made on the spot. He covered the frames with steel mesh, and sprayed concrete against the mesh until a water-tight hull was produced. Fellow engineers jeered at Williamson and said that a boat made of rock would surely sink. Someone asked Williamson, when his first barge bore up the heavy monitors and was successful, what he was going to name it. He painted the name in large letters on the barge "Ivory Soap, it floats."

We met in Costa Rica and Panama Arthur Herschel, city engineer of Kingston, Jamaica, who was responsible for the reconstruction of that city after the terrific earthquake of 1907. Herschel invited Spofford and me to stay with him on our way home, stopping off when we passed Jamaica. We did so, were delightfully entertained, and learned about engineering and rehabilitation after the most intense earthquake of all history.

The momentary intensity of the quake had been utterly without warning, as though two mountains had collided, and the masonry of the business section of Kingston crumbled almost instantaneously. A British major was walking along the main thoroughfare, carrying a heavy walking stick, when at the other end of the street, he noticed a commotion and thought it was a negro riot. The disturbance came toward him with a roar, and he saw clouds of dust rise from the street like a tornado and approach him. He felt the ground jolting, raised his stick, and decided to stand and fight it. The buildings right and left simply exploded, and he was fending off bricks and stones and timbers. His feet were half buried in rubble, and he sat down on a steel girder which had lunged out into the street behind him. The dust was suffocating, the noise was a traveling roar which went past him and on down the street behind him. He called to a black man to dig out his feet, but

the man rushed by with staring, crazy eyes. He heard screams and saw women running. It was some time before Red Cross stations were established and the army men rescued him.

The lesson taught by this earthquake, more intense than the one at Cartago, was that the wooden bungalows of the hilly suburbs on rocky ground stood the disaster better than even reinforced concrete in the congested waterfront district. The better built government buildings were preserved in part.

The Jamaica law of 1907 had established definite boundaries for wooden construction, limited to the suburbs, and made new and wider streets in the business district. It had also established rigorous fire insurance laws, and a city building code requiring specified construction for all masonry. The result was a marked ring of parkway separating the commercial center from the dwellings in the suburbs. The trouble with such legislation, the effect of which I saw in Kingston twenty-six years later, is that earthquakes are hopelessly discontinuous. With no more big earthquakes as testers, such laws become dead letter, a new generation remembers nothing, and an irresponsible and ignorant native population poses new problems of poverty and vice. Earthquake construction reform becomes an impractical dream. This is part of the unsatisfactory quality of earthquake science, where assistance to humanity is concerned.

So ends my expedition decade, 1901 to 1910, after a succession of studies in the field, which may be called Operation Pelée-Soufrière, Operation Vesuvius, Operation Aleutians, Operation Kilauea-Tarumai, and finally Operation Cartago. I did not think of these at the time as the strategic work of warring with a task force in geographical volcanology; but now as I look back on it, I can see in each expedition the organization of an institution and men, and progress of volcanic geology.

The Martinique event was destined, through many explorers, to reform geophysics. Vesuvius introduced me to the importance of superb photography as represented by Perret and Anderson. The Aleutian Islands introduced the question of nautical exploration and the importance of a field base laboratory for work in a land of adverse weather. The Japan-Hawaii expedition showed me the national seismometric work of Dr. Omori in the field, and laid the foundation for the Hawaiian Volcano Observatory.

Finally, the Costa Rica expedition introduced me to the complexity of seismological field work in a land of volcanoes, with the problems of engineering ably investigated, and afterwards published by Spofford. This decade thus logically leads into a totally different one, field experiment in geography and founding a volcano observatory in and on the most active volcano in the world, with a permanent dwelling on a crater.

CHAPTER IV

Living with Volcanoes

“He took his journey into a far country.”

The next decade began true experiments with volcanoes, when two organizations some 5,000 miles apart combined their resources. The Whitney Foundation created at Massachusetts Institute of Technology an endowment of \$25,000 for geophysical work on earthquakes and volcanoes, expressing a preference for work in Hawaii; and a group of businessmen in Honolulu, the Volcano Research Association, offered to pay my salary for five years.

When President Maclaurin and a group of professors at M. I. T. gave me a dinner at the University Club in Boston to celebrate my departure for Honolulu, the dinner table conversation turned to the terrors of the deep sea, the dangers of volcanoes, the awfulness of leprosy in Hawaii, and the heroism of giving up a secure teaching job in Boston. I replied that their pessimism reminded me of the last words of Daniel Webster, as quoted by a New England farmer, who said “Dan’l opened his eyes, took one look at the glass of whiskey on the table at his bedside, another at the pretty nurse, and said ‘I ain’t dead yet.’”

I had organized the funds available so that a pair of Bosch-Omori seismographs were shipped from Strassburg, and other seismographs were ordered from Omori’s instrument maker in Tokyo. I collected experimental instruments such as high temperature thermometers and chronographs, of the type used in experimental physiology. Vaguely, I was going to take the blood pressure and pulse of the globe. Also I obtained a full set of weather bureau instruments for temperature, rainfall, barometric pressure, and humidity, together with the electric pyrometers, range finders, and photographic apparatus used in my previous expeditions. And I had some small Japanese transits, as well as plane tables and alidades for topographic experiments.

I was unable to go to Hawaii until 1912, so I was delighted when Perret consented to go to Kilauea Volcano in company with E. S. Shepherd, gas

chemist of the Carnegie Geophysical Laboratory of Washington, in the summer of 1911. Dr. A. L. Day, director of the Carnegie laboratory, kindly supplied at our expense two Leeds and Northrop resistance pyrometers and the accompanying Wheatstone bridge, as well as thermocouples loaned from his equipment. Perret and Shepherd went to Kilauea Volcano House; and Perret built a hut at the edge of Halemaumau pit, where an inner lava lake was bubbling and maintaining an island some 200 feet below the rim. Kilauea is the big cauldron, Halemaumau is the firepit in its floor. "Kilauea" activity generally means Halemaumau. They have separate cliff margins.

L. A. Thurston, leading journalist and publicist of Hawaii and keen promoter of a proposed Hawaii National Park, did everything possible to help the scientists. Perret wrote weekly reports on the condition of Halemaumau lava, and sent in photographs to Mr. Thurston's newspaper, the *Pacific Commercial Advertiser*. Living and camping at the fire pit, Perret inaugurated something new for Hawaii, and set a standard for the Volcano Observatory. These continuous reports had been my dream for such volcanoes as Vesuvius, where publication had usually been in delayed annuals and gave no current news of what the volcano was doing. Furthermore, the Vesuvius observatory was at the foot of the peak.

I had ordered from the Lidgerwood Company an equipment of cables, including some containing electric wires. These were to span the 1,500 feet and to lower a thermometer into the pit of Halemaumau. Assisted by Alex Lancaster, the active little half-breed guide from Virginia, and by numerous laborers from the plantations, whose managers, spurred on by Thurston, took a great interest in the project, Perret and Shepherd erected two high A-frames on opposite sides of the fire pit and built a trolley on the cable stretched between them. Perret kept constant angular measurement of the changing height of the liquid lava, as the glowing slaggy pool rose and fell overflowing its banks. At one side of a triangular island was a point of ebullition called "Old Faithful" where gas bubbles burst in a fiery dome, irregularly, but approximately once a minute. The objective was to find the temperature of the liquid lava in the vicinity of the bubbling. This was achieved by actually dipping the electric pyrometers into the molten slag, then observing the precise temperature at the recording box, which was in

the hands of Dr. Shepherd, who remained on the pit rim at the upper end of the connecting wires.

Finally the day came, after numerous rehearsals, when the long steel tube, or terminal, on the end of the movable cable could be moved out by the trolley to a middle point over the pit, where it would make contact with bubbling liquid lava when lowered. This was an extremely ticklish procedure, for the lava was a heavy mat of self-crusting liquid rock with the crust forming hard slabs; few places kept up an appearance of bubbling porridge. No one had ever made contact before with the liquid of a fountain like "Old Faithful." It was fortunate that the apparatus, which was expensive, consisting of platinum wires imbedded in silica glass, was made in duplicate so that we had two of everything. The splashing liquid of "Old Faithful" looked as harmless as a kettle of boiling soup, but Perret and Shepherd were in for a surprise. When Shepherd lowered the terminal directly into the liquid, "Old Faithful" exploded, for the molten slag proved to be a suction whirlpool which threw tentacles of lava over the steel pipe. The apparatus went down to destruction "like a bass under a log," and the cable was bitten off like a piece of line. The entire terminal vanished into the vortex, leaving only a corroded wire.

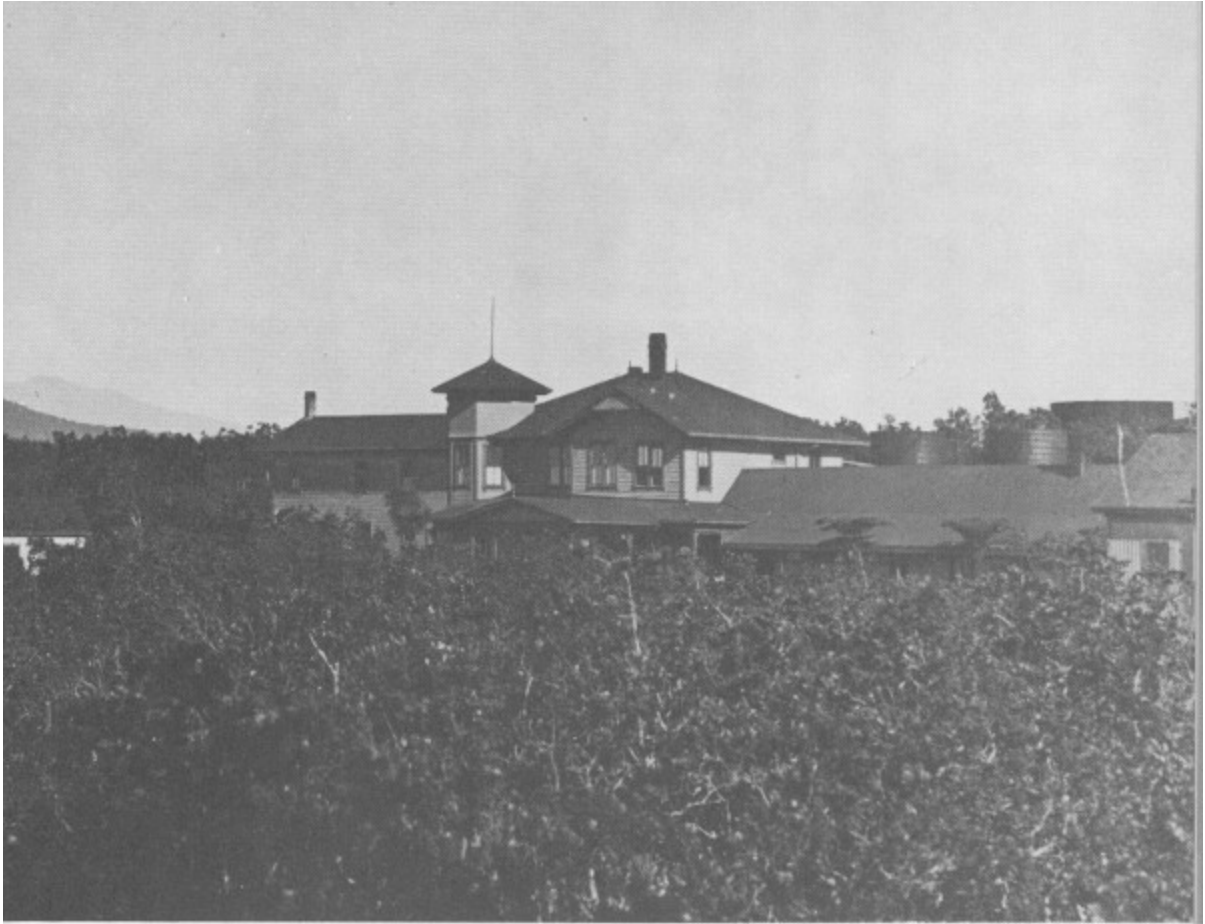
To shorten a long story, the second terminal was lowered into a seemingly safer liquid place. A wave of the melt slapped and strained the pipe, and though it was recovered, no electric resistance reading was obtained at any time with the box at the rim of the pit. Close to \$1,000 in equipment was lost. The resistance pyrometer is a sensitive tool in the laboratory, for giving precise degrees of temperature in the region of 1200° Centigrade, supposedly the melting point of basalt. But it was unsuited for the rugged bubbling of basalt slag, where flaming gases and chilling air play more important parts than mere melting.

Fortunately Shepherd and Perret were not at the end of their resources. There still remained the thermocouple, a simpler pair of wires of platinum and iridium encased in a steel tube. The connectors from these go to a simple galvanometer in the hands of the operator. The trolley could still be used, and the thermocouple pipe had no glass inside it to be shattered. A temperature of 1000° Centigrade was recorded in a bubbling area, and this was considered good enough for an approximation.

Another experiment was to lower an iron bucket into the liquid, and pull it up full and dripping with black lava glass. This was sent off to Washington for analysis. Afterwards the lava lake went down, no more experiments that year were possible, and Perret began the plotting of a curve of high and low in the rise and fall at the bottom of the pit.

It may seem extravagant to waste valuable apparatus on such seemingly small results; but as a matter of fact, the Shepherd-Perret journal of the summer of 1911 was epoch-making in the history of volcanology and in the work of the Hawaiian Volcano Observatory. It proved that skilled observers could dwell inside an active crater and there apply their skills in photography, chemistry, note-taking, and continuous publication. The substance of active lava lakes was proved to have viscosities and solidifications quite different from those implied by gases, and it was shown that different types of thermometers gave negative or positive results useful for the future. Above all, the notes on volcano chemistry by Shepherd and Perret demonstrated that engineering apparatus could be applied to the hottest and most continuously active pit in the world. Their success was at the relatively small expense of a journey and a few machines. Brun of Geneva had set an example of similar work, but Perret's curve of rise and fall added a more detailed record of the Kilauea pit from day to day than had ever been made before.

An observatory is a place of observation and measurement, whether the things observed are glaciers, rivers, stars, the weather, or volcanoes. The motive of observation in modern science is either the quality of what happens or the quantity expressed in lengths and degrees and rates of speed. Remembering the precedent of Vesuvius, I was confronted in Hawaii with the necessity of determining how a volcano should be observed, the need to measure changes in a single volcano, and the need for permanent records of what those changes are. We chose measuring instruments, photographic equipment, and thermometers, and I invented a note-taking system which was compiled into a single record book, from field notes taken uniformly by many different assistants.



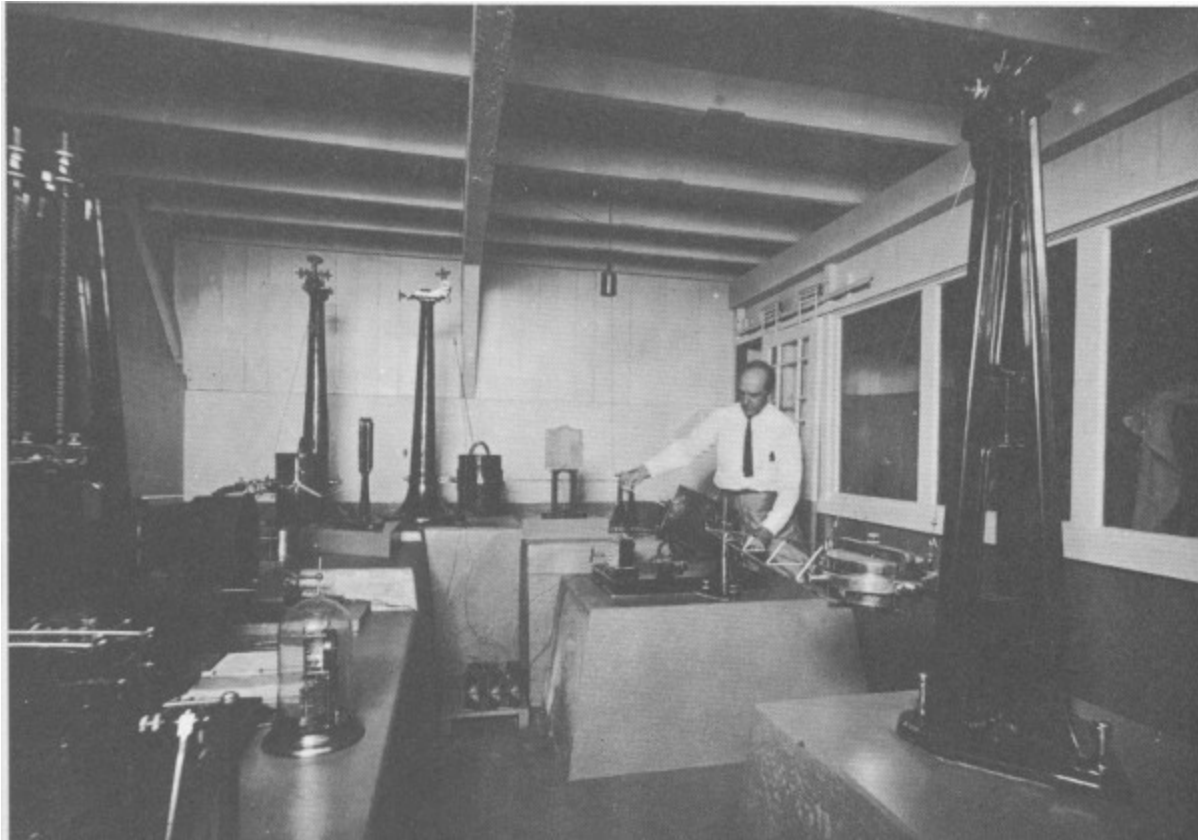
7. Volcano House from Observatory, 1913



8. Island in Halemaumau lava lake, 1911. Photo by Perret



9. Hawaiian Volcano Observatory, 1912



10. Jagger in seismograph vault beneath Volcano Observatory, 1916

The textbook needs for volcanology are records of the shape, height, number, distribution, temperature, and differences among volcanoes. How gaseous is lava? how radioactive is it? how often does it erupt? and how dangerous is it for human beings? With reference to the source, crack or crater, we need knowledge of how the earth crust is ruptured, how deep are the fractures, and how much accompanied by earthquake is the wedging upward of lava in those cracks.

My first job on arriving in Hawaii was to make contact with Mr. Thurston and his associates. The next was to get a good map made of Kilauea Volcano as a basis for measurement of changes in the fire pit. Governor Walter F. Frear came to my rescue and immediately sent Colonel Claude Birdseye and Captain Albert Burkland to make a topographic map of the proposed Hawaii National Park. These engineers brought into the field the topographic camp of the U.S. Geological Survey, and they were extremely sympathetic with my project, furnishing me with surveying

monuments, and sketching out methods wherewith to make an accurate base line for measurement of changes inside the pit.

A laboratory on the northeast edge of Kilauea Crater was quickly provided through the energy of the brilliant Demosthenes Lycurgus, hospitable Greek manager of the Volcano House, the hotel where I stayed. All the merchants of Hilo, thirty miles away, contributed funds and in a few weeks carpenters were at work, on land belonging to the Bishop Estate and sublet by the Volcano House. Furniture was paid for by the Whitney Fund.

A cellar for seismographs was blasted by Territorial prisoners in the hot rock under the laboratory, at the actual northeast edge of the greater crater of Kilauea. The lava pit Halemaumau, always smoking, was in full view two miles away. The cellar lined with concrete, which shut off the steam cracks, became a warm, dry place for instruments at a constant temperature of about 80° Fahrenheit. Concrete tables on the floor of the cellar held the pair of east-west and north-south horizontal pendulums, recording with delicate pens on smoked paper, stretched over a chronograph drum. These paper records, removed every day and fixed with shellac varnish, became the seismograms of the permanent files. Long belts of wavy lines on each paper exhibited seconds, minutes, and hours; and when a sharp zigzag in one of the lines occurred, it was evidence of either a local or a distant earthquake. H. O. Wood, who had been my assistant in field geology at Harvard and had had experience with Omori seismographs at the University of California, was summoned to the Observatory as seismologist.

Thus in the first six months of 1912 I became a resident of a volcano in Hawaii and had an adequate laboratory of eight rooms, and suitable porches, a darkroom for photography, and the beginnings of seismograph records in the basement. Horses and saddles were purchased, the necessary outer houses were built, and Alec Lancaster was employed as janitor and field man. Francis Dodge, athletic young Honolulu and son of a government surveyor, was appointed topographic assistant. He was a hardy cowboy, with some experience as rodman for the Geological Survey.

From the moment of my arrival I adopted uniform pocket scratch pads with detachable sheets for the use of all employees, insisting that anyone who went to the lava pit should write notes, inscribe the date and hour, tell what he saw, and hand the notes to me. Even Alec Lancaster, whose father

was a Cherokee Indian carpenter and whose mother was a mulatto, took notes and learned about the points of the compass and the names of the coves and blowholes of the lava lake in the bottom of the pit. Some of Alec's notes were very amusing, as when he wrote, "9:30 A.M. April 3, Old Faithful is on her job right sturdy." However, he quickly learned the correct technical expressions for surface streaming of the lava, brightness of the fountains at night, numbers of the bubble fountains, and places of smoke on the bottom of the pit. At all times Alec was a useful camp man, a good cook, and a fearless climber of cliffs. When it came to making and using rope ladders with hickory rungs for descent down a 200-foot cliff to the edge of the lava, Alec was the first to volunteer. He drove spikes into cracks in the rock and tested out the ladders, surrounded by smoke. This was done in June and December of 1912, when the gas chemists of the Carnegie Institution were conducted to the bottom to collect gases, by pumps and vacuum tubes, from flaming spatter cones.

I hope this introduction gives some idea of what the first year of the Observatory accomplished. Meanwhile problems of policy and of the publishing of results crowded upon me thick and fast. The notes of all employees had to be compiled; critical scientific visitors had to be convinced of the usefulness of the new effort; the Massachusetts Tech and Honolulu sponsors had to be given suitable reports; a permanent record book, reproducing surveys, notes, and photographs, had to be devised; and I had to make occasional journeys to California, Boston, and Washington for contact with the Government, with scientific societies, and with scientific magazines.

It was necessary to keep track of improvements in photographic plates, for the fire pit with its dark red heat and dark red rocks was a difficult subject for photography. Fortunately, the panchromatic plate had recently been invented by Dr. C. E. K. Mees, and was a godsend for experiments in recording liquid lava splashing at night. Dr. Mees, chief of research at Eastman Kodak Company in Rochester, has since been a visitor and good friend of the Observatory. Both surveying and photographing were difficult during 1912 because the inner pit sent up a dense column of fume which diminished only at those times when the liquid lava became hotter and developed fountaining. There was such smokeless development with hundreds of roaring fountains of liquid lava in January and July. The

intervening period showed a great deal of smoke, and in August there was a dense column of silently rising gray fume the full width of the pit, so that nothing of the bottom could be seen.

To determine the height of the bottom lava it was necessary to work from a fixed station with a transit, using a flashlight at night, and waiting for a view of a glowing spot or fountain. This involved reading vertical and horizontal angles, dependent on difficult determination from two stations, of the distance to the glow spot measured. Often in daytime one had to wait hours in order to get a view of the bottom through the fumes, from stations at the ends of a base line on the edge of the pit. At no time later, fortunately, were the fume conditions so bad as during 1912. A procedure was adopted of making a daily photograph of the smoke of the distant pit from the window of the observatory, and this proved of value when the inner lava lakes and crags rose to view in 1917.

Like Perret, I made reports to the newspapers in Honolulu; and gradually these reports took the form of a monthly bulletin, edited in Honolulu by Dr. Howard Ballou, who was the secretary of the Hawaiian Volcano Research Association. This association had occasional Directors' meetings, which I attended and before which I made reports and gave lectures. The report of the complete work done during the first few months of the year 1912 was published in Boston by Massachusetts Tech.

The earlier history of Hawaiian volcanoes had been recorded in excellent books by such travelers as the Misses Gordon-Cumming and Isabella Bird, William Lowthian Green, and Drs. C. H. Hitchcock and W. T. Brigham, and Professor James D. Dana of Yale. Dana had been furnished with data from 1840 to 1890 by a Hilo missionary, Titus Coan. When I arrived in Hawaii, two books on Kilauea's activity in 1909 had just been published, and a big monograph by Brun of Geneva who had determined that Kilauea lava was free from water vapor and was the hottest lava in the world.

Furthermore, R. A. Daly of Harvard had published his "Nature of volcanic action" on the basis of his summer at Kilauea in 1909. There was strong controversy against Brun on the water question, but the experts, including Day and Shepherd, came to the conclusion that lava eruption of the Kilauea type was actuated by such flaming gases as hydrogen, carbon monoxide, and sulfur; that these gases were in solution in some elemental

form deep down in the earth; and that the chemistry of their emission heated the lava on its way up. The lava lakes were hotter at the top than at the bottom. We shall see that all lava partly solidifies at its own bottom and stays liquid above.

The items of activity at Kilauea Volcano during the decade from 1911 to 1920 were marked fluctuation up and down in 1912–1913, with a notable low level in 1913, culminating in a strong earthquake in October. In 1914 the liquid lava came back into the bottom of Halemaumau pit, and in December Mauna Loa erupted in a fountain at its summit crater. The lava lakes of Kilauea grew bigger in 1915, and a triangular island appeared, lifting itself up from a shallow flat and even rotating or hinging horizontally. Its uplift was as a peaked escarpment of lava layers tilted in one direction, something very like Perret's island of 1911.

An affinity between Kilauea and Mauna Loa was obvious. In 1916 Mauna Loa completed its summit gushing by splitting open the mountain's southwest rift and making a lava flow into ranch and forest lands of South Kona. But just as Mauna Loa activity ended, the entire Halemaumau bottom thirty miles away lowered dramatically during one day, leaving a deep seething puddle of melt, surrounded by roaring red hot avalanches. The coincidence, along with appropriate earthquakes, was unmistakable.

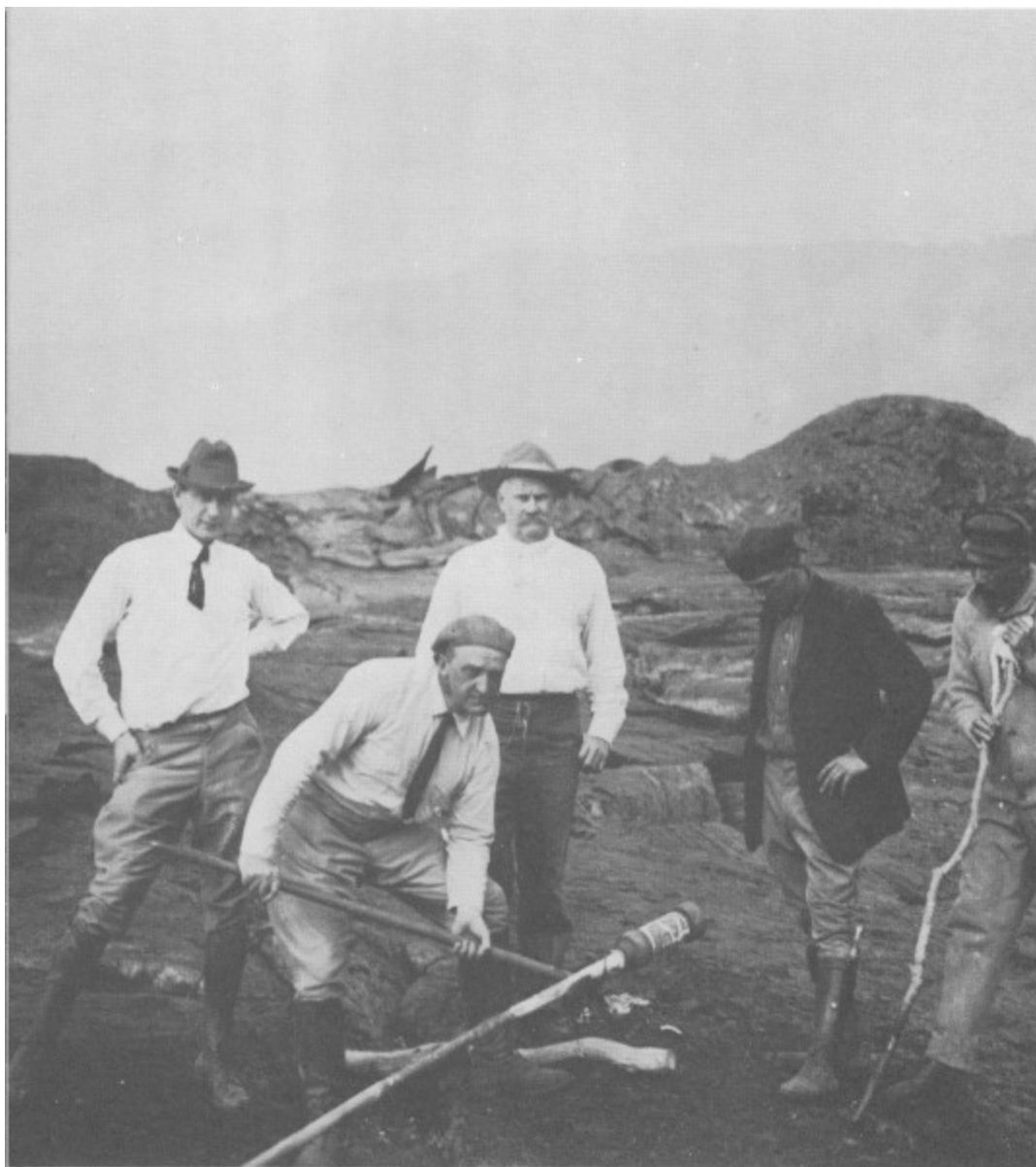
Immediately after the lowering, the liquid lava of Halemaumau welled up border wall cracks and cascaded through the talus to form an oval pool in the bottom funnel of broken rock. The lava column rose 600 feet in the next six months and a lobate lake developed, its coves separated by sectors of overflow lava which lifted slowly into crags in the center. In 1917 the lakes and crags inside Halemaumau were less than 100 feet down, the lake shores became accessible for experiments with iron pipes, and the crags came into view from the Observatory, fully justifying the daily photograph for comparing changes of the distant pit.



11. Lava lake, showing bench, March 30, 1917



12. Halemaumau, showing lava lake and crags, December 8, 1916



13. Jaggar holding pipe for sounding lava lake, 1917. Cylinder on end of pipe holds Seger cones for measuring lava temperature

By 1918 and 1919 the pit was full and overflowing the Kilauea floor. During the whole of 1919 Halemaumau, as a pit, was obliterated by its dome of fill. In autumn the south flank of Mauna Loa broke out again, into a flood of lava that reached the sea in South Kona. Remembering 1916, we predicted that, even though Halemaumau was full to the brim, the sinking away of Mauna Loa lava would pull down Kilauea lava suddenly, like a siphon. Exactly this happened on November 28, 1919. During the night the

crag, the clover-leaf lake, and the bulging dome of the lava fill above Halemaumau's edge went down as a cylinder to a depth of 400 feet in two or three hours leaving incandescent avalanching walls, a gratifying confirmation of theory.

As in 1916, the Halemaumau lava immediately returned to the bottom of the pit, and lifted itself thirty feet a day for three weeks, so that in December it was a violently boiling ring-shaped puddle, surrounding a horseshoe of crags with a quiet inner lagoon and resembling a coral atoll. The Kilauea floor, which is dome-shaped outside of Halemaumau, split open radially to the south, made floods of lava into the Kilauea Crater wall valley, and even escaped out into the Kau Desert. This was extended into a mountain crack, making flank lava flows of Kilauea Mountain, nine miles away to the southwest, something which had not happened since 1823 and 1868. Concentric craters like Kilauea caldera and Halemaumau pit are thus ring-in-ring, or cup-in-cup, structures by means of slag heapings over a deep fracture in the rock crust, the circularity determined by occasional central sinking.

This circularity has sometimes reached perfection. In 1894 and 1909 the liquid pool inside Halemaumau, by steady welling up about a central hole, became perfectly circular within a circumferential rampart of overflow. This is a rare condition dependent on steadiness of upwelling, temperature, and viscosity. It is important because it shows how the perfect circles, and rampart cauldrons, were made on the moon, where there are also angular calderas of subsidence like Kilauea Crater. Evidently gas heating and liquidity changed on the moon, just as it has done in Hawaii. The sources there are over cracks, as in Hawaii. The analogies are so complete in these and many other ways that I completely disbelieve in meteor impact for the moon craters. The moon awaits a complete comparison with active terrestrial basaltic lavas, by a modern volcanologist.

This is only a thumbnail sketch of the astonishing luck which met the photographers and note takers of the Hawaiian Volcano Observatory in its first decade. There were similar decades in the nineteenth century, and there were similar jagged crags rising as islands and shorelines around clover-leaf lakes in 1879 and at other times. There were undoubtedly earlier similar sympathetic movements whereby Kilauea had lowered following the end of Mauna Loa outbreaks. But none of this had ever before been measured from

day to day. Our staff from 1912 on occupied the trig stations, every day or night when the weather permitted, in order to measure within one foot the level of the live lava up or down. The lava was like the mercury in a barometer and needed incessant watching. This was done with a telescope, by people who dwelt on the edge of the vertical pipe. After 1913 the measurements clearly showed that sinkings were just as important as risings. They proved that the solid overflow matter and slide-rock slopes around the edges of lava lakes and coves measurably were a paste. This containing bank rose and fell at a different rate from that of the gassy liquid which streamed and fountained inside. The compiled results showed that the source of the liquid streaming was always at the west side of the pit bottom and that the streaming was toward fountaining grottos at the east. The liquid might at any time overflow its banks or sink down leaving inner cliffs, by failure of full supply up the west wall crack.

All of this may sound highly technical; but notes, photographs, seismograms, records of weather, and unceasing press releases and reports to the sponsors, while difficult for literary description, created a new technique. Science, when one is devising a new approach, consists of observation first, of experiment second, and of explanation or theory third. Something of that order has to be followed in the record of a scientist's life.

The surprising sympathetic lowering of Kilauea following the end of Mauna Loa eruptions was only one of numerous surprises during the first decade of the Observatory. For instance, the temperatures of hot cracks were repeatedly and systematically measured, and nothing sympathetic with lava motion was found. The same may be said about the weather. At the beginning it was supposed that rainfall, air temperature, barometric pressure, and possibly fluctuation of the trade wind, would affect the volcano. However, the only quickly evident effect was the visible vaporizing of many cracks on the Kilauea floor which dried up and diminished when the sunshine appeared, becoming dense and increasing in cold or wet weather. This obviously meant that the moisture content of the vaporizing cracks, some steam, but mostly moist hot air came from shallow rain water a short distance underground.

An effect that was more volcanic, but similar in principle, was the visible vapor inside of Halemaumau, close to the lava lakes, which always increased when the lava lowered and let the groundwater seep inward.

These visible vapors dwindled when the hot slag bubbled and rose, and acquired a brighter glow. No steam vapor rose from the glowing lakes. There was drying up of groundwater by increased volcanic heat, just as the cracks of the bigger crater had their moisture dried by the action of sunshine.

I shall have more to say on the subject of the seasons, the calendar effects on the plat of rising and falling lava, and especially the solar equinoxes and solstices. There appeared the hint of a daily tide-like rise and fall of the lava in the pit.

Finally, there arose the question of counting earthquakes, measuring their spacing in time and place, and seeing which belonged to Mauna Loa and which to Kilauea fault rifts. We had to plot earthquake frequency and size in relation to lowering lava, to day and night or to the seasons. The study of rhythmic swelling and creaking inside the great pasty mountains became an exciting quest. It gave promise of cycles from the hours of the day to the decades of the century.

We also discovered, by measuring vertical angles, that the inner floors rose and fell differently from the liquid lakes, hence the floors could be called the bench magma, as distinct from the liquid magma. This led to a bold experiment in 1917 when the liquid lava lakes became accessible, after a casual visitor, Mr. Walter Spalding of Honolulu, discovered an easy path down to the overflow floors at the edge of the north lake. Here the streaming slag rushed toward a glowing grotto, built up by spatter of a border fountain into a huge half-dome containing a glowing cavern hung with stalactites on the shore of the lake. The platform outside of the grotto was overflowed, and built up as the liquid lake rose, the platforms of overflow sloping away to the wall valley under the pit cliff. Thus the lake was at the top of an inner dome a thousand feet across, just as Halemaumau pit rim was at the top of an inner dome of Kilauea floor three miles across. The outer edge of Kilauea Crater is a big oval at the top of the outward sloping greater dome of Kilauea Mountain forty or fifty miles across.

When a little conelet formed on the northern or western floor platform inside Halemaumau, its slope around a splashing and fountaining crack would make a fourth innermost dome a few feet across in the series of progressively smaller cone-in-cone structures from the outer rim of the big

mountain inward to the Halemaumau centers of eruption. We saw such a conelet cave in just where I had stood and tested a flame the day before. Quietly the cone collapsed into a fountaining well of boiling lava beneath. The ring-in-ring conception must be held in mind with regard to any volcano, for one thing which we discovered is that cones are not only built up and collapsed but they are also swollen up by internal percolation of cracks and expansion of the hot stuff. This tumefying, or swelling, is concerned with the experiment now to be described.

Even after Perret described his “floating island” of 1911 and I saw the triangular islands appearing like shoals in a mud flat and gradually rising into crags in 1916 and 1917, I remained incredulous of the possibility of a basaltic island floating. When solid lava cracked off in pieces from inner cliffs around the lava lakes, the fragments immediately sank. Furthermore, when solid crusts formed on top of the foaming and streaming slag, the shells, when they got thick enough, cracked up, tilted up, and slid down and foundered in the melt beneath. It was obvious that lava rock is heavier than lava foam. Hence as an island is a rock, it would not float. This raised several questions. Where was the bottom of the lava lake on which it rested? Did the lava lake have a bottom, and if so how far down was the bottom when the same lake rose 600 feet in Halemaumau pit between June and December of 1916? In other words, was the lake 600 feet deep in December?

What would be the answer at any time if a stiff iron pipe were thrust down vertically into the liquid lake as a sounding rod? No one had ever raised the question. Cross-section drawings had always depicted the liquid as extending downward indefinitely within a vertical tube. When the lake became accessible in 1917, it seemed to me that a long steel pipe might be shoved over the border rampart, end on, and allowed to bend and sink, or to strike bottom. If the pipe could be recovered by dragging it back, fusible samples of known melting point might show the temperature of the depths.

For the experiment, 200 feet of one-inch iron pipe, which was screwed together in a single long piece, was laid across the north floor of Halemaumau. Ten assistants were distributed along the pipe twenty feet apart, and I stood on the rampart with Alec at the edge of the central portion of the lava lake. This was a high bank ten feet or more above the streaming liquid lava. The men were instructed to lift the entire long tube and walk

forward with it, so that it would plunge into the liquid lengthwise, arching down toward the center of the lake as it came past me. Alec helped guide the pipe over the bank, and the men came forward with it at a steady walk. The end of the pipe, covered with a screw cap, was plunged into the liquid lava, traveling toward the bottom at a good speed. The strong current toward the left dragged it somewhat, but not enough to prevent its sinking. After two and a half 20-foot joints of the pipe had plunged into the liquid at a slope of about fifty degrees, I could feel the pipe encountering the increasing resistance of a pasty bottom. Continued forward progress of the pipe caused it to stop and arch up, while the surface lava streamed past it, and its lower end was definitely stuck in the bottom substance of the lake.

I then gave the signal to the carriers to try to walk back to the place where they had started, with a view to pulling the pipe up and recovering the terminal length. The pipe trailed upward out of the lava lake like a red hot rope, then stuck and refused to come out. It came close against the bank where it was frozen solid in the stiff blankets of pahoehoe crust, which gripped it like hot iron.

The terminal length had been equipped internally with a spiral of spring steel, containing Seger cones which are used in the porcelain industry and which bear numbers indicating they melt at graded temperatures. This first thermometer by meltability was never recovered. The free lengths of pipe had to be unscrewed close to the bank, and four twenty-foot lengths were lost. In later tests we learned to keep the pipe oscillating back and forth so it would not freeze.

The epoch-making significance of this experiment was not understood until later. Calculation of the angle of slope of the pipe, where it went down into the liquid and hit on the bottom, showed that vertically the liquid was about fifty feet deep. With the aid of soldiers from the Kilauea Military Camp, this experiment was repeated several times; and each time the lake was found to be the same depth.

This conclusion was later verified by sudden subsidences of the liquid lava until the cliffs bordering the liquid were fifty feet high. The eastern grottos turned into cascades, with the liquid pouring down a well. The liquid lake had become a river pouring over a ledge of its own bottom, across from the western source wells to the eastern sinkholes. These latter

were fountaining grottos when the lakes were full, but they exhibited internal rectangular upright sinkholes when the lake level was down. This was verified repeatedly, and the phenomena of source wells at the west and cascading sinkholes at the east were confirmed and photographed. It thus became evident that the lava lakes were nothing more than convectional lava flows over pasty solidified substance of their own bottom sediment. Convection means rising foam, loss of gas, and sinking gas-free heavier liquid.

In other words, the bench magma capped with overflows on the marginal platforms was a paste, cooled from the top and bottom and sides and making the saucer of streaming liquid. It was this paste which constituted the swelling heart of the bench magma. The fountaining of gas bubbles escaping from solution robbed the lava of heat and caused it partially to solidify, always at a depth of about fifty feet. Thus there were necessarily three substances: The deep lava fizzing with self-heating gases (later proved to be inflammable hydrogen, carbon monoxide, sulfur, and inert nitrogen and argon), the streaming foam into which the deep lava expanded, and the semi-solidified refuse of the foam created at the bottoms and banks of the liquid lava when it cooled from bright yellow heat (about 1150° Centigrade) to a dark-red heat (about 900° Centigrade).

The streaming across the bottom from west to east meant that during six months of rising lava, some 600 feet in the last half of 1916, the lava column was a cylinder of semicooled lava, maintained by upward pressure of the deep lava bubbling up in the western crack between the cylinder and Halemaumau wall. Meanwhile, at all times, the lakes were nothing more than streams of foam fifty feet deep and skinned over on top, congealing on their bottoms and shores and cascading down sinkholes in the eastern wall cracks of the cylinder. A convectional circulation was what maintained the rising, foaming, heating, and cooling and the changes in density of the liquid as it lost its gas. Thus the entire fountaining phenomenon of the lava lakes was due to the self-heating of what is known as exothermic reaction of gas escaping from solution in molten basalt. Much of this is actually the burning of hydrogen in air, creating a convectional circulation wherever the deep lava can find an outlet.

Ordinarily these outlets are along cracks or rifts in the slope of the mountain, where they are seen to break out in gassy fountains 500 feet high,

and often to flow along the crack to a cavity where they cascade downward when less foamy and heavier. A lava flow is always solving a problem of foaming and liquefying, just as does champagne or beer.

There still remains the unsolved problems of how much of the deep lava is gas and whether it is mere pressure which holds the gases in solution, as in soda water. The alternative is for the deeper magma to be entirely gas, oozing up cracks in the globe, and reacting with oxygen from the air and solid rock, percolating from the core of the earth upward, and melting its walls.

In a sense, the entire decade to 1920 was an experiment. The results of that decade showed that the mountain swells and shrinks in tides with the passages of the sun and moon, but that Kilauea Mountain and Mauna Loa Mountain are all parts of what might be called Hawaii Island Mountain. The island of Hawaii is above an old ocean bottom 18,000 feet deep and is only the end of a ridge 1,700 miles long, which even at its lowest end, Midway and Ocean Islands, is still 12,000 feet high above the smooth mud-over-rock ball of the Pacific Ocean bottom. All the evidence shows the ridge to be a pile of lava flows over a crack, with a veneer of coral. If, then, the relatively small Kilauea dome is swelling and shrinking in sympathy with the sun, the long Hawaiian ridge is doing the same thing to a much greater degree.

Michelson has shown that the solid rock of the globe rises and falls in a tide about one foot every half day. As I have said, our daily measurements in 1912 showed that the lava in Halemaumau had a daily tide and that the larger movements reached maxima in June and December and minima in the intervening months, which proved it must be a solar effect. This was very exciting information and suggested a long train of experiments, which were to be successful in the next decade, based on the idea that the whole mountain swells as shown by leveling. This extends out to a radius of twenty miles from Kilauea Crater, and probably extends all the way to the seashore.

The actual measurement of a lava tide in Halemaumau was done during July and August 1919. R. H. Finch had just come from Washington to be my assistant. Oliver Emerson of Honolulu was another assistant, and two Harvard youths, Sumner Roberts and Charles Thorndike, who had been on

war missions in submarine chasers, sent word through their parents that they were anxious to do something dangerous around an active volcano. I jumped at the chance to employ them to help me measure the lava tide.

The north lake in Halemaumau was quite accessible, and we organized night and day shifts for surveying measurements from a canvas shelter on the actual bench lava near the lake. For twenty-minute periods, each observer critically measured a number of monuments on the bench magma and glowing places of the lake edge. Then a new measurement was started by leveling the transit. This sequence was kept up night and day for a lunar month, namely twenty-eight days. One of the monuments was a fixed Halemaumau benchmark, equipped at night with a lantern and used as a datum for the fluctuating lake points.

A second tent back from the Halemaumau rim was a camping base. Ford cars were kept running from the Volcano House for the changing of crew, Mrs. Jaggar looked out for the food, and I directed repeated surveys of the position of monuments and of the observation shelter.

Meantime the lava steadily rose during July, and at one time split open the Kilauea floor making an outflow back of the shelter. The vertical angles kept track of the movements of both the liquid and the semisolid lava. The instrument was planted on the lava column itself. On one occasion, Mrs. Jaggar's glove fell into a floor crack inside the shelter and burst into flame.

In all, there were more than 20,000 observations recorded. These were plotted on coordinate paper, and results were reduced to a smooth curve by overlapping averages. The actual curve of measurements was subjected to harmonic analysis at Yale University by Professor E. W. Brown, mathematician and specialist on motion of the moon and on lunar tides. The results showed a definite daily tide in both liquid lava and semisolid lava; of a few inches for the lunar tide, and of larger amounts for the solar effect. The curve plotted reached its greatest perfection of daily up-and-down waves during July at periods when the lava was steady. This became interrupted and ragged when accidents of drainage out on Kilauea floor pulled the liquid lava down.



14. River of Alikā flow, Mauna Loa, October 6, 1917



15. Lava streaming into a sinkhole in Halemaumau lava lake, July 7, 1917



16. Sakurajima Volcano, Japan, 1914



17. Fountain in lava lake, March 19, 1921

H. O. Wood, seismologist at the Volcano Observatory, was skilled in compiling the volcano's historical heights and depths of the nineteenth century and in plotting our curve of surveys of the liquid lava. He published a commentary on such plots for 1912–1913 in relation to solar curves of solstice and equinox, and to the oscillations of the global axis. He demonstrated a definite correlation between seasonal fluctuation of sun and moon and the seasonal rise and fall of the lava, presenting an extensive analysis of the rock tide in the globe and its application to Hawaiian volcanoes for a century. Perret had made a similar analysis for earthquakes and volcanoes in Italy.

These curves applied to the seasons, if compared with our lava tide applied to the hours of the day, left me with the conviction that the cyclical variations are a fact. They show correspondence between the swelling and shrinking of the globe and the movements of lava, when those movements are free and subject to surveying measurements. For few volcanoes are

surveys possible, and our measurements were the first in the world of any continuity.

Earthquakes, too, were studied. Dr. Arnold Romberg of the University of Texas—who has become a distinguished inventor in the world of seismology, magnetism, gravity, and oil prospecting—was Professor of Physics at the University of Hawaii about 1918 and for several summers came to the Hawaiian Observatory to assist me in experimental seismology.

From 1917 to 1920 I took the records of earthquakes and other seismic movements, as recorded by our Omori instruments, and Romberg remodelled these instruments. With his knowledge of the fundamental mathematics of pendulums, for at Harvard he had experimented with sensitive galvanometers, his facility for making instruments out of nothing but wire, solder, and old clockworks was wonderful and inspiring.

I spent many months measuring our smoked-paper seismograms of 1913 through 1918, with the assistance of Mrs. Jaggar, to whom I dictated. I measured types of local earthquakes, of volcanic tremors (some of which definitely accompany lava fountaining), and tilting of the ground, publishing the results in 1920. Tilt upswelling is shown in amount and direction by gradual change of the writing seismograph pens, and this is correlated with the recorded rise and fall of the lava.

In the course of three years, with Romberg's valuable advice, we changed the seismographs to record with little mirrors supported on silken fibers and with beams of light projected on photographic paper. And Romberg invented an ingenious improvement with a vane and a bath of oil, whereby a tilt-free seismograph for earthquakes only would keep the spacing of its lines uniform. Ground tilt crowds the lines.

We also experimented with a heavy cylinder which hung as a normal pendulum and which was capable of swinging in any direction, so that it threw a beam of light vertically upward to a chronograph covered with bromide paper. The chronograph was capable of being revolved and stopped, until the microseisms and microtremors reached their maximum of amplitude, for any given period of recording.

The permanent waviness of ground motion, the tremors with periods of about two-tenths of a second, and the microseisms with periods of about

five seconds showed their maxima of back-and-forth movement when the chronograph was revolved to a position where the pendulum swung northeast-southwest. This northeast-southwest tendency was found to be a characteristic of the seismograph cellar for many seismic measurements, including local earthquakes.

This was the direction at right angles to the edge of the cliff on which the Observatory stood. We concluded that this motion was characteristic of the upright flat slabs, with cracks behind them, which constitute the face of the crater cliff, and decided that any motion communicated to these slabs would tend to be a swaying toward the crater, rather than in the direction of stiffness parallel to the crater's edge. Omori has found a similar permanent tendency for Tokyo city, where the directions are northwest and southeast for maximum amplitude. This means that any spot on earth oscillates easiest in one direction.

These first ten years of the Observatory answered many questions and pointed the way for future experiment and study. It now appears that liquid lava is a gas froth, that Kilauea and Mauna Loa are all one system, that hydrogen is the most elemental gas in eruption, that a gas-free paste is the residue of flowing foam both in pits and lava flows, that earthquakes and vibrations are a function of this paste wedging up cracks and sinking back underground, and that the rise and fall is in tides and cycles, short and long. These things are not guesses, but measurements.

The earthquake problem at volcanoes is misunderstood in geology. The superstition that volcanic quakes are small is wrong. "Volcanic" in volcanology is not limited to volcanoes. Los Angeles, Charleston, Lisbon, and the deep ocean bottom are all volcanic, are all tremulous; and all have "lava" underneath. Kilauea and Midway Island are one, Rome and Etna are one, Iceland and St. Helena are one, Redlands and Mount Rainier are one, and the paste is underneath. These facts concern the globe, not a little bundle of wrinkles like the Alps.

We do not know what an earthquake is or what lava is. However, "lava" falling suddenly and rising slowly with big and few earthquakes accompanying fall, and little and many earthquakes appearing with rise, are facts observed at Kilauea. At Tokyo in 1923 the greatest quake in history centered at lowered lava and lowered sea-bottom next to Oshima Volcano

island. The Messina quake in 1908 made a hissing noise, and nearby Etna lava was low. There are long cracks in the earth shell somewhere deep down, and we know little about them except that volcanoes and faults are in lines. So long as the three-quarters of the globe under oceans are unexplored by man, with no rock specimens or even decent maps, and so long as there are no instruments planted on sea bottoms, we cannot use the term volcanic intelligently. Most volcanoes of the earth are undiscovered. Kilauea measurements whet the appetite for a new scientific frontier, the prospecting for ores, volcanoes, and mountains under the sea. The absence of core drilling and rock sampling over three-quarters of the earth is a disgrace to the oil-drilling and quarrying sciences of mankind.

The founding decade of the Hawaiian Observatory produced two effective expeditions, one to Japan and one to New Zealand.

The Research Association voted to send me to Kagoshima in Kyushu, the south island of Japan, where the volcano Sakurajima made earthquakes, explosions, and lava flows in January of 1914. About the same time Perret was sent to Sakurajima by Friedlaender of Naples, so we met in Japan.

Sakurajima, or Cherry Island, is a 4,000-foot cone in Kagoshima Sound, a deep inlet at the southernmost end of Kyushu. The volcano threatened 22,000 persons in villages on Sakurajima Island itself, and 70,000 in Kagoshima. It is a land of orange groves, fisherfolk, Satsuma porcelain, and maritime commerce, situated at the north end of the Okinawa-Ryukyu islands, a volcano chain extending north to Nagasaki.

Authorities in Kagoshima knew all about Pelée; and the army, navy, and governor wasted no time. Professor Omori, who had a seismograph at the weather station of Kagoshima, went at once to the volcano, and profiting from the lesson of Pelée, guided the lives of 90,000 persons.

The Sakurajima eruption began on a Saturday and Sunday with hundreds of earthquakes locally identified as coming from the volcano. Public and private vessels were called into service to move all the people of the island over to Kagoshima and beyond. With a general of the army in command, this was accomplished in two days. On Monday at ten o'clock the great, picturesque peak, quite like Pelée or Vesuvius, suddenly ejected vertically and quietly, from a crack in its flank, a column of "smoke" 30,000 feet high. This was answered by another, similar column on the opposite side of

the mountain; and the two columns joined above into a colossal arch of cauliflower clouds consisting of sand, dust, and boulders. The crack in the mountain which gave vent to all this opened with slight rumble and behaved like two radial ruptures meeting toward the peak, extending southwest and southeast. The sector of the mountain between them appeared to have been lifted like a piece of pie shoved up in the center. But the summit craters played no appreciable part in the eruption, unless it was a gush of steam on Sunday evening. The line of craterlets along the cracks and only half way up the mountain quickly developed lava flows, and these poured down, the one toward Kagoshima Strait, the other toward the narrow Osumi Strait, which separated the volcano from the wilder eastern mainland. This strait was filled up with heavy block lava, or aa, converting the island into a peninsula. A similar aa lava flow, fifty feet high in front, swept down to the beach on the Kagoshima side, with boulders as big as a house tumbling over its andesite front.

Tidal waves made by these two lava flows entering the sea were small but perceptible. The principal effect was thousands of white steam jets where the red hot blocks entered the ocean. Culmination of glowing heat came the second night, Tuesday. The flows continued for months, but the maximum of seismic effect had happened at six o'clock in the evening of the first day, Monday.

This was a really big earthquake damaging masonry and causing landslips from the cliff next to Kagoshima city and killing a number of people. The flux of refugees from the volcano villages on Monday was a dramatic event. When the lava outbreak occurred in the forenoon, the schools sent the children home. On their way, the children gazed entranced toward the terrific arch of cloud over the mountain, vomiting trajectories of stones. Shops closed, and the city was quiet while everybody sized up the crisis. As a schoolboy in English class wrote, "Monster rocks went horizontally from the down to the up, with smokes on their behind."

After the evening earthquake, however, when many buildings had shaken down, all except public officials were ordered to leave for the back country. Young men's clubs organized to receive the refugees along the roads which led into the interior of Satsuma province, while temples and schoolhouses were impressed into service to house them. The migration of more than 50,000 people with packs on their backs and with handcarts

bearing household goods, demonstrated how easily the Japanese people took to a nomad existence. This hegira came to an end on Wednesday, when Dr. Omori arrived from Tokyo, sized up the seismic record and the fiery crisis of Tuesday night, and took the grave responsibility of announcing that the population of Kagoshima might safely return. This was done, he was right, and no further damage beset the city.

Through all of this eruption, so different from Pelée in administrative control, no one was killed by the volcano, though one or two old people died of shock. One old lady who refused to leave her home on the island survived. Village roofs were bent down, crushed, and half buried under a heavy snowfall of ash, and it was notable that flat-roofed cottages were crushed, whereas those with steeper roofs were less damaged. Orange orchards were hopelessly destroyed.

At the west shore of Sakurajima in a place called Hakamagoshi, a fiery blast rushed down to the sea from the rift. Trees were stripped of limbs and bark, saplings were bent away from the volcano, and wood fiber on stumps was shredded by flying rocks. This blast was very short lived and never reached across to the city. It bore the marks of being similar to the downblasts of Mount Pelée. The lava flows kept on for a year and built new shore islands.

I had the remarkable experience of being rowed in a skiff over the submerged tongue of an eastern flow, trailing a thermometer in the increasingly boiling water. When the steaming water about us reached scalding temperature, we had the unpleasant thought that if we should capsize we would be cooked. We found boiled horses and cattle along the beaches, and thousands of dead fish. A climb near the eastern flank vent showed a portion of the moving lava flow pouring down the slope into a glowing cavern under a shell of its own bouldery texture.

The thousands of dollars of relief which came to Japan from America and elsewhere were handled with scrupulous honesty, and the inhabitants of the island were rehabilitated on Tanegashima, another island of the Ryukyu Archipelago.

Scientific investigations showed by leveling that the mountain had been lifted a few feet by the internal penetration of the lava, and reexamination of the benchmarks along roads extending out radially indicated that the

north end of the bay bottom and shore had definitely sunk, as though underground lava had been withdrawn from that region, to push up, swell, and overflow the mountain. This effect of subsidence outside was traced and shown to gradually lessen for a hundred miles from the place of greatest sinking. Investigation carried out by the geologist colleagues of Omori culminated in a monumental publication which demonstrates the solidarity of the Japanese methods of science. And both Omori and Professor Koto published books on Sakurajima in English, with maps, photographs, curves, and seismograms.

Omori, in 1910, had anticipated movement of the earth about a volcanic center as swelling up one place and sinking down in another while eruption was going on. At that time, he described Usu Volcano at the opposite end of Japan, where leveling instruments showed graded changes in height made by the Usu eruption. A remarkable physiographic character of Usu Mountain, and of the adjacent basin of Lake Toya, is that basin and dome appear complementary, just as Kagoshima Bay was compensated by Sakurajima. This same pairing of lake with volcano has been noted in other parts of Japan, as though tumefaction by lava penetration and lava eruption had robbed the underpinning of an adjacent piece of ground, which lowered and became a lake by filling up with groundwater.

From Sakurajima I went to Bandaisan, or Kobandai, a famous volcano in central Japan northwest of Tokyo and on the shore of a beautiful lake. It looks like an ordinary rocky peak, but its fame was made by a steam explosion from its flank which blew out the side of the mountain and left avast sulfurous quarry with numerous solfataras and hot springs. Bandai was known by geologists to be one of a chain of volcanoes, but prior to 1885 its activity was in question. One morning the sky was darkened by the overwhelming explosion, and vast volumes of rock from the outbreak poured down as a landslide and completely dammed a river system. It left extraordinary little heaps in the new dammed up lake. These appeared to be individual blocks of rock against which heaps of debris were piled so as to leave pyramidal humps scattered over the surface of the impounded water near the volcano. An excellent report in English on this eruption was published at the time, and the eruption became the type of what geologists call a phreatic explosion, meaning pure steam. There was doubt as to whether any fragments of new lava were thrown up.

I took with me to Bandaisan a photographer-guide. We camped in a mountain inn with thatched roof, visited a hot spring resort, and hiked to the crater where we measured temperatures and took photographs. It was a vast flat-floored shelf, dug out of the side of the mountain, with steam jets and puddles of boiling water at the back. Looking out at the new water-filled valley with its many islands at the base of the slope below the crater, we could see shoreline levels higher than the present beach, where the damming had produced the highest stand of the water. The eruption and landslide overwhelmed villages and killed many people, though it lasted only a few days. It was on the side of the mountain remote from the older lake. In clambering over the broken debris, which looked more like glacial deposits than volcanic agglomerate, I picked up some pieces of vesicular basalt that were definitely lava. Wada, a Japanese geologist, had found the same thing, and we both concluded that these were an internal live basalt blown to fragments in the Bandaisan eruption, but that most of the material was from the shattered old mountain.

My interpretation of Bandaisan is that it is an old volcano in the line of Asama and other volcanoes of central Japan, and that the line is a deep crack always full of lava in the depths, which is selective of outlet, depending upon what part of the crack opens as the path of least resistance. Eruption may be occasioned by lava wedging upward at one volcano, or by lava sinking downward at another volcano, according to the way the medial rift of continental Honshu is warped and stressed by the earthquake forces. One part of a volcano chain is always sinking, with lava withdrawn. Another part is always swelling up, with lava penetrating the cracks under active crater pits, like that of Asama.

Asama is the Vesuvius of central Japan near the village of Karuizawa, famous as the resort of American missionaries. Bandaisan is one of a line of volcanic peaks north of Asama, all of which have hot springs and solfataras. The explosion of Bandaisan, where the big natural lake represents the groundwater level of abundant rainfall, occurred when the underground lava column suddenly sank rapidly by the gaping open of the deep rift. The water poured into red hot cavities, while the lava was rising and erupting by frothing up in the depths of one of the other volcanoes. The results of Bandai's explosion were first, earthquake collapse, which was assisted by

vast outjets of boiling steam from groundwater, and then the blowing out of the mountainside.

Of special interest is the spacing, twenty to forty miles, between volcanoes along such a system as Asama-Bandai. The underlying cracks must be in echelon arrangement, and the spacing is a function of the thickness of the upper earth crust and its capacity through the ages of producing spaced-out widenings or bends in the crack, above whatever shell confines the lava. The same spacing of the new and old volcanoes is true in the Caribbees and in the Costa Rica-Mexico line. There an old peak might make a Bandaisan by unforeseen breakage and steam development.

This applies also to the Ryukyu-Sakurajima line. I visited Kaimon at the extreme south end of Kyushu, a steep dome blocked on top by a lava plug. South of here to Suwanose Island, an active volcano, the spacing of islands is similar to the northward spacing of Sakurajima, Kirishima, and Asosan, following the same law of selected vents and offset cracks. Kirishima thirty miles north of Sakurajima is a treacherous and dangerous volcano that made a bad explosion just prior to the Sakurajima eruption. I saw on the rim of its summit cavity a breadcrust bomb, a triangular block of rock eight feet long, with its surface beautifully tessellated with gaping cracks. This breadcrust fracture indicates that the fragment of glowing andesite was thrown up while pasty, then congealed on its surface to smooth glass and continued to swell evenly with internal gas, so as to rupture the glassy surface as expanding dough.

At Aso Volcano farther north I entered a natural gateway into a cauldron nine miles across, surrounded by a wall, and with a hilly country inside, from which a river escaped through the gateway. The summit peak in this landscape proved to contain an active pit on top. The pit was steaming and the source of the steam was boiling puddles of mud at the bottom. This was the "Halemaumau" of Asosan, which has had a record of many eruptions near the city of Kumamoto. The chain of Kyushu volcanoes ends, after the usual spacing, with a volcano at Nagasaki.

From Shimonoseki Strait, going northeastward, new belts of volcanic fissures have built the mountains of central Japan, cut across northwest of Tokyo by what Naumann called the fossa magna or big trench. This is famous in the history of Japanese geology, for which this German geologist

laid the foundations. The fossa magna extends northwest and southeast, through Fujiyama and Oshima Volcanoes to the Ogasawara Islands and the Bonin Islands, scene of volcanoes making and disappearing, from craters under the ocean.

Omori had discovered historical similarities between the eruptions of this chain and those of the Ryukyu chain. This is significant, because as we go from the small spacing of the individual volcanoes, we come to some deeper and larger fracturing of the whole crust of the earth that determines a spacing of hundreds of miles between such larger arcs of rupture as those of Kyushu and the Bonin Islands. As all are volcanic and have been so since the birth of the globe, it is unthinkable to me that they are anything but deep fractures which go down to the earth's core. The surface geology of marine strata is a mere veneer compared with the deep and ancient igneous rocks.

I went to New Zealand in 1920, taking with me in manuscript form the Hawaiian Observatory results of the past decade. Notable among geologists there was Dr. Allan Thomson, director of the Dominion Museum in Wellington. Dr. Thomson and his distinguished father, the Honorable William Thomson, guided Mrs. Jaggar and me all the way from Auckland to Dunedin. It was my task to give lectures on volcano research, to show lantern slides of Mount Pelée and Kilauea, to tell about seismographs and cycles, and to urge upon New Zealand science the importance of establishing a volcano observatory system in the Taupo Belt of volcanoes.

Here, in 1886, had occurred the terrific eruption of Tarawera. Here are spaced out volcanoes extending north into the islands of Tonga. Here, possibly, along the Cook Channel between the North and the South Islands, is a transition from volcanoes to earthquakes, and quite possibly another fossa magna worthy of comparison with Japan. Off to the east lies the profound linear Tonga Deep, compensating the New Zealand volcanic uplift. This is analogous to the Tuscarora Deep east of Japan.

We were fortunate to procure accommodations in Rotorua, the boiling geyser district, at the time of the visit of the Prince of Wales, later King Edward VIII, and to see the hakas, or dances, of an encampment of 5,000 Maoris, gathered to honor British royalty.

I was interested in the relics of liquid basalt collected on the lip of the great rift through Tarawera Mountain. The rupture extends the length of

Rotomahana Lake, which sank away as a groundwater phenomenon in 1886. This, like Bandaisan, was one of the great steamblast eruptions of history. It was right on the line of volcano spacing extending from White Island in the Bay of Plenty, to Ngauruhoe and Ruapehu Volcanoes, beyond Lake Taupo at the south. Here was a land of echelons of deep cracks, building up along scores of miles from submarine eruptions such as Falcon Island in the Tonga group. Farther south is the dangerous looking White Island close to the New Zealand shoreline, resembling Bogoslof, and so on to the lava volcanoes at the south. Big earthquakes have been characteristic, along with uplift, of both shorelines of Cook Strait.

This kind of gradation is certainly like the transitions from submarine eruption to continental uplift, crowned with volcanoes, so characteristic of Japan, the Aleutians, California, and Italy. It is impossible to think of it, when we consider water depths of 4,000 fathoms, and a step upward to such altitudes as the New Zealand alps, all linear for a distance of several hundred miles, except in terms of the faulted deep earth crust. And seismologists tell us that that crust is 1,800 miles deep.

The associations made on this trip were destined to have far-reaching effect in meetings with New Zealand scientists at later dates. I met Professor Bartrum of Auckland; the officials of the New Zealand Geological Survey; Dr. Ernest Marsden, distinguished physicist who had worked with Rutherford in England; and Dr. C. A. Cotton, physical geographer and author. Cotton showed us the elevated shorelines of Wellington associated with the big earthquakes of 1851. Other personages were Professor Speight, geologist of Christchurch College, and in Dunedin, Professor R. L. Jack, physicist of Otago University and our host. Dr. C. E. Adams, government astronomer of Wellington, we were to meet again on Tin Can Island in 1930, during the United States Eclipse Expedition. Dr. J. MacMillan-Brown, chancellor of the University of New Zealand, and his daughter entertained us in Christchurch; and he later visited us several times in Hawaii in the course of his extensive travels.

I was glad to stimulate volcanology in New Zealand and pleased when there eventually appeared the splendid work of Dr. L. I. Grange, on the "Rotorua District," with a project for geophysical surveys made imperative by the Napier earthquake disaster.

Before this chapter is closed, some personalities of the first decade of the Observatory should be mentioned. Foremost was L. A. Thurston, founder of the Volcano Research Association and its president for many years. It was his interest and enthusiasm coupled with that of the other members of the Association that made the Observatory possible. Prominent among those members was L. W. de Vis-Norton, for many years secretary of the Association and a devoted apostle of volcanology.

Mrs. Isabel Jaggar, from 1917, was my helper not only as wife and amanuensis, but as general assistant at the Observatory. She could operate instruments, take notes at the pit, keep the record books, and act as buffer against an overinquisitive public.

There was Demosthenes Lycurgus, genial Greek host of the Volcano House, who did all in his power to help us, by grants of lands, raising money, and personally promoting science with all the vigor of his wonderful personality. He went home to Greece to be married, and alas, died during his honeymoon. Later came my good friend George Lycurgus, who still operates the Volcano House.

Colleagues of the founding decade included H. O. Wood, who came from Berkeley in 1912, acted as seismologist and geological assistant, and established a seismological bulletin. He left to enter the army in 1917. In years to come Wood established in Pasadena under the Carnegie Institution one of the great seismographic laboratories of the world, and his name became coupled with a California Institute of Technology physicist to name the Wood-Anderson seismograph. Later came R. H. Finch who had worked with Dr. Humphreys of the Weather Bureau in Washington and had been a flight meteorologist in Ireland during the first World War. He was assigned by Marvin to me as assistant in 1919, when the Congress took over our work for the U.S. Weather Bureau.

Finally, I should like to name the numerous workers of the U.S. Geological Survey in topography and geology, notably Birdseye, Burkland, Stearns, Wilson, Clark, Meinzer, and Macdonald. These men brought to reality my Geological Survey estimate of 1899, when I recommended to Walcott a survey of the Hawaiian Islands.

The Hawaii geologic survey included investigations of water, highways, and minerals, and was to map lavas, volcanic processes, and island growth.

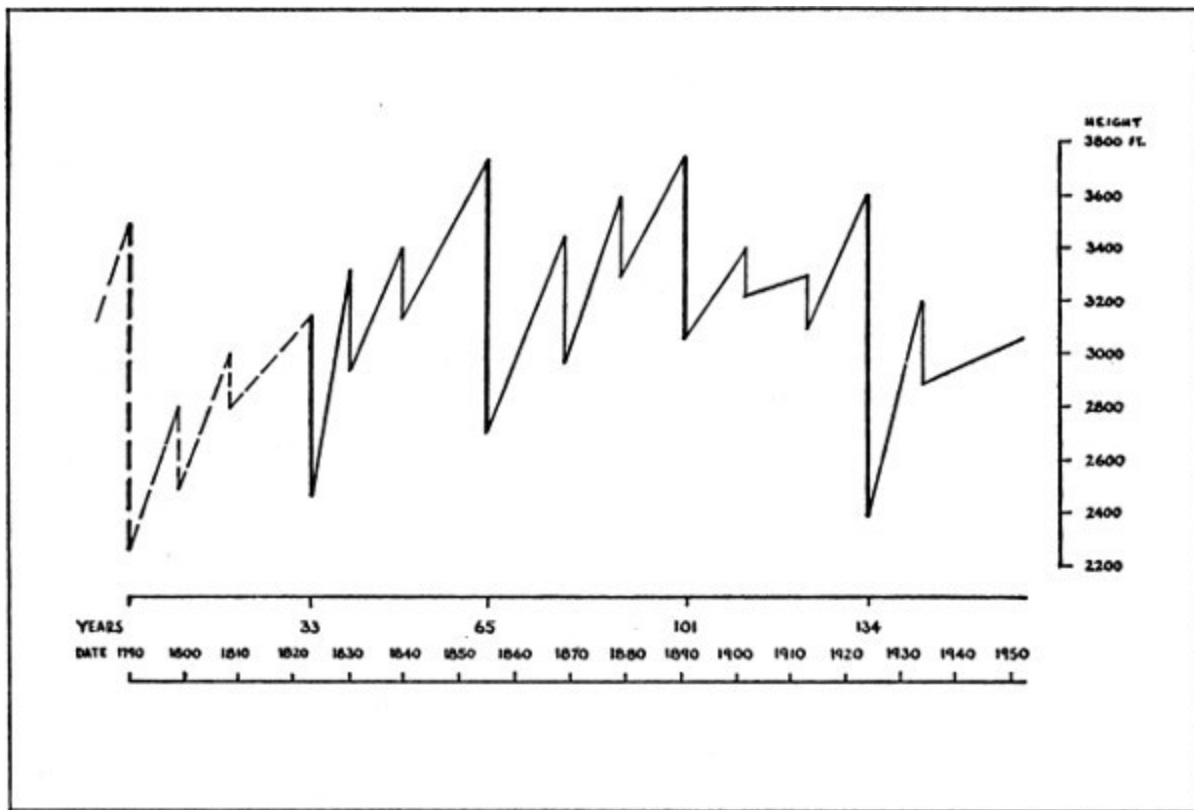
The annual cost of the work had been estimated at \$22,000, including \$6,300 for salaries in geology and \$10,000 for the total cost of topography study, or \$90,000 for five years. The project was begun in 1909 in cooperation with the Territory of Hawaii. In 1951 the mapping was completed and the cost had been many times the original estimate.

Among visitors who contributed to the Observatory work were Sidney Powers, a voluntary observer who had been one of my students in Boston. He explored and published on many volcanoes around the world and followed me in Sakurajima and the Aleutian Islands. He later became an outstanding petroleum geologist of the Amerada Company in Tulsa. Arthur Hannon, an architect from Cleveland, acted as a volunteer mapper, and for months aided with sketches of the changes in Halemaumau. William Twigg-Smith, an artist from New Zealand, joined us in the lava-sounding experiment and made numerous sketches and paintings. He later became the illustrator and photographer for the Hawaiian Sugar Planters' Association. Dr. A. L. Day of the Geophysical Laboratory visited us repeatedly, in association with gas chemist E. S. Shepherd. He wrote important monographs, along with E. H. Allen the chemist for the Carnegie Institution, on the Yellowstone and Lassen National Parks, and on Geyserville. Allen came to the Observatory for critical analysis of the steam of Sulphur Bank.

Among other visitors were geologists, geodesists, and biologists of the Pacific Science Congress, held in the spring of 1920. These included H. E. Gregory, Griffith Taylor, Frederick Wood-Jones, William Bowie, T. W. Vaughan, E. O. Hovey, E. C. Andrews, F. Omori, H. S. Washington, and Dr. Chilton of Christchurch, who had been one of our inspirers in the New Zealand trip. This Honolulu world congress assigned one meeting to Kilauea Volcano, which enabled me to summarize results before a cosmopolitan group of scientists.

The Washington executives who at this time promoted the Observatory were Secretary of Agriculture David F. Houston, Director George Otis Smith of Geological Survey, Chief Charles Marvin of the Weather Bureau, and Charles D. Walcott, Secretary of the Smithsonian. Later came W. C. Mendenhall, firm friend of the Observatory, and Director of the Survey.

It was my good fortune that between 1914 and 1919 Mauna Loa and Kilauea were building up lava toward a fiery crisis, and that the sugar business of Hawaii boomed at the same time. When the 1920 science congress convened there was much fresh lava to be seen, and our Research Association was so prosperous that M. I. T. in Boston kept up its financial interest. The *American Journal of Science* under Edward Dana of Yale published our results. This was fitting, as Dana's father, J. D. Dana, had published much about Hawaiian volcanoes. Consequently the end of the foundation decade made easier the financing of the next five years. Just at this time the Geological Survey spurted ahead, the National Park was opened, the Army built a recreation camp and a trail up Mauna Loa, the Inter-island Steamship Company took over the Volcano House, and a Promotion Committee was bringing many tourists.



Fluctuations of Halemaumau lava from 1790 to 1952, the verticals indicating maximum lowering preceding repose periods; minor fluctuations not shown.

CHAPTER V

Expansion Decade

“There shall be famines and earthquakes in divers places.”

The decade from 1921 through 1930 was a period of tremendous events and of experimentation at Kilauea and Mauna Loa. It was also an expansion decade for the Observatory, and for me. Additional funds made possible new buildings and equipment on Hawaii; observatory activity was established at Lassen Volcano in California; and expeditionary work included a study of the 1923 Tokyo earthquakes, explorations on Alaska volcanoes in 1927, and a visit to Niuafoou in Tonga, part of the great New Zealand-Tonga volcanic chain.

Increased government aid was largely due to the help of the Honorable Louis C. Cramton of Michigan, Republican floor leader of Congress, who took great interest in extending activities within national parks. After we moved from Weather Bureau control to Geological Survey in 1924, Cramton visited our Observatory, concluded that it was an orphan child of the government, and asked me what I wanted. I told him that I needed men and machines, and I suggested expanding our studies to California and the Aleutians.

Meantime, the Research Association was persuaded that we needed a fire-resistant iron building to house library accumulation, record books, and photographic negatives, as well as seismograms and lava specimens. These were precious relics of the very active overflows and experiments of the 1912–1921 period. With the advice of Walter F. Dillingham and Engineer John Mason Young of the University of Hawaii, I built a sheet-iron house with concrete floor and wire-glass skylights, and installed steel furniture. This became an invaluable office, drafting room, and workroom, as well as a place for files.

The Volcano Research Association, in cooperation with Hawaii National Park, built a trail side museum and lecture hall atop the high western bluff of Kilauea Crater. Later, when the drive was extended completely around the greater crater, the museum was on the road to Halemaumau. This

museum had a plate glass front, concrete floor, skylight illumination, and an esplanade looking down on the caldera and across the vast panorama of Mauna Loa, Mauna Kea, and the Kau Desert. The building protected the lookout platform from the trade winds.

We housed in the museum a gleaming, nickel-plated seismograph from Japan, suitable photographs, and the best of our specimens for visitors to see. This combined with the magnificent views to instruct the public in volcanology as nothing else could have done. At the same time, I equipped machine shops and added a first class mechanic to the staff.

It was during this decade and after my New Zealand trip that such persons as Omori and Nakamura, in Japan, and geologists in Seattle, Berkeley, and Pasadena began to take an interest in the volcano problem as dominant in the study of earthquakes.

There were conflicting theories about the earth crust. Earlier, in Hawaii, Wood was a disciple of the tectonic or contracting theories of the earth, whereas I increasingly believed volcanism to be profound, crustal, oceanic, and ancient. It is more fundamental than the strata and mountain folds of continents.

This conflict extended to the water question in volcanology. I was inclined to believe the waters of eruption to be oxidized hydrogen, whereas such physical chemists as Day, Shepherd, and Allen believed water vapor, like carbon dioxide, to be fundamental in magma.

The whole question of the origin of oxygen—the most abundant element of the rocks, air, and water—is a matter of startling doubt in geology. Where oxides are known to exist in lava, flames of oxidation make the gaseous fires; and underground water full of oxygen plays a part in steamblast eruption. All the waters of glaciers and oceans are oxides, and prove that the volcanic oxidation of hydrogen was the most primitive of the volcanic processes. Dr. E. H. Allen found water vapor dominant in the Sulphur Bank gas at Kilauea, whereas Day and Shepherd, who opposed Brun, thought water dominant in the gases of live lava. Its great preponderance in geological theory for such eruptions as Vesuvius led Allen to review theories and publish a long paper designed to refute my notion that oxidizing hydrogen is the primary volcanic ingredient.

As to earthquakes and so-called tectonic faults, the whole of geology has its thinking so warped by continents, the dwelling place of mankind, and so diverted from the great linear trenches and the ridges of the ocean crowned with volcanoes parallel to the deeps, that I became incredulous, along with Willis and Oldham, about the textbook cause of earthquakes.

The fascination offered by fossils, by ages of shellfish and reptiles, and by mountains of folded strata like the Alps and the Himalaya makes the votaries of evolutionary science neglect the mud-covered rocks and oceanic mountain ranges of almost three-quarters of the surface of the globe. This seventy-two percent they have never seen, nor collected hard rock specimens from, nor even mapped topographically. They are not acquainted with it by exploration, and their theories about it are a blank, except that gravity pendulums indicate it to be basalt.

The so-called geosyncline, or continental basin of sediments, filled with shells and strata as is the Mediterranean, is at the heart of all the theories of continents and mountains; and geology expressly excludes the geosyncline and its strata from the probabilities of deep ocean valleys. The most interesting subjects of continental geology are simply banished from conjecture. Interest in deep-ocean geology is lacking because science has made no field effort to bore or blast into it, and so extend engineering science to the deep ocean bottoms.

Earthquakes made a theme wherein I instinctively distrusted the word "tectonic." For generations the geological mind thought the earth losing heat, contracting internally, and wrinkling a crust in bumps, with vast overthrusts of broken strata, thus folding the Appalachians and the Andes. All sorts of accommodations to a thin crust thirty miles deep were invented; by Dana and Geikie, by Suess and Wiechert, and finally by one who should have been the foremost block faulting expert, Dutton. Hawaii convinced him that volcanoes are only skin deep and that the thin crust is so sensitive that a shift of the weight of river muds and sands is enough to push down the great valley of California, while an underflow pushes up the Sierra Nevada. This is the doctrine of "isostasy." It agrees with the Stübel idea of shallow remnant reservoirs for the lava of volcanoes.

Isostasy was devised by Dutton and pounced upon by the mathematicians, until they had gravity proving the whole world thin-crust

over an understratum of plastic lava. The seismologists on continents agreed, finding a density change, but with no evidence of fluidity. The world became, mathematically and petrologically, a sphere built of layers all the way down to the heavy fluid hot core, which was conveniently imagined to consist of iron and nickel, because some bolides of the solar system made of those metals occasionally fall on the earth.

All my experience of volcanoes and of deep oceans militated against a thin crust, a shallow underlayer of basalt to feed volcanoes, and a nickel iron core. The core is heavy, and sixty-two elements are heavier than iron. All reason seemed against the notion that the vast volcanic sea bottoms are a thin crust wrinkling under contraction. Reason found every evidence on both earth and moon for a thick peridotite or olivine crust, broken into ancient blocks, bounded by long lines of fracture, the blocks variously settling and scraping against each other from time immemorial, actuated by volcanic forces from the core. The whole of volcanology points toward sinking and down-faulted ocean basins, alongside the remnant upstanding continents which are the minor feature of the primitive earth surface. Water condensed and filled hollows. The processes of the core that made all this were volcanism—mother of air, ocean, seabottom, land, and life. The crust was thick enough to make cracks 2,000 miles long on a globe 8,000 miles in diameter. If there was a balancing of weights as in “isostasy,” it was between high silica in continental lava and low silica basalt that spread under the oceans. This is not static, but is a continuing process of a kinetic, or changing, earth.

This excursion into theory is intentional, so that in the middle of this book the geologically trained reader will understand that experience of volcanoes in Hawaii, the Caribbean, New Zealand, Alaska, Italy, and Japan had made me a rebel against conventional geology. The reason is that the great submerged mountain range of the long Hawaiian Archipelago is different from the mountain ranges of Europe and Asia and must be accounted for in global history. How would the three decades 1921 to 1950 confirm expectation that the deep ocean bottom is the most important and volcanic thing in geology, just as it is the biggest thing?

Routine observation and photography at Halemaumau pit reached a climax of recording brilliant fiery events in March 1921, and it changed to the recording of explosive steam in May 1924. The first of these fireworks,

after lava flows from a rift in the Kau Desert, draining the pit and fluctuating with the ups and downs of the pit lava, occurred in 1919–1920. This was a return to Halemaumau of effervescence in frothy volumes, so that the pit was overflowing on five sides. On March 20, 1921, occurred the most intense display of brilliancy, culminating the gradual rising of the lava column to outflow following 1918.

Then came, in the later months of 1921, a sinking away and recovery of the lava. In 1922 came a sinking again, with the lava breaking out in the Chain of Craters of the eastern rift, as though it had been blocked by freezing in the southwest rift and was forced over to split open the old cracks of the mountain to the east. This was confirmed in 1923 by another outbreak in the forest adjacent to the sixth crater, Makaopuhi, which with Napau pit beyond, had been the scene of the 1922 outflows.

This action was all extended in April 1924 to the shoreline end of the eastern rift, thirty miles away from Halemaumau, when the Kapoho country cracked open with many earthquakes, and a block of the mountain settled beneath sea level. Coconut palms at the beach were left in a lagoon of sea water eight feet deep. Seventy-five earthquakes in a day frightened away Filipino plantation laborers; railway and roads were ruptured, with new cliffs forming nine feet high; and all of this followed a monumental sinking of Halemaumau bottom, from a vast sea of lava to a tumble of debris in two months.

It was evident that between 1920 and 1924 the fracture of the long curved rift athwart Kilauea cauldron from the Kau Desert to the east point of the island was draining the lava out under the ocean to the east. Forty miles from the shore, the submarine slope is covered by 18,000 feet of water.

What is the result? The whole of Kilauea Mountain is charged with groundwater, which trickles warm through the beach at Pohoiki and partly warms ponds near Kapoho. Obviously this groundwater of the southeastern lobe of the island mountain surrounds the shaft of Halemaumau at some undefined depth, and the rising and falling glassy lava in the shaft ordinarily glazes itself with a water-tight skin, and may be thought of as a crusted tube. About this tube the groundwater shows only as the lazy steam of the little vents of the pit margins. On May 10, 1924, came the collapse of the

Halemaumau pit walls, introducing an explosive steam eruption such as had not been seen by five generations of Hawaiians.

The adventures of this period were glorious ones for the scientists. First should be mentioned the amazing subsidence which occurred suddenly at 2 A.M. November 28, 1919, just as Mrs. Jaggar looked across Kilauea Crater at the outline of crags and lava lakes making a glowing dome where Halemaumau pit should have been. We felt a lot of little earthquakes and saw the dome of lava heapings, with glowing lakes on top, sink slowly and majestically and leave the old familiar glowing pit. For almost the whole of 1919 this had been a dome, with overflows, now here, now there. At ten o'clock only the evening before, old Alec had conducted tourists to the top of the dome, where they looked down at the clover-leaf lakes. If it had started to go down while they were there—and any of us might have been there—it is awesome to think of the inevitable fiery engulfment.

After watching the sinking, which was followed by puffs of dust and smoke and some avalanche noise, we took a car to the pit at once. And when we got there in the early morning hours we found the pit enlarged to 2,000 feet across, with the pattern of the lava lakes still apparent at the bottom, indicating that the entire cylinder had lowered as a unit to a depth of about 700 feet. Red hot avalanches were tumbling inward with a roar, from the veneer of lava plastered on the wall. By the forenoon of that day the liquid lava started to pour up and inward as a ring of bubbling fountains all around the edges. What this ring represented was the wall crack between the subsided cylinder of semisolid lava, now pushing upward, and the funnel of rock wall outside. This V-shaped filling grew wider as the uprising progressed, and so the ring lake became wider, while the top of the harder column became a ring of crags and the space inside became a quiet lava puddle supplied by inflow from the ring lake. The whole column of ring crags with the lagoon inside and the brilliantly fountaining lake outside rose with unheard of rapidity during the next three weeks.

In mid-December, I took Mrs. Jaggar and a woman friend down to inspect this amazing corolla, or lily, of hard crags which had blossomed up in less than a month, so that the outer ring of boiling fluid was less than a hundred feet below us. We stood at the rift in the Kilauea floor which heads toward the southwest cliff, and suddenly we felt slight earthquakes and saw the face of that cliff crumbling in a visible tumble of rocks. The mountain

was quietly breaking open athwart the Kilauea caldera floor, and while we watched we saw forty or fifty low lava fountains in a straight line burst up along a floor crack between us and the cliff.

Remember that this crack traversed the downslope between Halemaumau edge and Kilauea wall. Looking back at the ring lake, we saw it beginning to lower and leave a shoreline of black plastering spatter. When we looked into the rift crack at our feet, only one or two feet gaping open, the liquid lava showed about twenty feet down. We were standing on the side of the crack away from the motor car terminus, and floods of lava on the Kilauea floor were spreading right and left from the straight line of vents between us and Kilauea wall. We had to get away from there pronto, as no one could tell what ground might erupt between us and our car.

I carefully instructed our friend to be deliberate and step across the fissure; but the girl felt sure that crossing a red hot crack called for a leap. She stepped on a loose slab at the edge of the narrow chasm and slipped into the crack, where she was wedged until we pulled her out. We then stepped across the fissure, for the live lava was far below, and made our way back to the car without further trouble.

The lake lowered only apportionately to the slowing black outflow on the south floor, which was short-lived. This was the beginning of a splitting open of the main Kilauea Mountain flank southwest and outside the crater which continued for months.

Another adventure, and an important one, happened with the outflooding of lava in the Kau Desert, where terrace upon terrace of pahoe-hoe lava was building up. This finally became a hill over the rift, two miles long and 200 feet high, which we called Mauna Iki, or little Mauna Loa. The exploration, day after day, of the extending quiet lava outwelling along this rift made it necessary to find new trails from the Pahala roadway and across the desert to the lengthening hillock.

Following the new Mauna Iki trail, Mr. Finch noticed that the ancient ash beds, two or three feet thick, had surfaces as hard as Portland cement. And on one of these he, like Robinson Crusoe, found the print of a naked foot, made when the old ash was a mud. On the trail across these old surfaces many more hardened, ancient footprints were found, of men, women, children, and pigs headed both up and down the mountain.



18. Isabel and Tom Jaggar in woods on Kilauea Volcano on their return from viewing 1923 eruption in Napau Crater



19. Lava lake, fountains, and crags, March 20, 1921



20. Footprints in ash west of Mauna Iki, said to have been made by Keoua's army during Halemaumau eruption of 1790

These prints recalled the story of Keoua's army when there was a big explosive eruption of Halemaumau in 1790 and the mud rains of the period were from ash which had been baked by the volcanic fires. If roasted and moistened, the chemical composition of powdered basalt is that of weak cement, and these surfaces were in hollows which had resisted erosion wash for 130 years. Part of the slopes closer to Halemaumau had been eroded bare, but they also showed footprints. Later the trail was followed up the mountain close to Kilauea Crater and down toward Pahala, and the ash of 1790 was found to be made up of pisolites, or fossil raindrops, in many places. Evidently the eruption had been accompanied by torrential thunder storms, and the natives had walked through the deposits of mud, which had in a century been dried by the sun into a resistant surface. These fossil footprints were to become one of the attractions of a tourist trail in the National Park.

One night in 1922, after some earthquakes of the evening, we were awakened by friends who told us that a glow like a forest fire could be seen from the high cliffs of Kilauea in the easterly direction of Makaopuhi. This big crater had a platform at one end and a pit at the other. We aroused Mr. Finch, then traveled by car as far as we could go on the truck trail, got lost, and with flashlights made our way on foot toward the glow and fume in a rugged wilderness, over cracked ground and old aa lava and obstructing vegetation. We were chilled by a cold drizzle and not at all sure where we would emerge.

Fortunately, the country is sufficiently open so that we could see the “pillar of fire by night.” It turned out that the new fire was in the deep end of Makaopuhi itself. From the western edge of Makaopuhi pit we looked down on ten or fifteen ribbons of lava, made by a line of spouting fountains at the top of the talus heap, and pouring from the top of the big slide-rock slope. We spent the night on the edge in much discomfort, and watched the puddle of accumulation in the bottom of the funnel and the glowing streaks which fed it. It was evident that the eastern rift of Kilauea Mountain had opened, and the lava outflow was found to extend to Napau Crater, a shallow saucer pit farther east. At the same time the lava in Halemaumau went down, enlarging the pit, and cauliflower dust clouds arose from much internal avalanching. This anticipated and resembled the avalanche steam blasts of 1924.

The adventures of the 1924 explosive eruption were too numerous and complicated to elaborate here. However, it was a tremendous event in the history of Hawaii and was totally unforeseeable on the basis of earlier experience. Mrs. Jaggar and I were in New York writing magazine articles and I was giving lectures, when word came from Finch and the newspapers that Halemaumau was caving in and throwing up rocks. We traveled with all haste to Honolulu, where the Navy agreed to send me by plane to Hilo, though they refused to take Mrs. Jaggar.

The Admiral’s car took us to Pearl Harbor, where a seaplane was ready and Mr. Thurston was waiting to see me off, accompanied by a motion picture cameraman. Then pilot Chourré took me into the sky over Diamond Head on my first flight. A companion plane was piloted by Lieutenant Sinton, who had radio communication with Pearl Harbor. Crossing high above the Molokai Channel, I looked down at the beautiful pattern of trade-

wind formed whitecaps, and was surprised after a half hour to observe that the wave crests were farther apart. I was even more surprised to see Sinton's plane far above us. The mechanic in the forward cockpit had been putting up his fingers repeatedly during our flight, to indicate, I later learned, how many cylinders were missing in the Liberty engine supported above us. Our plane was getting closer and closer to the waves and flying fish raced beside us. Finally we felt the bump of wave after wave on the bottoms of the pontoons, and the pilot brought the seaplane to a squelching stop, close to the surf of the Molokai reef.

We found ourselves in fifteen feet of water, the coral reef visible below. I was deputed to throw out an anchor and make the line fast to a cleat, while pilot and mechanic climbed up to the engine, which had been losing compression and could not keep up the requisite speed. Lieutenant Sinton's pilot plane came down and circled above us until he saw we were safe, then went on to Maui. Meanwhile, I watched the water with great interest, for sharks. When our boys got the engine going with a roar, I pulled up the anchor and we took off against wind and wave, with the pontoons going bang, bang, bang, against the tops of the waves. But finally we were airborne and out above the blue water.

Then the engine gave out again and we came down. This time the men rigged a sea anchor made of buckets with a line attached to the bow, to hold the ship's nose up to the wind, and battened the hatches with canvas covers. We clambered up on top of the upper wing to wait for rescue. The wind was blowing a gale, the whitecaps hissed by us, and we lay on our bellies. The aviators told me that this was the first forced landing they had had. The word landing seemed to me inapplicable.

We drifted for five hours, moving slowly down the wind, before a white motor boat appeared, coming from Molokai. At the same time smoke showed from two rescue vessels in the Pearl Harbor and Maui directions respectively. Sinton, who had radioed for help, flew back and circled above us, reminding me of the goonies soaring over a wounded bird on a fish line which I had seen in Alaskan waters. The Molokai boat reached us first, picked up our sea anchor and towed us into Kaunakakai. We pitched so and took such a pounding from the gigantic trade-wind waves that it didn't seem possible that the mahogany hull and the two lateral pontoons could hold together. However, we made the harbor and tied up to the buoy.

I hoped and prayed that the commercial packet, the *Mauna Kea*, might take me to Hilo. But no, the navy tug *Navaho* came from Lahaina, and Captain Green put up his megaphone and announced that the Admiral's instructions were that he was to take Dr. Jaggard to Hawaii. My heart sank because I knew what a seaway would be running against that little tub. The second rescue ship proved to be the *Pelican* equipped with a crane to swing the plane on board and take it back to Pearl Harbor.

On board the *Navaho* I was assigned a canvas camp cot in the lower, circular wheelhouse at the bow; and all night long waves broke over the bow and a foot of water sloshed back and forth under my cot. The pitching was so heavy and our speed so reduced that it took us all night to get across the Hawaii Channel, and we didn't make Hilo until 2 P.M. the second day. After that wet and seasick night, I found wry humor in our reception at Hilo Wharf, where we were met by Frank Cody with his motion picture camera and a bunch of hula girls and leis. Instead of five hours, the journey took thirty and quite failed to make me air-minded. Furthermore, I arrived at Kilauea Volcano in time for only the final stages of the explosive eruption.

Finch had organized volunteers, including Oliver Emerson as photographer, and even our collie dog, Teddy, who could hear and feel an explosion coming before we had any other warning. All observers wrote notes and fondled the seismographs during the three weeks of steam blast and cavings in of the pit, which had enlarged itself by collapse 700 feet outward radially in all directions. When I got there it was 3,500 by 3,000 feet in diameters and 1,300 feet deep, the bottom a funnel of converging taluses, made of avalanches from the pit walls. The taluses were wet and steaming vigorously in vertical lines, and at night showed red hot avalanches from the north and west walls, where two intrusive bodies of hard rock were red hot inside. The talus below stayed hot, and slides occurred for only a few seconds. The incandescent matter was not flowing in any sense, but was, rather, the peeling of a rocky boss of reddish color at the west and a canoe-shaped ledge at the north about 600 feet below the rim.

This showed the cross section of old screes, revealed above it, and horizontal basalt flows overlapping above that. It was a beautiful section of an ancient pit, of the same quality as Halemaumau itself, and the incandescent canoe sill at the bottom appeared to be an intrusion of fine-

grained gabbro, which had pushed its way in under an older talus funnel, similar to the present talus cup of Halemaumau, the bottom of which was 700 feet lower.

On the opposite wall of the pit the Kau Desert rift was displayed as a vertical cavern or arcade, merging into a group of dikes higher up and tapering to zero thinness at the top. These same dikes, less conspicuous, cut the canoe sill on the northeast wall, to indicate that the ring of the pit was fractured vertically from below. This fracture is the main deep rift of the mountain which crosses under Kilauea, bending in the direction of Kilauea Iki, and this it was which had opened as a curved chasm to let the lava down. Lava had gone down in a succession of flank outflows, with intervening rises, from the Kau Desert in 1920 to the final drainage under the sea at the east. This drainage had let in the groundwater, made a steam boiler, and so caused the explosive eruption and engulfment of Halemaumau walls as the mountain yawned open.

A. L. Day made one of his return excursions to Kilauea at this time and thus saw the extraordinary phenomenon of the hard basaltic intrusive bodies half way down the walls, caving to a red hot talus. The explosions, which started with two-hour intervals, gradually decreased, coming at four hours and eight hours; and on May 18 came the culminating cauliflower clouds with torrents downward of broken rock, some of it showing low red heat. At all times the motive power was steam jets 10 to 15 thousand feet high, which plastered the pahoehoe of the pit edge with broken wall rock fragments of every size.

There was no sign of pasty lava or glassy bombs in the ejecta, and the red incandescence seen at night in some of the explosions was the avalanche material of the western boss and the canoe sill.

It took the pit less than two months, to mid-July, to recover its liquid lava, which poured through the talus and made aa puddles, to form a new pattern of cone source and short-lived flows. Then everything came to rest, and lava activity was not resumed there until the summer of 1927. However, in 1926 Mauna Loa went into action on its southwestern rift, and sent an aa flow into the sea at South Kona, destroying the village of Hoopuloa.

Here was history in the island lava column of majestic decline and recovery from 1914 onward. Outflow in Mauna Loa crater at 13,000 feet in 1914 extended to outflow from the southwest rift in 1916 and 1919 at 8,000 feet. Next, in 1920, came outflow in the Kau Desert from Kilauea, at 3,000 feet. There were alternating spurts upward within Halemaumau pit, acting as a crater similar to Mauna Loa's at the lower Kilauea level of 3,700 feet.

Then this whole progress downward moved over to the Chain of Craters at 2,500 feet, and finally to the ruptured earthquake rift of Kapoho on the east point of Hawaii and at beach level. Some miles farther east, on the same rift beneath the sea, the gigantic submarine mountain of Hawaii drained the last lava from Halemaumau pit and let in groundwater which caused steam explosions.

July 1924 saw the deep lava recovering in the crack and sealing off the water, so as to bubble up in the bottom debris of Halemaumau and push its way upward into the crevices of the island. It reached the top of Mauna Loa in 1926 and reactivated outflow at the center of the island. This migration of vents from top to bottom and back again took twelve years of fracturing, and it relieved from lava this big piece of the Hawaiian ridge. In reaching the groundwater and steamblast phase, it accomplished something which had not happened since 1790, making a supercycle of 134 years.

The decade after explosion at Halemaumau was marked by small lava gushes in the bottom of the pit, bringing the depth from 1,300 feet in 1924 to 750 feet in 1934. The layers were something less than 100 feet each, and they were fed by pahoehoe conelets at the slide-rock margin. As usual, the lava was gushing up the western wall crack along the margin of the bottom magma cylinder. There was no trace of recurrence of steam blasts.

Despite the excitement of actual events, experimentation continued; and I continued working on inventions for the experiments. Two approaches to our problems concerned seismic recorders which could be put in the hands of amateurs, and range finders for improving pit surveys. I had been convinced for many years that the three-component seismograph was too elaborate to be operated by volunteer school teachers or telephone operators who have other things to do. Such a seismograph records with photographic paper the north-south, east-west, and up-down motion of the ground, on a chronograph which keeps accurate time and registers a wavy line every

second, so that the recording paper has to be changed and developed every day. Moreover, these instruments are for measuring distance to earthquake origins by physics of wave motion, and they have become hopelessly mathematical. Such mathematics makes for assumptions of uniformity about a rock crust which is not uniform. Qualitative science wants to know what happens at a specific rock location and wants the motion recorded by the simplest possible mechanical device. It also wants a value in number at each location, for size and direction of the first motion. This is for an earthquake, identified as one incident, over such an island as Hawaii, where the rock units are many and different. This is especially true of long periods of time when there may be no earthquakes to record.

I devised a simple shock recorder, consisting of a horizontal boom of very light wood attached to a hinged weight which swung like a door, so that the boom scratched a line on a circular card which was rotated and moved along by a common alarm clock. The result was a spiral mark on the card, such that an earthquake interposed would write a zigzag opposite a place on the clock face appropriate to the time of day. All that was necessary was to remove and date the card, wind the clock once a day, and measure the zigzag.

Mr. Ingalls of *Scientific American* read an article by me in which I described my shock recorder and thought it would lead amateur machinists to devise their own machines and to record the vibrations about them. Numerous amateurs did send in designs for instruments, and Ingalls believed that the seismograph hobby would become as popular as the amateur astronomical telescope hobby. But it failed because the amateurs were waiting for earthquakes, which didn't happen. They were not content with vibrations from trucks, railroad trains, waterfalls, surf on rocks, artillery practice, or wind storms.

My improved shock recorder gained some use later in New Zealand and Montserrat, after big earthquakes in those places stirred the authorities to build simple instruments. However, popular seismoscope simply doesn't exist.

The range finder I had been working on since my teaching days in Massachusetts Tech, where I had made an optical device with a traveling index mirror which moved along an upright scale of centimeters, and a

sextant telescope. The idea was a transit, with self-contained base line close to the operator. My theory was that in such measurements of distance as we had to use—to about a thousand feet or less, to the lava fountains in the bottom of Halemaumau pit—we might read off the vertical distance from a single station, when all other stations were enclosed in smoke.

In the Aleutian Islands and elsewhere I experimented with a Zeiss stereoscopic rangefinder designed for artillery ranges, but it was not accurate enough for short distances. Everything in my instrument depended on moving a telescope parallel to itself with superlative precision, on a scale within the instrument. I finally hit upon using a track of taut piano wire, probably the straightest line in all mechanics.

If one first looked at an object twenty miles away (infinite distance), the telescope could be moved along right and left and the image would remain immovable on a vertical hair. If it were now focussed on an object 1,000 feet away, the displacement of the telescope on the centimeter scale would measure the distance with a high degree of accuracy. This was the stadia principle inverted to contain the rod at the observing position.

I also made several graphic devices for surveying Halemaumau daily from the rim benchmarks. However, when lava overtopped the rim and destroyed the datum posts, mapping became difficult.

Drilling temperature wells into the floor and rim of Kilauea Crater was a project I had anticipated when Mr. John Brooks Henderson of Washington came to Hawaii and offered to help finance it. We had taken the temperature of hot cracks in many places, and found them to range from 320° Centigrade at the Postal Card Crack close to Halemaumau, down to 96° Centigrade at Sulphur Bank, and then on to lower temperatures at many cracks which yielded visible vapor in damp weather but no vapor at all in sunshine. A spectacle for tourists was a crack on the Sulphur Bank flat, where a cigar to windward or the exhaust gas of a car would nucleate the invisible vapor and cause a big puff of white “steam” to show. This phenomenon, which depends on smoke particles condensing invisible water vapor, is well known at Solfatara near Naples.

The experimental approach to finding out what the temperature of the ground really is, is to drill a hole and keep the temperature measured repeatedly with a thermometer, and to find out the thermal gradient change

vertically if possible. This means to measure how much the temperature changes with depth. The whole problem concerns how much unusual heat energy is released at a place like Kilauea Crater.

With the aid of Hobart, a drilling engineer, I started at Sulphur Bank with a churn drill. We quickly discovered that we were going through intensely hard basalt, containing metallic sulfide which appeared to be pyrite but turned out to be marcasite. After drilling for several years—with four holes at Sulphur Bank, one sixty-foot hole under the observatory shop, and about twenty-five holes in the eastern part of the Kilauea floor over a surveyed map pattern—we changed to a rotary core drill using steel shot, and then changed again to a percussion drill actuated by compressed air, for shallow holes to show cross-country temperatures. Unfortunately core drilling requires large quantities of water, which we did not have, for the Hawaii National Park depends upon rainwater collected from roofs in redwood tanks. Without water cooling, rotary bits heat and expand in hot rock, stick, and are often lost.

Two seventy-foot holes, one at Sulphur Bank and the other in the middle of Kilauea floor, showed no definite thermal gradient; and in general it turned out that drill holes were dependent on steam in the cracks for their temperatures.

Heat was brought up by vapor, and in a number of ten-foot holes scattered over the Kilauea floor, the hottest were at the edge of the floor. The Postal Card Crack, near the edge of Halemaumau and 600 feet above red hot intrusives, was exceptionally hot, and it is not at all clear how the water made contact with the hot intrusive rock underneath. This place completely caved in and was lost forever within the enlarged pit of 1924. Sulphur Bank itself is at the edge of an old Kilauea floor on a shelf at a high level. The extra heat at floor edges means a wall crack between the crater fill and the confining funnel, so that hot gas comes up from intrusive lava somewhere deep down toward the center.

Thus when a mercurial thermometer was lowered, ten-foot holes would show a hot place half way down and cold rock at the bottom. Some holes had no heat at all, which meant that an inclined steam crack was cut across by the hole or that no steam crack was present. The heat supply was dependent upon vapor channels from heated rainwater, but we were never

able, owing to lack of funds, to drill a hole deep enough to find the water supply which made the steam. It is a remarkable fact that the casings on three wells at Sulphur Bank emit continuously a column of steam exactly at the theoretical boiling point for this altitude, as though the groundwater were boiling only a short distance below. Dr. Allen by his analyses proved that Sulphur Bank vapor was ninety-nine percent steam and that the remainder contained fractions of a percent of sulfur and carbon dioxide, but this sulfur was enough in the course of months to coat the interior of our casings with yellow crystals over black iron sulfide. It coated the Sulphur Bank with yellow crystals of sulfur and soaked the rock below to generate brassy iron sulfide.

The result of these experiments was to exhibit the complexity of any solfatar in its relation to underground lava, and to the soakage of a volcanic country by rainfall. This is especially important for Martinique and Montserrat.

Another experiment was conducted by Emerson, who was equipped by the Observatory with chemical apparatus to make qualitative analyses of numerous Kilauea products, and he also did critical photographic work, including some photography in the infra-red. In one valuable experiment, he melted Kilauea lava in a refractory crucible at a temperature of about 1200° Centigrade until it was as fluid as honey. Allowed to chill and harden naturally, it was shiny glass like pahoehoe lava. If stirred with an iron rod, it made sprouted black needles, crystallized all through, like aa lava. Thus he proved that stirring made Hawaiian lava crystallize and sprout like fudge, or like the solidification of such metals as silver and bismuth. Sudden outbreak and stirring anywhere will convert pahoehoe to aa; but never does aa become converted physically to pahoehoe, unless flame melts it. The standing pinnacles in the midst of an aa flow, which breaks up into boulders, give evidence of the stirring process.

When Emerson's discovery is applied to basaltic lava flows, it appears that the glassy lava of a source, when stirred by gas fountaining or by flowing, will change from its glassy condition to a sprouting and crystallizing condition. All flows are glassy pahoehoe pumice fountains at source vents, and a quarter of a mile away they become aa. Later the source pahoehoe preserves itself within a glassy skin and pours forward under glassy shells and frontal toes.

R. M. Wilson's work supplied proof of a swelling mountain. Wilson was one of the three leading members of the topographic party of the Geological Survey. The other two were C. Birdseye and A. Burkland. Wilson, whom I had known as a student in M. I. T., was a product of Spofford's Civil Engineering department and was to become the chief computer of the Survey in Washington. As levelman in Birdseye's organization, he became topographic engineer of the Observatory, and produced by precise leveling and triangulation the brilliant experiment which showed [the swelling and shrinking of the mountain during fifteen years.

By close cooperation with the U.S. Coast and Geodetic Survey we placed a tide gauge at Hilo, both for a sea level base and to record tidal waves. Wilson also, in 1921, ran a level line from Hilo to the Volcano House benchmark, where the Geological Survey had run levels in 1911. The county roadway was marked with bronze plates inscribed with leveling heights, and Wilson's results showed the edge of Kilauea Crater to be three feet higher in 1921 than it had been in 1911.

Wilson's determination of heights above sea level certified that the whole mountain swelled up during the ten years prior to 1921. About 1918, lava and seismographs had proved rising overflow at the center, while the edge of Kilauea Crater was being tilted away from the center. This went on during the massive rising of the interior lava of Halemaumau into a dome where the pit had been, and it proved that Kilauea Mountain was being injected along cracks, not only under the pit, but along the rifts, as indicated by outflow on the southwest and east in the years 1920 and 1924.

But this was not all of Wilson's work. He revisited all surveying stations after the big collapse of Halemaumau that accompanied the explosive eruption of May 1924, and found that the Volcano House benchmark lowered a little more than three feet during May 1924 and that places close to Halemaumau dropped nearly fifteen feet. This lowering of the mountain was graduated outward twenty miles from the center at trig stations, or concrete posts, in the Kau Desert and at stations along the road to Hilo. These stations changed altitude to show that the big mountain tumefied or swelled up to that distance of twenty miles during the big intrusion of cracks at the overflowing time, as though the mountain dome were a tumor forty miles in diameter with Halemaumau at the center. Of course there is no certainty that the shore line in Puna, or even the Hilo tide gauge itself,

did not go down with the slumping of the mountain, for the thing called sea level is nothing but an average of tide gauge readings at a fixed wharf. Remember that the east point of Hawaii sank eight feet on the Kilauea rift during the April crisis.

Wilson also surveyed by horizontal triangulation in 1921, determining that stations around Halemaumau had moved inward toward the center, by a specified number of feet, different at each station, and that other stations outside of Kilauea Crater had changed position horizontally on the map, as though the mountain were shrinking. This entire series of measurements of change between 1911 and 1926 jibed with the seismograph's measurements of the tilting of the ground. The seismograph picked out 1918, when Halemaumau overflowed, as the swelling year. In 1924 the tilt reversed itself, turning inward toward Halemaumau, and became tremendous when the pit collapsed and exploded.

It is impossible to accent sufficiently the importance of the discovery of a measured swelling and slumping of a volcano throughout a lava crisis occupying fifteen years. It was so tremendous that critical engineers in Washington refused to believe Wilson's results. However, his findings were verified by the contemporaneous lava measurement results, earthquake enumerations, and tilt meter results. These showed that earthquake frequency increased when Kilauea slumped and that a lava mountain had swelled until it was three feet higher at the summit in ten years and had contracted by a larger amount during the years of an explosive eruption period immediately thereafter. This all agrees with the excellent results in Omori's volcanic and seismic events, obtained by Japanese army and navy engineers at several volcanoes and earthquakes. It also agrees with the positive facts of Vesuvius and the Canary Islands, starting with the controversy about "elevation craters" started by Leopold von Buch in the first half of the nineteenth century and carried forward by Mercalli on Vesuvius in 1894 when a lava hill was seen to swell up. There, too, others would not believe. The opposition always insisted that a volcano was built by heaped material, that it could not possibly swell.

Wilson's results are far-reaching, for the whole of geology depends on uplift of continents and downsinking of sedimentary basins. Most geologists account for these things by the theory of weighting and underflow at a thin crust (isostasy), refusing to grant that volcanic heat and

tumefaction yield intrusive power everywhere through cracks in the deeper crust.

I wish that I could describe adequately the high adventure of this fruitful time. We built a vehicle from a model T Ford with a Ruckstell axle, stripped of mudguards and equipped with balloon tires doubled at the rear, so as to travel and carry loads over the smooth pahoehoe of the Kilauea floor. We found that a powerful light rig of this type, with excessively low gear, could climb up on lava lobes one to two feet high. But this called for experienced driving and special methods. Sending a man on foot ahead to pick a way and to drag a crowbar which scratched a track, we could drive anywhere on the lava. And we used this rig to haul drums of water and drill apparatus. I once drove artillery officers out over the rough floor of the crater, and afterwards saw similar cars used by the army in the first World War as cross-country transportation for the doctors and wounded in No Man's Land.

Before a roadway encircled Kilauea Crater, Mr. Finch and I, carrying two-inch planks for bridging cracks, made the complete circuit of the crater by way of the rifted Kau Desert in our special vehicle, which has now been succeeded by the jeep, the most universal vehicle of World War II. Volcanology prospected the field of war in more ways than one, so I named my popular book "Volcanoes declare war."

Inventions led to expeditions both in Hawaii and in distant lands during the decade of the twenties, some by invitation, some to offer assistance at disasters, and some for the natural extension of my own work. On September 1, 1923, came the big earthquake at Tokyo. With Mrs. Jaggar, I was permitted to land in Japan and make a study of the effects of the disaster. The destruction of Tokyo and Yokohama was a final, sad tragedy for Omori, who for years had worked to protect the Emperor and Japan by studying earthquake forecasts for Tokyo and by conducting research in earthquake-proof engineering. It was a cruel commentary that the disaster came while he was attending a science congress in Australia, particularly as the great destruction of life was occasioned by fire and typhoon winds. But Omori's organization handled the seismic event admirably. Omori returned at once, but almost immediately died.

We steamed into Yokohama harbor, were welcomed by Captain Gatesford Lincoln U. S. N. and his destroyer flotilla. We went on board his flagship, and were sent in his launch to the broken jetties of Yokohama, where we found no custom house or police. We walked up to the camp of United States marines, amid the wreckage of the United States Consulate, where the Consul had been killed.

Yokohama, which I had known well in 1909 and 1914, was a tumble of ruins; and the long Bund with its splendid waterfront structures, including the Grand Hotel, was a heap of rubble. My classmate Purington, a mining geologist who had been staying at the Grand with his family escaped with one child and went back to rescue his wife. A second shock brought down more masonry and crushed him.

We were given a tent and allowed to mess with the marines, and next day we crowded into a train for Tokyo. It was packed to the doors and had people seated on the roof. We were warned by Americans to dress as roughly as possible, as the populace was on edge, and foreigners must not appear to be tourists. By great good luck we got into the Imperial Hotel which had withstood the shake and fire, though it was considerably damaged.

We visited the Honjo district of the river bottom, where the damage was at maximum, and we saw the remains of a pile of corpses, clothes, and household goods in one small yard where 30,000 people had been incinerated. Fire had closed in from all sides, and the shrieking mob of men, women, and children piled up on top of each other, amid handcarts and clothing bundles—kindling which added fuel to the horror.

The mayor of Tokyo sent us in a small steamer to the island of Oshima, on which is the volcano Mihara, close to the earthquake center. We climbed up and looked down into a glowing pit which was making no lava outflow at the time, though Mihara is famous for basalt flows.

Water soundings showed 900 feet of subsidence in Sagami Bay opposite Oshima, and there were changed depths elsewhere, some of them shallowing by underwater land slips. We went to the Boshu Peninsula east of Tokyo, where the beach had been rising for many years, and where the earthquake rising left wharves high and dry. The principal effect of the earthquake, occurring at noon just when all charcoal braziers were lighted

for luncheon in the flimsy Japanese houses of wood and paper, was to set fires in an area of hundreds of square miles and a score of towns. Water reservoirs were destroyed, there was no adequate fire department to care for a conflagration, and a high wind was blowing in the bright sunshine. A characteristic of Japanese cities was the absence of open parkways for refugees, hence the crushing, burning, drowning, suffocation, and annihilation of hundreds of thousands of people and the destruction of factories, railroad trains, water supplies, power plants, and every essential utility in a great metropolitan area with a population of many millions. The horizontal movement of the ground in the shock was about eight inches, and aftershocks kept on for many months.

We explored Yokohama, clambering up to the Bluff where everything was wrecked and where land slips had tumbled down the precipice. We visited what remained of a beautiful English type villa with a slate roof, which had been occupied by two missionary ladies and their numerous parrots. These people were encamped, along with their parrots, in a shack built by their yardman, for the residence had tumbled down like a house of cards. One woman had been imprisoned between her bed and the wall, and was quite uninjured when the gardener dug her out through cracks of the roof.

Scientifically, what happened to the ground in this earthquake was not explained by any single fault. Whatever happened to the bottom of Sagami Bay was not communicated across the beach to the coast as any great rift. Small faults were identified in a number of places, the shoreline in one place lifted a few feet and lowered in another; but no such movement as the big subsidence of the bottom of the bay crossed the contact of sea and land. It appeared as though the margin of the bay itself outlined an area of sudden slumping, somehow related to Mihara Volcano on Oshima; but the shoreline of that island was not seriously affected. The great mountains of the foothills of Fujiyama, and the Hakone district, were shaken to a hash of broken railway tunnels and land slips, but the topography was not altered.

A resurvey of trig stations west of Tokyo revealed movements that indicated the country had been spirally twisted. However, it has always seemed a mystery to me that all the motion on land was so small, when change on the bottom of the bay was so great.

There was a local tidal wave at the bay shore of Kamakura, but no great tidal wave from the deep ocean came to Tokyo. Some volcanic effort of deep lava had wedged open and jolted the sea bottom, but how it acted is entirely obscure. It was quite different from the San Francisco quake, with its side slip of twenty-one feet and a crack 400 miles long.

When we returned to Japan in 1926 with the Pacific Science Congress, the restoration of Tokyo was practically complete and a magnificent greater city had been built with wide and large parkways. The government was lavish in its entertainment of the scientists attending the congress. Dr. Lacroix and I were sent to Osaka to lecture, and to show lantern slides of Mount Pelée; and expeditions all over Japan were arranged for the visitors. I had an opportunity to see for the first time the large basaltic lava fields of the lake district at the base of Fujiyama, and I was astonished at the similarity of the basaltic pahoehoe to our Hawaiian outflows and the freshness of the lavas and the lava caverns. I had never thought of Fujiyama as a “lava flow” volcano.

My next expedition was in the autumn of 1924, when I was invited by H. E. Gregory, Director of Bishop Museum, to go on an expedition on the USS *Whippoorwill*, Commander Samuel King, to Howland and Baker Islands. Others on the expedition were C. Montague Cooke (malacologist), George Munro (ornithologist), Erling Christophersen (botanist), Ted Dranga (marine shell collector), George Collins (Museum Trustee), and Bruce Cartwright (naturalist). These men were invited to make up one of several Bishop Museum parties which were sent out to south sea islands for collection and report.

As geologist, my job was to carry a portable seismograph and record earthquakes or microseisms and to take photographs. We had made up at the Observatory a one-component horizontal pendulum, in which the chronograph drum used smoked paper. In camp I lowered the box containing the seismograph into a hole in the sand under my cot, with a view to finding out what tremors occurred on these flat coral islands. However, no movements were detected during the period of our stay, within the sensitivity limit of the small seismograph.

Howland and Baker are coral islets, not atolls, close to the equator, with no lagoons and with deep water all around them. Howland later became

famous in the tragedy of Amelia Earhart, for whom the Coast Guard prepared an airfield on the island. These islands had been guano diggings for parties from Honolulu fifty years earlier, and we found old cisterns and tracks. The islands were inhabited by thousands of goonies (gannets), man-of-war birds, and terns. In some places they covered the ground with their nests, eggs, and young, rising noisily in terrifying swarms as we walked among them. The land was perfectly flat brown guano and red weeds, with beaches of coral boulders and *Tridacna*, or giant clams, the highest bit ridges on the windward side. The easterly trade winds blew a powerful gale most of the time, and our ship had to land us on the leeward beaches, where we made our camp in a line of tents. The staff was divided into pairs for each tent, and Filipino mess boys did the cooking.

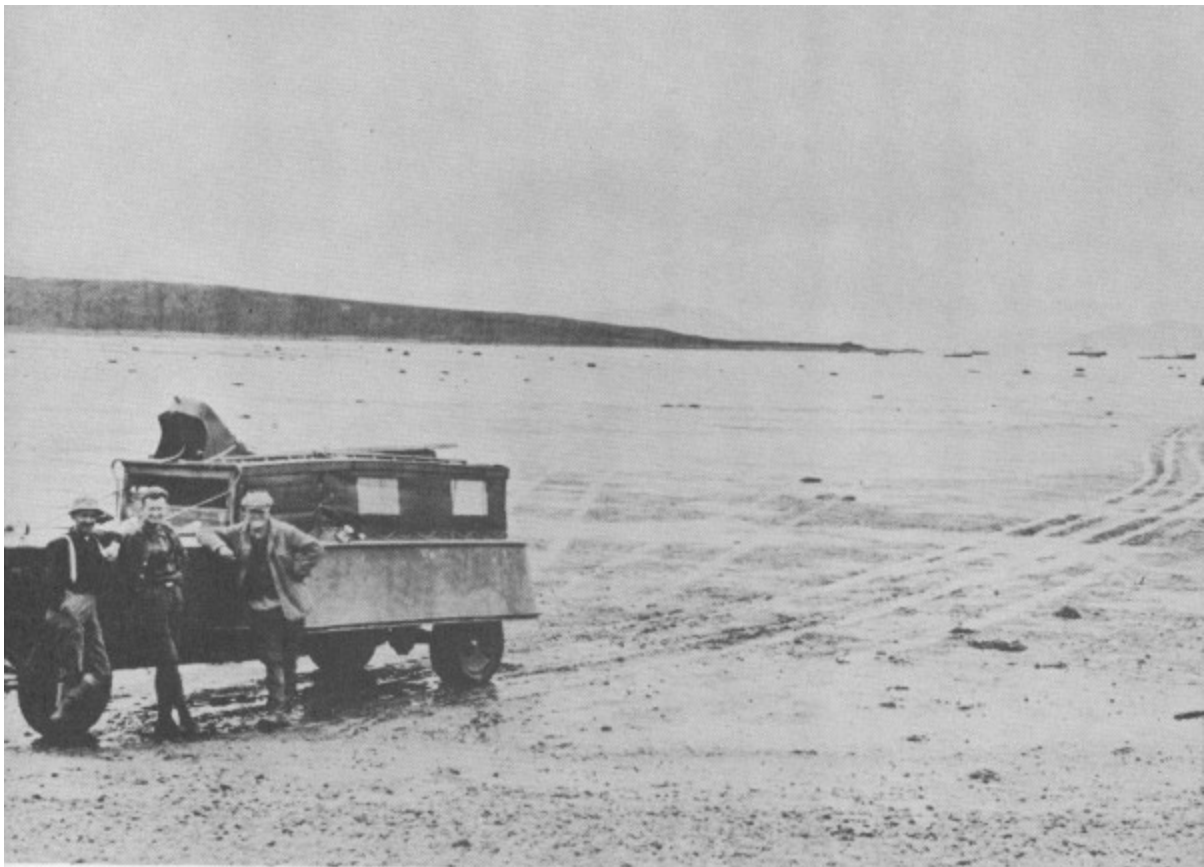
Landing was arduous, for there was heavy surf, even on the leeward side, and it was necessary to have a man swim in with a line in his teeth. The swimmer, Ted Dranga, made the line fast between a buoy and the shore, then built a signal fire while the ship stood off. Men and baggage were loaded into a skiff and hauled ashore by the sailor in the bow, who pulled, hand over hand, on the rope from the buoy, when the waves were favorable. The ship had to drift away each night and come back, as there was no anchorage. A few stunted kou trees still survived from guano-digging days, and numerous grasses and fleshy-leaved salt weeds grew. The beaches were covered with rats, hermit crabs, and some white ghost crabs. The ghost crabs were seen at night flittering down into the water when a flashlight was turned on the waves.

The hermit crabs, with borrowed shells, came clanking over the canvas floor under our cots at night; and as one walked along the beach with a flashlight, Polynesian rats pattered away in all directions. They had been brought by the guano schooners and doubtless lived on shellfish, birds, eggs, and fledglings.

The principal products of this expedition were notes, pictures, maps, and collections.

Within the next few years we were to combine expeditions with experimentation in the organization of new observatories in California and Alaska.

California volcanoes as a field of observatory study were an obvious choice when Judge Cramton proposed enlargement of the volcano enterprise. He succeeded in getting me a Section of Volcanology in the Geological Survey, and I sent R. H. Finch to Lassen Volcanic National Park, where he made his headquarters at Mineral. Lassen had made steamblast explosions in 1912 through 1914 which had rushed down into the forest with such horizontal destruction as occurred at Mount Pelée. It was not realized that this blast was terrible, for it was in the backwoods on top of the Sierra Nevada and little known. The national park there was created later. It is an area with a recent (1871?) cinder cone and rocky lava flow, boiling lakes and mud pots, numerous solfataras and hot springs, and a lava cavern much like those on Hawaii.



21. *The Honukai on Alaska beach, 1928. Jaggard on the right*



22. The Ohiki, first amphibian truck, with passengers Isabel Jaggar, Tahara, L. A. Thurston, Jaggar, and Ted Dranga, 1928



23. Lava flow entering village of Hoopuloa, 1926



24. Lava flow of 1926 Mauna Loa eruption approaching Hoopuloa Village, which was destroyed.
Photo section U.S. Army Air Force

Lassen Peak is the southernmost volcano on the line where the Cascade Range merges with the Sierra Nevada. The line of volcanoes extends beyond Mount Baker into Canada. North of Lassen is the Glass Mountain region where there are obsidian lava flows. Like Mount Shasta, Lassen is a volcano of very few recent eruptions, but there were at least two outbreaks in the nineteenth century. These two volcanoes resemble Pelée and Soufrière. Their linear quality implies a long ragged rift in the earth's crust, and south of Lassen there is suggested an offset rift at Mount St. Helena, near the famous superheated steam of Geyserville. This is near the northern end of the great San Andreas rift, which extends many hundreds of miles southeast of San Francisco. The rift shifted in a north-south direction during the earthquake of 1906, and is one of the many evidences that the north-south faults of California are all a part of the faulting up, over lava, of the Cordillera, relative to downsunken Pacific Ocean slabs.

I put Finch in charge of Aleutian Islands seismographs, as well as the one he was to establish at Mineral. With Wilson as seismologist and instrument designer in Hawaii, we started constructing horizontal pendulums, like those used in Hawaii, making the weights out of large iron pipes, to be filled with sand at the place of operation. These were two-component seismographs, recording on a single chronograph drum. We sent one to the Coast Survey station at Sitka and built two more for Kodiak and Unalaska. Finch built and set up his own seismograph at Mineral. He started systematic surveys of the temperatures of hot springs and steam jets in different parts of Lassen Park and kept close contact with the Geological Department of the University of California at Berkeley. Lassen was the subject of geological surveys by Anderson and Finch, and later the park area was studied by Howel Williams.

I went to Washington to see government authorities, particularly Professor Charles F. Marvin, Chief of the Weather Bureau, and Dr. G. O. Smith, Director of the Geological Survey. I can never express my indebtedness to Marvin, a good designer who built an inverted pendulum seismograph in Washington. Finch had worked with Marvin when he was weather observer in airplanes based on Ireland during the first World War. Hence methods of government contact and reports, in the early days of our Observatory, were kindly guided by Marvin. The Weather Bureau was a place of self-recording instruments, something new for geology, and much needed for volcano observation. For weather is a matter of present changes, whereas geology had long been a matter of ancient specimens.

Director Smith was instrumental in calling a meeting in Washington, of scientists of all bureaus interested in the Aleutian Islands. I was selected to lead the symposium, which included representatives of climatology, biology and fisheries, geology and geochemistry, oceanography and geodesy, hydrographic charting, gravity, and magnetism. There proved to be great interest in the Alaskan Peninsula and the islands, and the Survey published a special bulletin on the symposium.

W. C. Mendenhall, who had written a monograph on the volcano of Mount Wrangell in the great bend of the continent around the Gulf of Alaska, became director of the Geological Survey and one of my best friends.

In 1927 I was ready with cross-country cars and a seismograph to explore once more the volcanoes of Alaska. Organizing an expensive expedition which called for a special ship was obviously out, but in the years after the Technology Expedition of 1907 I had learned many economies which I wanted to try out. Also I had two experimental and mechanical tests to make. The first was to set up in Alaska a seismograph, the second was to test Alaskan beaches with a cross-country car, with a view to building an amphibian boat. I had read in several languages on the subject of motor vehicles with boat bodies, and my 1907 experience of finding no anchorage on Umnak Island had convinced me of the need for a vessel on wheels which could climb up an Alaskan beach and be converted into a camp. So I started from Seattle with a low gear Ford runabout. I unloaded it first in Kodiak village, where there were only one or two cars, and made tests of driving it along beaches.

At Kodiak the Agricultural Experiment Station allowed me to set up the seismograph in a vacant basement, and I arranged with a local housewife to operate the instrument. Aided by a sheet of instructions, she made tests, changed the smoked papers, varnished them, mailing them to Hawaii, and kept notes on earthquakes which were felt.

The roadster and I then traveled by the local mailboat steamer *Starr*, Captain Johanssen, and sailed along the south shore of the Alaskan Peninsula to King Cove, visiting Bradford on the way. Disembarking at King Cove, I made runs on the beach with the car. With the aid of the cannery mechanic, I tried attaching winch spools to drive wheels, in order to haul the car up to grassy land behind the beach. No car had ever landed at the cannery, there were no roads, and the problem of getting from the wharf to the tundra, and from the tundra to the beach and back again, posed practical mechanical problems, the solution of which was to be useful later. We ran along the beach as far as a rocky headland, until we needed an amphibian boat in which to round the point and rejoin the stony beach at some place beyond. How that boat body should be constructed was planned from this experience.

The superintendent, the physician, and the boatbuilders of the large King Cove cannery planned an exploration for me, with John Gardner as boatman and Peter Yatchmeneff as his mate. These two were on their way to hunt bears for an eastern museum and were going to Pavlof Volcano, the

Vesuvius of the Alaskan Peninsula. I transferred my baggage to their motor sloop, the *Plug Ugly*, and we headed for Pavlof Bay.

At Volcano Bay we landed for a bear hunt, which was very exciting for me. When we found bear tracks in an amphitheater under big mountains, we climbed toward the divide at the head; but we could find no pass over it. From the high ground we looked across the river at clumps of alders. John borrowed my field glass, handed it back, and pointed out a black spot far away under the bushes. "I just saw it move," he said, "that spot is a big brown bear where he has been holed up."

I remained watching while John and Pete, with their 25-caliber Savage carbines, crept across the valley bottom, keeping down the wind from the bear in the shelter of bushes. I saw that they were getting very close to the game, lost sight of them for a few minutes, then heard two sharp cracks and saw the bear in violent motion, thrashing around and tearing up the ground, then quickly subsiding. I made my way across the valley and found they had neatly shot a year old Alaskan brown bear. The rest of the day was given to skinning it, and we sank the skull, tied to a fish line from the sloop, to the bottom of the bay where marine organisms would eat away remaining flesh and leave the bone clean.

Next we sailed up to the head of Pavlof Bay and camped at a barabara, or sod hut, preparatory to a trek to a small volcano that lies near a shallow lake on the north side of the magnificent pair of snowy volcano cones known as Pavlof and Pavlof Sister. We were early in the season and could see a glacier extending down from Pavlof Crater, which is a cup containing a conelet at the side of the summit. The crater is like a collar, the conelet like the knot of a necktie, while the glacier is the ribbon of the necktie, itself, extending down to a jumble of snowy hills with rocky moraines at the edge of the lake. We made camp and ran into some adverse weather, and also into a party of mainland sportsmen. We gave up further hunting and returned to King Cove, for John had his bear and that was enough. The curved beauty of the Pavlof cones, with a sweep of lava flows to the west of them, heavily mantled with snow, was exquisite and a knowledge of the cones was useful when plans were made for a later expedition.

Mrs. Jaggar, after a trip by way of the Yukon into the interior of Alaska waited for me at Kodiak while I took Captain Johanssen's SS *Starr* to

Unalaska where I saw my friends of the Coast Guard and received an invitation to go later on the *Unalga* to Attu. I stayed on the *Starr* to Bristol Bay on the Bering Sea side, in order to see the Alaskan Peninsula from the north.

A rewarding view showed me the almost inaccessible Aghileen Pinnacles, a marvelous mountain west of Pavlof, consisting of dozens of upright spires, all covered with ice, and looking like a cluster of cathedrals in a snow storm. At the head of Bristol Bay I saw one of the government Indian schools, met some of the teachers, and met trappers who came on board with interesting collections of fox furs. They told me about Naknek Lake, which gives access to Katmai from that side by dog sled in winter. The necessary husky dogs were tied out in the fields around a mission station.

A rumpus on deck between a storekeeper of the district and the United States Marshall arose over a feud between two villages which were quarreling about the placing of a United States post office. There was no shooting, though it looked bad for a few minutes, and I realized the far north was a replica of the far west.

On my return to Unalaska, Coast Guard officers and I were invited to a dinner on board the German cruiser and training ship *Emden*. I had nothing to wear but a hunting coat, whereas the others were in dress uniforms, but the Germans didn't mind. I greatly enjoyed the *Emden's* officers, whom I heard from later, including Captain Foerster, an acquaintance of my son in Seattle.

On board the *Unalga* I was given the Captain's cabin, for he was absent on sick leave. Executive Officer Perkins, who acted as skipper, preferred to live in his own quarters. Another guest on the trip to Attu was Jack McCord, whose interests were sheep herding and whaling, two industries which were making experimental progress in the islands. We saw a sheep ranch in the western part of Unalaska Island and learned that a recent landing on Bogoslof had found the conditions much like those I had seen in 1907 when I noted the smoking cone, the millions of murre, the three islands, the connecting beaches, the warm lagoon, and the dozens of sea lions.

At Nikolski on the west end of Umnak Island, a flat land where sedimentary rocks appeared, we had to mine and blow up a schooner

recently sunk in the harbor. Going westward, we passed cones in groups or on individual islands, and we met the usual fogs and gales. The officers were interested in Adak Harbor, but our plan to enter it was defeated by storms.

We anchored off Chugul, where two Aleutian men and a boy had been marooned for months by the non-return of the wrecked schooner. A trader had leased the island and left them to collect blue foxes for him. When their supplies gave out, they lived on fish, vegetation, eggs, and sea birds. They had matches left but no ammunition, so they had loaded cartridges by assembling match ends. However, they were sheltered in a sod hut at one side of the grassy volcano, and were living proof that an Aleut cannot starve. They were fat and healthy and had a good load of furs. When we transported them to the village on Attu, the first thing one of these men did was to marry an Attu girl, with the aid of the local priest.

Chugul was the last of the shapely volcanic cones. Attu geology was different, with old metamorphic and sedimentary rocks and ancient lavas, but without any sign of fresh volcanoes. It is a mountainous island with deep fjords, and we crossed a divide in order to look down on Sarana Bay, made famous by World War II. McCord and I walked out on the peninsula west of the village of Chernofski, and saw snowy ranges beyond the next bay to the west. The Aleutian uplands are covered with luxurious grasses, many flowers, and much mossy swamp; and there are signs of terracing in places, as though made by old elevated beaches. The country is too wet and stormy to be attractive for raising livestock. However, when we landed on Amchitka Island on the south side of the chain, we found it drier with fine grassy uplands. We found also the usual shore cliffs and foxes.

We returned to Unalaska, where I was attracted by the empty hotel building and wharves at Dutch Harbor, deserted by the Alaska Commercial Company after the booming maritime trade of the Cape Nome gold days. I talked to Company officers about using the buildings as a scientific station. An old powder house would be suitable for a seismograph cellar; the wireless station was nearby; and there was water, lumber, and housing for every possible purpose. It was ideal for an Aleutian geophysical station, if financing and collaboration could be had. Later, in Seattle, I addressed the Chamber of Commerce and published in our Bulletin a proposal for an Aleutian Geographical Observatory, but nothing came of it at that time. The

Aleutian Islands became a center for landing craft, airfields, and defense forces during World War II, and eventually our men Howard Powers and Austin Jones were employed there.

In 1928, Gilbert Grosvenor of the National Geographic Society, in cooperation with the Geological Survey, equipped me with an expedition to map, photograph, and survey 2,500 square miles in the vicinity of Pavlof Volcano. Again I had John Gardner and Pete as camp men. McKinley, our topographer, brought pack animals and Alex Bradford transported us to our base camp in Canoe Bay. I slept during summer in the *Honukai*, a twin-screw steel amphibian boat, which was manufactured in Chicago, after a preliminary vessel made of wood and impelled by paddle wheels had been constructed at our Hawaiian Observatory shop and tried out over a 400-mile course along the shores of Hawaii.

The trial of the preliminary vessel, which we called *Ohiki*, Hawaiian for ghost crab, took place during the spring of 1928. The entire staff of our Observatory were engaged in it, with Mrs. Jaggar as stewardess, as usual. Mr. Thurston went along as a passenger and publicity man on the trip up the west coast of Hawaii, where I tested out Kona beaches and checked on the craft's seaworthiness.

We had misadventure at the start, in that the driving wheels tended to dig in on soft beaches; and we found it necessary to build washboards to raise the gunwhale amidships to avoid shipping water in choppy seas. In the cross country trek from Kilauea, using the boat as a truck, Mr. Thurston was overwhelmed with admiration for the twenty-one foot work skiff, thundering down the steep hills of Kona on wheels, controlled by the low gears of a Ford. Its boat body excited all the roadside kids to wild antics of delight. My excellent truck builder, Boyrie, used the same Ford which had run along the beaches in Alaska, reconstructing it in the observatory machine shop.

Wilson's photograph of the *Ohiki*, with Mr. Thurston on board, became the frontispiece of a top secret publication on amphibians of World War II's joint army staffs in London. The amphibian war of the Pacific Ocean and Normandy was to develop dozens of different designs of landing craft, but war use was unforeseen by me at the time of our experiments.

With a crew of four we cruised from Kailua to Kawaihae along the west coast of Hawaii, landing on beaches and lava flows, and camping at Makalawena, Kiholo, and Puako. We encountered real grief at Kawaihae against the front of a soft submerged bank in shallow water, where the front wheels made too much resistance and the rear wheels dug into a mud bottom. We needed front wheel pull, but we finally got the craft up the beach by power hauling with gypsy and cable and a tree. More grief developed on our way up to Waimea when we fractured wooden rear axle attachments. We went gratefully into the Parker Ranch shop for some days, until we were able to return to Hilo and the volcano, completing the circuit of the island.

The National Geographic vessel was built by George Powell who advertised a Ford “mobileboat,” designed for the use of fishermen to enter midcontinent lakes. He had started on a larger model, which Grosvenor accepted for the National Geographic Expedition. Powell and I tried it out on Bellingham roads and lakes and on beaches of Puget Sound. We provided everything extra, for Alaska had no roadside filling stations. A wheeled vehicle on the peninsula was unheard of. We had elongate steel mats to give traction across the upper sands of a beach, and this plus bow winch, levers, and manpower enabled us to abandon beaches and enter the tundra. Our planning paid off, for in the 400 miles along the coast of Alaska from Shumagin Islands to King Cove, over water, beaches, and tundra, we did not even have to pump up the tires. The *Honukai*’s numerous excessively low gears even enabled us to drive to the snowline and bring out the heavy fur and bones of a bear that I had shot on a snowy volcano, Mount Dana.

The expedition was very productive. McKinley made an excellent topographic map; we corrected errors in old maps; we obtained many photographs through Richard Stewart, who carried still, color, and movie cameras; and we obtained minerals, fossils, geologic notes, and many plants which I collected. McKinley used a panorama camera for his topographic work and his wide photographs were invaluable as a record of the country.

Meanwhile, I kept in touch with the seismograph station at Kodiak. The steamers of the Pacific Commercial Company, which owned several of the canneries and had headquarters in Bellingham, transported us from Puget Sound to King Cove, and the many tugs for the canneries’ salmon traps

enabled me to make local explorations along the southern coast. At one trap the fishermen had a tame baby seal, who would eat nothing but little trout caught for him from the brook. He lived in a box, and went off to sea by himself at night; but he always came home next morning.

In 1929 Finch sent Austin Jones, a seismologist, to construct and establish a hut at the Dutch Harbor radio station for a second Alaskan seismograph, of the Hawaiian type designed by R. M. Wilson. Jones taught the wife of a radio operator to manipulate the station and transmit the seismograms. The women in charge of the two stations at Kodiak and Dutch Harbor kept their work up for several years, and kept in constant correspondence with me. Though in the winter time they had to dig the stations out of snowdrifts, and to cope with all kinds of damage from rain and storm, they courageously and faithfully visited the instruments. It is a hellish country for weather.

Although both stations were within fifty miles of active volcanoes, earthquakes were not numerous, and the story was very different from that told by our records made at the edge of Kilauea caldera, only two miles from an active lava center. Thus we have demonstrated that the only way to study an active volcano is to live close to the crater itself, even if a shelter has to be built underground.

In concluding this story of our Alaskan expeditions of the twenties, in contrast to my windjamming experience of 1907, I must underscore the importance of water transportation and credit those who have provided it. In fact, all transportation was by water until aircraft became supplemental. I feel that the U.S. Coast Guard, which takes care of the Pribilof Island seals, is the supreme achievement of our government in policing these stormy waters. Their 60-foot motor cruiser, equipped with sails has come to be standard for such government bureaus as the Biological Survey and has replaced the earlier, 80-foot sealing schooner among the traders.

The canneries maintain big boatbuilding yards and operate large and powerful tugs for visiting the salmon traps. The traps are heavy weirs made of northwest pine logs, which are battered to pieces by the winter storms and must be rebuilt with pile drivers every spring. Thus a by product of cannery activities, and a godsend for trappers, fishermen, Aleuts, and campers is the pine lumber distributed all along the beaches from the annual

wreckage of salmon traps. It is the only firewood and construction material of the country to be found anywhere west of Kodiak, for the land has no forests.

Our contribution to the boating problem was the exhibition of what an amphibian landing truck will do on Alaskan beaches and its usefulness along those beaches where a boat may be in difficulties from stormy weather.

I returned to my Hawaii headquarters in the fall of 1928. The year 1929 was marked by an earthquake crisis which began in mid-September with an unusual number of shocks in the vicinity of Hualalai Volcano, a place hitherto notably free from earthquakes. This was of interest because events on Mauna Loa had shown higher and higher lava sources and quake centers for the south rift. The 1926 outflow had begun by splitting northward across the summit crater, and there making a considerable flow eastward toward Wood Valley while Wingate and his topographic party were in camp close to the summit. Therefore, when the 1929 quakes began near Puuwaawaa, it looked as though Mauna Loa eruptions might begin again at the northwest.

A very strong quake of September 25 was felt all over the island, and in our seismograph cellar was a peculiar swaying movement that set all the instruments jiggling, dismantled recording pens, and produced a queer feeling that the building was floating like a boat in a whirlpool. Immediately came word that North Kona had suffered heavily, particularly at Puuwaawaa Ranch near the cone of that name, where the 1859 Mauna Loa flow had swept past.

I motored at once with Mrs. Jaggar to Puuwaawaa, where we were hospitably entertained by the family of Mrs. Robert Hind. The damage all about was fantastic, with houses pulled apart, stone walls flung down in a seaward direction, redwood water tanks wrecked, and shops on the lower side of the highway moved toward the sea leaving a chasm between them and the road. Resting in our bedroom, we could hear the window frames ticking like clocks for long periods of time, then coming to a sudden wrench which felt as though a lifting wave had passed through the mountain under us.

I returned to the Observatory to get a shock recorder for use at the ranch porch to count these strong motion shocks. Meanwhile residents in Kona

jotted down times of the shocks, which were coming by hundreds. On October 5 at about 6 P.M., as I was returning through North Kona in my car, I noticed a little unexplained excitement among people by the roadside. I stopped at the residence of Frank Greenwell, whose wife was a faithful counter of quakes, to find Mrs. Greenwell and her daughter on the veranda in tears. They had just been through fearful earthquakes, which in a moving car I had not felt. Flower vases were overturned, furniture was disarranged, dishes were flung off the dining room table, and kitchen utensils and milk were in a jumble. It was hard to believe that anything so terrific could have happened without my feeling it.

I found even more dire catastrophe at Puuwaawaa. The stone chimney was overturned, breakage of china and of glass in the preserve closet in the basement was severe, a stone bench was flung down and broken on the lawn, and one side of the cellar was caved in. We took to living in automobiles, for there had been land slips on the mountain. This earthquake had been worse than that of September 25. Even hillside cottages were split apart.

I set up the shock recorder, which registered about 3,000 earthquakes during the next three months, until mid-December. The intensity and frequency of these quakes declined, as is usual with aftershocks of a big earthquake, recalling 1868 and the south end of the island. At that time both Mauna Loa and Kilauea had had rift outflows, and as the seismographic center of the new earthquakes was close to the 1800 and 1859 flows from Hualalai and Mauna Loa, everybody expected a lava flow; but none came. Armine von Tempski who was a visitor during this period was inspired to write "Lava." She added a Hualalai lava flow using material that I gave her to describe it. Her description is magnificent although she, herself, had never seen a lava flow.

The October 5 shock was bad on the west flank of Mauna Kea, where water tanks were overturned and the high wireless station was damaged, and at Kamuela, where plumbing pipes were fractured. Parker Ranch was damaged, and the constant racking along the entire length of the Kona settlements caused land slips and broken masonry in many places, always damaging north-south stone fences more than those at right angles to the seashore.

This three months of northwest earthquakes, a condition unknown since 1801, the year when Hualalai lava flowed into the sea, indicated that lava was coming north of Mauna Loa. This had not happened since 1899, for the flows on the southwest rift, always beginning near the summit crater, had been during 1903, 1907, 1914, 1916, 1919, and 1926.

Belief was that the southwestern rift of the mountain was filling progressively higher with solidified redhot cement, not brittle enough to fracture open easily, whereas the northern rifts—such as the sources of 1859, 1881, and 1899—were now hard and brittle and ready for fracture. The fracturing took the form of northwest cracking and this was lava wedging, confirmed by the summit and northern outflows which were to come in 1933 and 1935.

July of 1929 produced a new influx of lava into Halemaumau, nineteen degrees north of the equator. And a curiously simultaneous event occurred on nearly the same date 2,000 miles away at Tin Can Island (Niuafuou) in Tonga where the influx broke into basaltic eruption fifteen degrees south of the equator. Apparently a stress lagging behind the solstice time had acted on the equatorial protuberance to release the wedging open of lava fractures on both sides of the equator.

I was pleased when the U.S. Naval Observatory invited me to go to Niuafuou in 1930 as the geologist on an expedition going to study the total eclipse of the sun. The expedition, under Captain C. H. C. Keppler, used the Naval Station at Samoa as a base. Mrs. Jaggar accompanied me as far as Pago Pago and made trips to Western Samoa, Fiji, and the Tonga Islands. With other wives of expedition members, she was allowed to make a short visit to Tin Can Island at the time of the eclipse in October. Spending some time in Samoa, she listened to the Congressional hearings under Senator Hiram Bingham, which were to investigate civil versus naval government. We were delighted to find our old friend Captain Lincoln, of Tokyo earthquake relief, in command of the Navy at Samoa, and I also renewed acquaintance with the pilot of my companion plane in the Molokai forced landing of 1924, Lieutenant Bill Sinton, and his family, whom we were to meet again in Honolulu. Prominent on Captain Keppler's staff was Lieutenant-Commander Kellers, physician and naturalist, whose enterprise on the Niuafuou expedition, like mine, dealt with sciences other than astronomy.

From the sea, Niuafuou looks like a hat in shape. It is about five miles in diameter with eleven villages, mostly along the eastern shores, and at that time had a population of about a thousand people. In the center is a circular lake, bordered by cliffs, and much like Crater Lake in Oregon. Standing about seventy feet above sea level and 250 feet deep, it has slightly brackish water. The naval camp was established at Angaha on the north side of the island, and here a new village housed the refugees from Futu to the northwest, destroyed in 1929 by an aa lava flow. This flow came from erupting cracks trending north and south, along the west side of the ring ridge around the crater lake. These lava flows had been liquid pahoehoe at the source; had poured into the sea in many places; and had made striking tree molds around coconut palms, which were left as stone trees when the wood burned and the liquid lava lowered. The western source crack extends to the south end of the island and has accounted for most of the earlier eruptions known to history. Futu had been the only western settlement left.

Angaha came nearest to being a harbor, but was really on an open roadstead, with a rocky boat landing and copra chute below the village which stood on a cliff above.

Copra, the only commercial product, is bought and warehoused by two Australian firms. The two grown sons of the manager of one assisted me in tramping and photographing all over the island. The landing at Angaha brought about the name Tin Can Island, for the visiting steamers stopped a mile off shore and incoming mail, soldered into large biscuit tins by the steamship engineer, was lowered into the sea, tied together. The tins were towed in by the village policeman. Outgoing mail was carried out in paper packages tied on top of sticks and held aloft by hardy swimmers with hau wood poles, which they held under their arms as floats. A short time after our trip a shark got a swimmer, and canoes were adopted.

Thanks to the infrequent visits of vessels, the natives were unspoiled, splendid specimens of the Polynesian race. The laws of Tonga required every youth to cultivate an area of coconut trees and vegetables, and the island was traversed by lovely trails. The houses and churches were exquisite arched structures with thatched roofs, the beams tied with coconut-fiber cords. There were native ministers, and the choirs were superb. Services often started at 4 A.M.

My jobs were to take photographs with three cameras and make a geological map. Northeast of the crater lake is a cluster of sand hills, relics of an unusual explosive eruption in 1878, another Hawaiian eruption date. This eruption was confined to one side of the crater and came up the wall crack, between the encircling cliff and the top of the lava plug under the lake. Its description is very reminiscent of the Kilauea steamblasts of 1924.

We found a remarkable inhabitant of the sand in the malau bird, a small partridge with big feet, with which it dug a deep hole in the sand for its large egg which was then covered up. The sun's heat did the rest, with the warm sand acting as incubator. The young bird scratched its way to freedom and flight without aid from its mother. Another item of Dr. Kellers' natural history was the flying fox, a giant bat with a high singing note and odoriferous rookeries in the tops of trees. It had a heavy flight like an eagle's. A third item was the tiny black crab, the size of a ten-cent piece, which lived in the midst of limey flats at one side of the lake, where there were crusts that suggested calcareous algae. The little black crabs, which lived by thousands in the midst of the crust, resembled compact spiders.

An artificial feature of great convenience was a trail following the top of the ring ridge, all around the crater. The Quensell boys had a rowboat on the lake, and Dr. Kellers and I were guided by them to all parts of the island, making the acquaintance of the people in the villages along the eastern trade-wind shore. Just as in Hawaii, the trade wind is a controlling feature; and the surf erodes cliffs on the east, whereas beaches are more common along the lava flows of the western strand line. These are sheltered from wind but are remote from habitations. The entire island is made of lava and ash deposits, and is evidently the top of a volcano cone extending far below sea level. The lava activity, as shown by the arrangement of the old and new source cracks, depends on concentric cracking around the caldera, which makes concentric rifts, rather than the long radial ones found in Hawaii. The crack along the west side—which had vented the succession of flows from south to north, ending with the Futu flow of 1929—indicated that the next flow might threaten Angaha. This is just what happened during the next decade, forcing the island population to be evacuated.

My geology photographs and pictures of people, ships, and dwellings were developed in a darkroom tent, which I set up in a copra shed, so as to keep the development of negatives abreast of the exposures. Copra bugs

crawling over me in the dark and getting into developer added excitement, and the eternal smell of copra began to tinge my dreams.

The routine of our work was broken by two good fights, a fist fight between a Filipino steward and a sailor, and a knock-down and drag-out between two native women of Angaha. The real fun was the row between the two women. A younger woman who was a loose, shrill character, disliked by the villagers and the sailors, attempted to attack an older woman who was a big husky dame. There was screaming and hair pulling and fisticuffs, while the Navy men stood around and cheered them on. The younger woman made most of the noise, while the older woman laughed and ripped off the other's clothes. Finally the young woman, in tears and with clothing in tatters, retreated and disappeared.

But to get back to the eclipse, telescope lenses were mounted on high scaffolds, the ladies arrived in October, and the total eclipse of the sun happened and was photographed at the time anticipated.

When the time came for us to return to Samoa, some of us were fortunate enough to get a place on the Flood Brothers' copra ship *Carisso*, out of San Francisco. Along with the family of a Navy officer, we went ashore at Niuatoputapu (Keppel Island) after climbing down a rope ladder to a bobbing whale boat. We found beautiful mats, which are the wealth of the people throughout Tonga. The village men and women who had mats to sell were not so much interested in coins or trinkets and merchandise as they were in the clothing we wore. I literally divested myself of a shirt and a suit for a beautiful fringed mat ornamented with clusters of shells, made to be given to Queen Charlotte on her next visit. We were fortunate in reaching Samoa in good weather, but a big storm after we got there wrought havoc with the *Tanager* carrying astronomical photo plates and bundles of Polynesian mats which were much damaged by sea water.

CHAPTER VI

Prophecy and Hope

“For we know in part, and we prophesy in part.”

The fifth decade of my sixty years of geology, 1931 through 1940, was a time of culmination at Kilauea; the ending of an eleven-year cycle on Mauna Loa; and the introduction, in 1940, of a new Mauna Loa cycle. This new cycle resembled strikingly the one which followed 1843 because of the similarity of places—notably the north side of Mauna Loa toward Humuula, followed by the northeast side toward Hilo—and the intervals of eruption. Kilauea behaved differently in the nineteenth century, because in 1840 it rent open the east flank to make a flood of lava into the ocean, though afterwards it restored its lava to Halemaumau.

In 1934, on the other hand, Halemaumau went to sleep, after adding one more extra thick filling in the bottom of Halemaumau pit in September, when it gushed up behind wall slabs 300 feet high, cascading down the talus in twenty-five ribbons of lava. This proved that effervescence in a small crack can rise far above the level of the lava lake in the pit. It made a marvelous display in early morning darkness, and the new lava lake rose rapidly within the flatly funneling talus. This lay at thirty degrees, so that the outward spread enlarged the lake and reached beyond the foot of the talus.

The slide-rock slope that was conspicuous had been fed by avalanches, and it rested against the half-circle of wall slab, behind which had risen the cascades. This source migrated around the slab to the north and developed the biggest fountaining jets there. By the outward and upward spread of the new lake these jets became lake fountains, while the cascading ribbons at the slope stopped. The lake rose and crusted over. The northern fountains became a small oval pond and center of accumulation and upward doming, while pahoehoe lava radiated down a slope to the edges of the floor heap, south and east. Surveys at this time placed the northern lava pond definitely at the top of an inner heap. The fountains in the pond changed to conelets with their own craterlets. These, after a month, developed gas explosions,

flinging up lava shreds to 800 feet and sometimes higher than the edge of the pit.

This started a slumping of the exploding cones. The explosions were a symptom of increasing viscosity of lava under the bottom heap, and the viscosity was revealed in stiff lava welling up around the edges of the floor. This filled up the wall valley, and so compensated the slump that the bottom became level. Then the activity ceased. It all demonstrated how an inner dome in Halemaumau could become filled with an intrusive lens, which by welling out around the edges, could restore the dome to horizontality. It was like the “laccoliths” of the Black Hills. After the 1934 eruption, Kilauea simply went out of business for eighteen years. Halemaumau lava returned in 1952.

Mauna Loa activity was renewed meanwhile, with summit crater inflows in 1933 and with intense seismic activity under the northeast rift. Depths of seismic centers were at first seventeen miles down, and thereafter five miles down, as reported by seismologist Hugh Waesche. He worked with the formulae of distance, direction, and depth established by Austin Jones, using preliminary tremor, comparative excursions of lines, and a model of the island. These had become precise by mathematical triangulation of the island of Hawaii, with seismograph records from Kilauea, Hilo, and Kona stations. The distance from each station was interpreted from the duration of the preliminary tremor; and the meeting point of the several distances within the island model located the seismic focus inside Mauna Loa Mountain where the lava was splitting it open. The epicenter, or point over the focus, when the lava in Mauna Loa’s summit crater was stiffening, lay on the northeast rift in 1933. Therefore the eruption was expected at an old cone, whence had come the first outbreak of 1843. This came to pass in 1935.

E. G. Wingate, who had become superintendent of Hawaii National Park, agreed with my dictum, made on the basis of seismograms and history, that the next outflow would come at the north within two years and would endanger Hilo. This I discussed at a public meeting of the Hilo Chamber of Commerce in January 1934, and the report was published under the title “The coming lava flow.” The prediction was fulfilled in December 1935, when the flow came as it had in 1843. The eruption broke out on top and

traveled down to Humuula, the saddle between Mauna Loa and Mauna Kea, then pooled in the saddle and turned toward Hilo.



25. Jaggar in office of Observatory in "Tin House," 1937



26. Bomb bursting on lava flow, December 27, 1935. Photo by Eleventh Photo Section, A.C., Wheeler Field, T.H.

The 1843 flow had reached the saddle and turned toward Kona, and the solid remnant of that lava bank deflected the 1935 puddle to the east. It was traveling toward Hilo at the rate of a mile a day. This was the signal to try stopping it by bombing from airplanes, a procedure which had been proposed from experience with flows in tunnels of their own crust, where a person on the Kilauea floor could look through a caved-in hole in the roof and see the glowing river inside. Thurston and I had discussed blasting such

a roof to cool off the lava and pile it up, thus forcing it to a new outlet and stopping the frontal flow. It was Guido Giacometti of Olaa who suggested bombing rather than dynamiting. I called on the Army Air Force, and a conference was held in Hilo. With Colonel Delos C. Emmons, Wing Commander, I flew over the source tunnel. This was at 9,000 feet on the north side of Mauna Loa, where a gleaming silvery ribbon of pahoehoe emerged from a hole in the north slope. This was a crusted lava river, and the fliers were instructed to smash it with 600-pound demolition bombs of TNT.

The forenoon of December 27 was fixed for the bombing; and by invitation of Herbert Shipman, Mrs. Jaggar and I went to Puu Oo Ranch on Mauna Kea to watch what happened. The day was clear, and I saw one explosion send up a column of incandescent liquid lava hundreds of feet high, looking like a geyser of blood. In the foreground was the front of the flow, which we watched as it moved toward Hilo. At the same time we were receiving reports from cowboys on its rapidly diminishing speed. For about a week the liquid lava remaining in the tunnels kept spilling forward, and then it stopped. The front was in the headwaters of the Wailuku River, Hilo's water supply.

We afterwards visited the bomb craters in the source region, to find that there had been numerous hits on the lava tunnel and that the cooling off had solidified the source lava back into the mountain rift. The remainder of the eruption expended itself with internal fountaining in the summit wells at the top end of the flank rift. From the coincidence of the times of bombing and the slowing down of frontal flow, there appeared no question that the smashing of the source tunnel was effective and had saved Hilo. We had not anticipated that active fountaining would be forced back to the summit well from the 9,000-foot craterlet, but summit smoke continuing for two months verified that this had happened. This showed the physical chemistry of bubbling slag to be in delicate adjustment and a lava eruption once started to be more sensitive to shock than anyone had dreamed. This conclusion was reconfirmed by the bombing of the 1942 flow.

During this period, changes in Observatory personnel led to new researches. Wingate, who succeeded Wilson as engineer, set up triangulation monuments in Puna to test further motion on the Kapoho rift of 1924. He also devised and set up three tilt instruments in three cellars

which were blasted out of the lava around Halemaumau pit. Howard Powers came from Harvard as petrologist and collected and mapped many rock specimens in Kona, on Hualalai, and in Olaa. He also made curves of the tilt records for the first twenty years of the Observatory. Hugh Waesche was transferred from the Park Service to the position of geologist at the Observatory. A skilled radio amateur, he took over seismological work. In 1938 he dealt with an important group of earthquakes along the Chain of Craters east of Kilauea. These were accompanied by faulting, which made cracks, chasms, and humps in the road, and some new hot places. This indicated a reaction underground, back toward Halemaumau from the submarine outflow of April 1924.

Finch from his headquarters at Lassen reported regularly in the *Volcano Letter*, on hot spring temperatures and earthquakes. He conducted two expeditions to Alaska, inspecting the seismographs and making volcano explorations and maps on Akutan Island. Another expedition was to Shishaldin Volcano, at the west end of the big Aleutian island of Unimak during one of its eruptive spells.

Throughout this time and earlier H. T. Stearns represented the Geological Survey and the Territory of Hawaii in publications on geology and water supply on all the islands. The island of Hawaii was made the subject of a splendid geological map in color by Stearns and Gordon Macdonald, petrographer, with a book on the geological history of Hawaii, profusely illustrated with photographs and diagrams. Their book is practically a modern textbook on the geology of active lava volcanoes.

Richmond Hodges, sent by the Geological Survey from Washington, was trained in the technique of government filing and relieved me of work with correspondence and routine. He also took over the editing of the *Volcano Letter* and assisted Mr. Wilson with the writing of articles when I was away in Alaska. My secretaries after Hodges were Ruth Baker and Sutejiro Sato, and Miss Baker's work extended into the 1940's.

Tilt studies made at the three cellars around the rim of Halemaumau did not produce the anticipated results, but they answered our questions. The three cellars were placed at 120 degrees to each other, with reference to a meridian crossing the pit, one at the north, one east-southeast and one west-southwest.

It was thought, when these tiltscopes were set up, that the Kilauea floor would swell or shrink as an inner dome, with the pit at its center. But nothing of the kind was revealed. The tilting was found to be more or less at right angles to the long western wall of Kilauea Crater, itself an extension from the southwestern rift of Kilauea Mountain. The rift extends under Halemaumau pit, as was proved in 1920, when the Kau Desert outflows from the rift cracks kept pace with the lowering of Halemaumau lava. This means that the ring of Halemaumau's rock wall is in two pieces, divided by the rift dikes trending northeast-southwest, and that the tilting over upward pressure from below is not radial but is northwest and southeast. Wilson's leveling results that showed the whole mountain swelling up were based on isolated benchmarks relative to sea level, and this swelling was probably unsymmetrical, just as the southwest rift and the eastern rift of the Chain of Craters make a bend in plan and are unsymmetrical. The mountain is not a uniform elliptical dome.

I have said that the decade of the 1930's was a time of culmination for Kilauea. It was also a period of financial depression and stress for all of us. The Volcano House burned down, the new hotel was placed on the Observatory site, and the Observatory administration barely survived. The Hawaiian Volcano Research Association did much to keep the Observatory alive, but one year we all went on half pay. By dint of this half-pay episode and because everybody insisted that volcano records must not be permitted to lapse, the Secretary of the Interior transferred the Observatory in 1935 to the better financed National Park Service.

With Wingate as superintendent of Hawaii National Park, we were assured of loyal support and were able to combine scientific aims with National Park activities. Thus, the Volcano Observatory regained its status. We were also assisted by the publication of the economic success of the Mauna Loa bombing, in face of the threat to Hilo which involved some 51 million dollars of buildings and harbor. This threat Wingate and I studied carefully in the light of history, and we succeeded in getting \$10,000 from Congress for an investigation by U.S. Engineers of the possibility of a construction to protect Hilo from a disastrous lava flow. Colonel Bermel appointed civil engineer Belcher to Hilo, and Belcher worked for a year in 1938 on my design of a lava diversion channel and earthworks, to extend for seven miles from the Wailuku River gorge above Hilo to the airport.

This was to take care of another such lava flow as that of 1881 by deflecting it with the natural valleys southward from the congested district. A critical design was made of the channel, the height of the obstruction, and the openings needed for waterways and public roads. The plan was not to block the passage of lava, but merely to deflect it by means of an artificial barrier to channel it downhill. This would send it along the natural grades, diagonally forcing a lava stream away from the business district, the harbor, the factories, and the airport.

The design was approved by a reviewing board in Washington as effective for the purpose intended. However, with this project went a redesign of Hilo breakwater and a plan for dredging the harbor which took into consideration the possibility of a severe tidal wave. Unfortunately the appropriation estimate was considered too large and was turned down in Washington. When the great tidal wave came in 1946 it proved that such an extended breakwater attached to the northern shore of Hilo harbor would have lessened the terrible destruction and loss of life.

A diversion in the lives of Mrs. Jaggar and myself was an invitation in 1936 from the Royal Society of London, to go to Montserrat in the West Indies where for three years they had been having bad earthquakes. Sir Gerald Lenox-Conyngham, whom we had met at the Japan congress, wrote me asking for my help because Montserrat's dormant hot volcano was making excessive hydrogen sulfide gas at its two solfataras. The smell sickened and alarmed the inhabitants of the port of Plymouth, and the gas was blackening the paint of white steamships. The earthquakes had come in spasms culminating in big damage to masonry from 1934 onward. Perret had flown over from Martinique and tried to help by applying sound theories to prediction of seasonal tidal controls of the volcano, but he was scoffed at as a voodoo soothsayer by a British Navy captain. The scientific commission appointed was Dr. C. F. Powell of Bristol, now Nobel Prize physicist, and Dr. A. G. MacGregor of the Geological Survey, besides Dr. Lenox-Conyngham, formerly Director of the Geodetic Survey of India. Dr. Powell used adaptations of my shock recorder, both horizontal and vertical, built by the Kew Observatory. Designs had been obtained from instruments I sent to Dr. Marsden in New Zealand, after the Napier earthquake.

When I received the invitation to go to Montserrat, I packed up such instruments as I could find, and telephoned Mrs. Jaggar in Honolulu to be

ready to go with me to Los Angeles the following Saturday. She was always ready to act as secretary on a new adventure, and with much bustle and scramble we packed her things. Later I joined her at the steamer, a Danish freighter which was to take us through the Canal to the Caribbean. Boarding as we did on such short notice, we were given a steward's room in the bowels of the ship; but we had the run of the first cabin. It was a delightful trip through Panama and Jamaica, both of which I was happy to see again, twenty-six years after my 1910 experience with the canal engineers. Great changes had been wrought, and it was a thrill to see the ship pulled through step-up after step-up of canal locks, by the "iron mules" of that marvelous machinery.

We left the delightful freighter people at Charlotte Amalie in the Virgin Islands where we stayed at Blue Beard's Castle. After a wait of some days, we got a small Dutch island freighter to go to Montserrat. We stopped at St. Martin, an astonishing place, French at one end and Dutch at the other, with practically no custom house to mark the boundary, though the wines and the language changed in the middle of the island.

Saba is a startling extinct volcano rising as a steep rocky cone directly from the water, with no harbor but a stop opposite a gully that leads up to the crater. After landing in small boats, we climbed up the gulch to the settlement, a picturesque place, with masonry houses and many flowers, where the government is Dutch but all talk English, and its history goes back to the buccaneers. The village is on a flat in the lowest part of a cup crater, the top of our climb, but the name of the settlement is The Bottoms.

Our little ship joined the main line of the leeward volcanoes at St. Kitts, where we made connections for Antigua and Montserrat. In Montserrat we stayed with Miss Gillie at the Rainbow House and joined the Englishman Powell and the Scot MacGregor. I met Perret at Antigua, and we compared notes on the similarity of the earthquakes and the rotten-egg smell (sulfuretted hydrogen) at Montserrat to the eruptions of Pelée in Martinique, where these phenomena were followed by explosions and lava. The Montserrat authorities justly feared what was coming.

Perret had for two years kept track of events at Montserrat in relation to equinox and solstice. He had built a hut there at the dangerous solfatara near town, had made an instrument shelter with a thermograph, and on a

pedestal close to a nearby residence had set up an ingenious earthquake accumulator, which had recorded at the end of twenty-four hours the total expenditure of seismic energy in each direction. As there were hundreds of strong shocks, the instruments recorded total seismic energy per day and its dominant direction.

I found that Powell had set up my shock recorders among volunteers on the island, and a seismograph at the agricultural station. A new form of the Jaggard shock recorder had the weight attached to horizontal flat springs so as to oscillate up and down. I was especially pleased with the earthquake records kept by a Mr. English living in the countryside. Assisted by his wife, he had carefully listed the times and intensities of hundreds of shocks, with notes on important events.

Much help was furnished by the Agricultural Experiment Station, which provided an assistant to take us to many geologic places and to the second solfatara, consisting of hot springs and sulfur in a southern valley of the volcano. The volcano of Montserrat is at the south end of the island, while the northern part consists of older hills. The summit crater is a remote and inaccessible forested area among peaks. The volcano is much like Pelée in size and appearance.

We were allowed to take a steamer to St. Vincent and Barbados, stopping at Dominica. There the Governor kindly entertained us for a few hours, sending the government launch and driving us up the valley on a fête day when the negro women were all in picturesque costume. We saw his summer place with lovely gardens. We had tea with his wife, and I discussed with him the earthquake problem. On the drive we saw a remarkable cliff of hexagonal columns, some of them curved like a fan, representing the old lavas of Dominica.

The administrative problems of the British islands involved not only hurricanes and earthquakes, but tactful handling of the dominant negro, Carib Indian, and mulatto population, which is very ticklish, for there have been riots and labor troubles. I was astonished in several of the islands to learn that distinguished Englishmen in government and planter classes were partly colored. In the society club of Montserrat we met a leading lawyer who was coal black, and we saw London-educated negroes dancing with

English girls. We found the same customs in St. Vincent, and to a much lesser extent, in Barbados.

In St. Vincent Mr. Abbot, MacDonald's secretary, took us to see my old friend T. M. MacDonald the planter, at Chateau Belair on the west side of Soufrière, where Hovey, Curtis, and I had climbed in 1902. We traveled up the west coast by automobile, and saw one of the primitive sugar mills, where the juice is boiled down to a syrup to be shipped to lumber mills in Canada. Nothing could be in greater contrast to the modern sugar factories in Hawaii, and the negro labor gives the industry an entirely different aspect. To get to Chateau Belair we had to motor up a canyon far into the interior, around hairpin turns over vertical cliffs and along a narrow ridge, and then return to shore on the other side of the valley. We rode along the beach under the volcano, and saw the rehabilitated Richmond plantation, with the west flank of Soufrière Volcano under heavy clouds. Owing to torrents of rain, we had to make part of the return to Kingstown in a rowboat.

Later we drove over an excellent road up the east shore to Georgetown, and beyond that on the foot of the volcano slope, where a group of plantations had been purchased after the eruption of 1902 by Mr. Barnard, who with his charming wife, entertained us. Hundreds of acres of coconut trees, arrowroot, and sugar cane had replaced the utter devastation of 1902. Barnard showed us a modern still for making rum from sugar cane, and I was astonished to see that the product is just as clear as alcohol, the rum color being artificial. We rode horseback most of the way to the crater of Soufrière, over a trail through forests and across streams, very different from hiking in horrible desolation and fog up bare ridges covered with volcanic bombs, such as Hovey and Curtis and I had encountered on this same slope at the time of the eruptions.

The trail still followed knife-edge divides with perilous slopes on both sides of the path, but now concealed with mountain growth. We rode nearly to the edge of the crater, now a very different picture, with a large lake only a few hundred feet below, as it had been before the eruption of 1902. Two sturdy native women coming from Chateau Belair appeared with baskets of fruit on their heads, tramping a 3,000 foot height to deliver their goods to Georgetown on the east side of the island. This is an old story for these straight-backed natives, and these treks across mountains were equally

characteristic of the creoles in Martinique and the northern islands. These people would spend the night near their market on the opposite side of the islands.

In Kingstown we were shown the elaborate process by which arrowroot is made into edible starch, the powdery product being critically graded by delicate shades of color. This corm, which makes inconspicuous fields of low growing pointed *Canna* leaves and small white flowers, is quite different from the cassava, or manioc, which I had known on my first visits to the West Indies. Arrowroot has been developed by the agricultural experiment stations of the British, who for many years searched for a new commercial product. The St. Vincent arrowroot is now a major industry which has spread to the other islands and is cultivated by small planters.

In the volcano islands I interviewed government people to call attention to the crisis in Montserrat, using it as an illustration of the need at the numerous vents for the development of observatory methods, particularly in geology, chemistry, oceanography, and seismology, including measurements of ground surface movements and tilt. I had recommended this for Martinique and St. Vincent in 1902; and Perret, with some support by the French government, had gone to live in St. Pierre and make a museum, stimulated by the Pelée outbreak of 1929. So far as geophysics is concerned, the governments of St. Vincent and Jamaica have gone to sleep since the volcano disaster of 1902 and the earthquake building reforms of 1907. It is discouraging to a scientist to know that the science of economic geophysics and geography in such a magnificent field as the West Indian volcanoes has to be awakened by such disasters as were now occurring in Montserrat, with no forecasting at all. The whole Montserrat episode was like our unforeseen Hualalai earthquakes of 1929, and in both places my shock recorder was called in to help.

We went on to Barbados, a flat non-seismic land, where in 1902 I had interviewed the *Roraima* victims. We returned by way of St. Lucia, where we drove to the solfatara, which as usual is in a valley with sulfur and hot springs, near sea level, and not in a crater.

We returned to Montserrat, where the earthquakes and bad gases had died down after 1936. The investigations of the Commission (Lenox-Conyngham made his visit after I left), came to publication in Powell and

MacGregor's reports on the seismic analysis and the geology. I sent in a report with photographs and charts on the whole chain of volcanoes, in relation to the Montserrat crisis, by comparison with other volcanoes. Lenox-Conyngham wrote an article for *Nature*. MacGregor later published a critical analysis of modern data on the probabilities of eruption in all of the West Indian volcanoes. Perret published a large monograph on Montserrat, illustrated with his beautiful photographs.

We passed Martinique by sea, and I saw the huge pile of lava the 1929 eruption had added, to make an entirely new summit to Mount Pelée. Vegetation and habitation had reappeared at St. Pierre, but the mountain was bare.

We returned to Hawaii by way of Bermuda, Boston, and Washington, where the temperature was hotter than we had felt in the tropics. Reviewing the journey, I was encouraged to perceive that geology had changed a great deal since the struggle that Hovey and I, after our experience at Mount Pelée, had had to make geological societies realize that changes in the field must be constantly measured. The real obstacles to getting field measurements permanently manned as pure science are lack of money and the fashions of education. Perret and I have been two isolated enthusiasts crying in the wilderness.

Any young scientist with photographic skill who will give his life to living with and reporting upon a single volcano group can make a great contribution to science. He must have suitable financial backers and a publication agency and instruments not dependent upon frequent eruptions. What volcano science needs most is permanent dwellers, using all the resources of sensitive geophysics and chemistry and dwelling close to craters or solfataras. Such lands as the Taupo District of New Zealand are ideal, but not when observed at a distance. Wairaki is now under investigation for commercial power. Hilo is being critically examined for a lava diversion scheme. But these projects are not what I mean, and are not pure science. The personal devotion of a lifetime, as in the cases of Pasteur or Schweitzer, is what produces the emergent evolution of true science.

I have called this chapter Prophecy and Hope because of six fruitful prognostications and hope for the future of volcanology. Of the prognostications, one was the threat to Hilo which came true in 1934. Two,

the forecast to the effect that bombing would stop a lava flow came true. Three, the belief that a volcano observatory would be productive of instruments came true. Four, the prediction of danger to Hilo produced definite defensive plans by U.S. Engineers. Five, predictions of time and place of Mauna Loa outbreaks, seismically and historically proved practical. Six, the prediction of Kilauea sinking lava, based on sinking at Mauna Loa, had repeatedly been fulfilled.

When my government service as Volcanologist ended in 1940 and R. H. Finch had been appointed my successor, substantial recognition of the Observatory had come from Washington, New Zealand, and Great Britain. Great help had come from Presidents Arthur L. Dean and David L. Crawford of the University of Hawaii in Honolulu, and new assistance came from President Gregg M. Sinclair. This was to lead to my employment by the University as Research Associate in 1940. Thus I was to continue, during the next decade, the publishing of Volcano Observatory results.

Over and over again Hawaiian volcanology demonstrated the need of advertisement, occasionally reaching such men as Everett Morss, M.I.T. trustee in Boston; Lorin Thurston, business leader in Honolulu; Henderson, Washington financier, for our borings; and Cramton, leader of Congress. The Volcano Research Association in Honolulu is a devoted group of businessmen keeping up a small fund of \$6,000 per annum, trivial compared to the big laboratories of commerce and astronomy. A pure science of volcanology, with world-wide laboratories is now needed to catch the eye and ear of imaginative men of business. Friedlaender in Naples, Perret on Mount Pelée, and Omori in Tokyo almost created enough imaginative stimulus to real exploration of volcanoes and of the inner earth. They were battered down by natural catastrophe and by wars.

The 1940's were enriched by three good friends Vern Hinkley, Stanley Porteus, and Frank Rieber; respectively journalist, psychologist, and physicist-inventor. They all took a keen interest in my writing and mechanical inventions, and Hinkley assisted in the Observatory work during the explosive eruption and wrote "that was the top experience of my newspaper career."

Hinkley, who had edited the *Hilo Tribune Herald*, became managing editor of the *Honolulu Star-Bulletin* and published a series of my radio

addresses on Kilauea. He also sent his photographer to photograph our laboratories, thereby keeping the public informed about volcano study. And he worked up a history of my navy monographs and hardness testing instruments. He was a lovable fellow whose publicity instinct was a great asset to volcano science. He did not think of a volcano as something sensational, but remained moderate about it and informed his public accurately. Through him, the Volcano Observatory reports came to be accepted as desirable routine, and he was elected a director of the Volcano Research Association. His many friends were desolated by his early and sudden death.

Porteus is an Australian man of science who conducted expeditions among the Australian blacks and the primitive Africans of Kalihari and specialized in the mental outlook of primitive peoples. He devised a famous maze for intelligence tests. He has published numerous books about Hawaii and several novels, including "Restless voyage," the life of Archibald Campbell, who lived with Kamehameha the great and survived amputation of both legs.

With Guido Giacometti, who suggested airplane bombing of the volcano lava flows, Porteus and I foregathered at the crater frequently to discuss the constitution of earth interior. Porteus differed with my belief on the evolution of mind as a mutation of evolution. Like Hinkley he became a member of the Board of Directors of our Research Association. He is a judge of the juvenile court, skilled in curing delinquency. Porteus is a world thinker, who agrees with me in thinking of altruism as a form of energy. Porteus invented the title of this present book.

Rieber started from the University of California where he became interested in making an echo from underground strata to locate oil. He moved to Los Angeles, where his father was a professor of classical languages and a college dean. Frank invented a complex recording seismograph carried on a motor truck, wherewith he set off explosive bombs and registered echo earthquakes from every important underground layer. These layers identified oil-bearing strata, so that the marks on a revolving drum practically mapped a section underground for a guide to oil drilling. He moved to New York and established war inventions, among them phonograph disks for repeating whole conferences of many talkers. He founded Geovision Ltd., a company which greatly abbreviated the

scanning of echoes for subterranean mapping. Then he died suddenly, like Hinkley, in the full flower of a brilliant mind. Rieber and I corresponded for years on invention gadgets, comparing notes by letters, and meeting all too rarely. To me he was one of our most productive physicists, always inspiring. He was convinced that discovery of petroleum will endlessly increase and will become automatic. He and I looked downward into the shell of the globe.

This decade I devoted primarily to writing and publication, some of the writing voluminous and still unpublished. In 1941 I moved into an office in Hawaii Hall of the University of Hawaii in Honolulu. My paper work consisted primarily in completing, revising, and illustrating a memoir on "Origin and development of craters," in cooperation with the Geological Society of America. The censor chosen by the Society was Dr. Howel Williams of the University of California, who cordially endorsed the book.

The Society subscribed \$350 from its Penrose Fund to assist with drafting and clerical work on the substantial results of our observations of Hawaiian craters in the twentieth century. The groundwork had long been laid, for beginning under Alexander Agassiz at the Museum of Comparative Zoology in Cambridge and during my visit to Vesuvius in 1906, I planned a book on volcanology. Later, in 1910 after careful study of the work of Dana, Hitchcock, and Brigham on Hawaiian volcanoes, I started analysis of Kilauea Volcano in the nineteenth century. Thus this one large volume with photogravures, maps, and diagrams covers the history of observations and conclusions from Hawaiian Volcano Observatory work for thirty years.

My thesis is that there must be some order in time and space for what is obviously 1,700 miles of submarine volcanic upbuilding in the Hawaiian chain. Active volcanoes are hot and erupting in Hawaii; sunken ones are covered with coral at Midway Island; and intermediate ones, half coral and half lava, are in the middle of the chain. Disregarding the ocean water, all of these are gigantic mountains below sea level. On the island of Hawaii I found symmetry, which I called "The cross of Hawaii" in an address to the Honolulu Chamber of Commerce in 1912. I noted that Mauna Kea forms the top of a cross on the map; the upright extends along the southwest rift of Mauna Loa, and two symmetrical curved arms extend to Hualalai summit and Kilauea summit. The lava flows from Mauna Loa north and south arrange themselves symmetrically about this design, with every evidence

that Mauna Loa dome was piled up in a spoon underlaid by Hualalai, Mauna Kea, and Kilauea. It is obvious on the map that Mauna Loa upbuilding was obstructed by grandpa Mauna Kea and that it has been forced off to the southwest by the two daughters, to build the elongate point of the island. Kilauea is old on the Haleakala, Kohala, Kea line; and Hualalai is old on a right angle line at Kea.

From my training in physiography under W. M. Davis of Harvard, I was convinced when I first saw Hawaii and studied the books about it that downward faulting toward the sea bottom, of sliding island blocks, is conspicuous. It shows in the V-shaped fracture of Haleakala Crater, a broken sector, and in the straight fracture of the north half of Molokai volcano, leaving the mighty cliffs there. It shows in the eastern half of Kohala volcano leaving the fault facets and hanging valleys of Waimanu, and in the Mohokea embayment of the southeast end of Mauna Loa. The embayment shows evidence of the breakdown of an ancient crater as described by Hitchcock. Moreover, Kilauea, Wood Valley, Mohokea, and Waiohinu amphitheater are four old calderas of faulting in a line. This seemed to me confirmed by the down-faulted steps of the southeast side of Kilauea Mountain, and the observed down-breaking there of the shoreline during earthquakes. This, in 1868, drowned coconut trees below the sea and caused big earthquakes on a submerged fault in 1868 and 1952.

Such action was further confirmed by our experience of a down-faulted block during earthquakes at Kapoho in April 1924, before Halemaumau exploded, confirming the view that the active volcanoes break downward in slices along shorelines, even when they swell upward around craters. Harold Stearns always combatted the idea of faulting and made Mohokea, Haleakala, and Waipio erosion forms; but this I cannot accept.

The logic to the effect that in the long and large the old volcanoes from Hawaii to Midway have been on slices of the earth's crust faulted downward below sea level through the ages seems incontrovertible. The fault planes are diagonals across the main volcanic rift trend and make the channels between the islands at an angle in plan to the trend of the island chain. These channels are very deep. All of this philosophy developed in my mind before I came to Hawaii.

Also I thought that the origin of life might have been from volcanic gas, owing to the prominence of carbon dioxide, water vapor, hydrogen, sulfur, and nitrogen, all ingredients of both protein and volcanoes. I put this up to R. T. Jackson, who taught me phylogeny when I was studying fossils and he was studying genetics. Knowing the sulfurous quality of an egg yolk, I asked him if it wasn't possible that as evolution goes back behind the embryo, we should find volcanic traces chemically. Phylogeny means that the history of the embryo reenacts the history of the race, and I merely extended this back to the inorganic. I was laughed at for carrying biological origin back to gases of volcanoes; but Shepherd and I collected gases from flaming Kilauea lava, and found the five elemental constituents: carbon, oxygen, nitrogen, hydrogen, and sulfur. These also make up the aminoacids of protein, so my philosophy of origin still seemed to me to be reasonable. Volcanoes erupted through the ocean, and life came out of the unexplored deeps of the sea.

Thus in 1910 I began a book on craters which came to fruition in a Memoir of the Geological Society. This was not published until 1947, but I was working on it, drawing the diagrams, dictating the typescript to Sato, and selecting for illustration the best of our photographs during the thirties.

One of the diagrams shows eleven-year cycles, beginning with 1790 and ending with 1935. I adopted this after finding in Hitchcock a tabulation for Halemaumau, indicating big lowerings of lava in 1790, 1823, 1855, and 1891, to which we added 1924 from our own experience. These were approximately thirty-three years apart, as I found when I plotted the data on a curve of Hitchcock's table. Taking other major sinkings as punctuation points—such as the outflows and collapse of Halemaumau in 1832, 1840, 1868, and 1931—there developed a correspondence in the subsidence times treated as repose periods, with the years having the least numbers of sunspots at average intervals of 11.1 years. The intervening times of maxima of sunspots all occurred in the intermediate times of rising lava.

The curve as a whole from 1823 to 1924 shows a notable crest from 1855 to 1890, and a crest of the greatest volume of Mauna Loa gushing occurred between 1855 and 1877. Stearns and Macdonald object to this diagram as not showing all the little intermediate events, but what I have taken are the actual peaks and depressions above sea level and those which correspond to the sunspot interval of 11.1 years. This is an average even for

sunspots, which had long intervals at the beginning of the nineteenth century, a time when no reports were made for Kilauea.

I have guessed a drop of Kilauea lava as dating from about A. D. 1800, corresponding to the notable expulsion of Mauna Loa lava through Hualalai, and an imaginary unreported lowering eleven years thereafter, as it is improbable the island was wholly dead in the first twenty years of the century. The explosive eruptions of 1790 certainly produced a big collapse at Kilauea.

My faith in this diagram is based on the fact that our own eruption sinkings at eleven year intervals (1902, 1913, 1924, and 1935) agree so well with an eleven-year theory that we are justified in looking backward for eleven year averages. Perret has found intervals of about a decade for Vesuvius. All my experience of Hawaiian lava leads to the belief, shown by our lava tides and several short-term diagrams, that rhythmic periods of a volcanic system are related to gravitational control of the sun and moon. There are rhythmic controls of the globe by the gravitational control of the sun and moon. There are rhythmic controls of the globe by the sun, and rhythmic controls of very deep volcanic cracks by the globe, and rhythmic controls of individual groups of volcanoes by the long volcanic chains over cracks. Our experimental data are limited by the little groups of volcanoes, and so the big rhythmic movements seem inaccessible to science, mostly because we have no record of relationships of single volcanoes 500 miles apart in such a place as Alaska.

We raise no question about night and day or about the oceanic tides or about the moon's phases. We know there is a rock tide in the earth, that there is a hot earth core of about 2200° Centigrade which appears to seismology to be a very massive liquid 1,800 miles down. Gravitation is the controlling force of the solar system, the galaxy and the universe, and it works by rhythms, from the orbits of the planets in years, to the outermost spiral nebulae in millions of centuries. We are ourselves controlled by it in locomotion and in the circulation of the blood. Therefore to think of volcanoes as anything but periodic and gravitational in their relation to the globe would, to me, make the science of volcanoes entirely uninteresting. All science lives on rhythmic action.

A second manuscript entitled “Steamblast eruptions,” was based on Mount Pelée in Martinique and a comparison with the 1924 steamblast of Kilauea. This last had conclusively shown outflow under the sea, and inflow of groundwater, to change lava surging to blasts from a steam boiler. A paper published in 1940 was a study of the gas collections from flaming basalt on Kilauea and Mauna Loa, made by E. S. Shepherd and me. In this I plotted curves of relative excellence of collection in relation to the amount of the volcanic gases, in contrast to the non-volcanic aqueous and oceanic gases. These latter, notably water vapor, decreased in proportion to the manipulative excellence of the handling of vacuum tubes; and the volcanic gases increased, notably hydrogen and the carbon gases. This convinced me that the deep gas of volcanoes is hydrogen, associated with carbon dioxide and nitrogen.

In this decade, too, war brought new demands on my time and experience and had its effect on the Kilauea Observatory. Major James Snedeker of the Marine Corps, legal officer for the Commanding General in Honolulu, having heard of our experience with motorcar amphibians, told me that the Pacific Ocean war would depend on amphibian landing craft. And a letter from Admiral Bloch urged me to send to the Navy details of our experience with amphibians. As this involved geology of beaches around the Pacific Ocean, I set to work on twelve monographs for the Navy dealing with the mechanism of amphibians and the problems they posed on beaches in Hawaii, Puget Sound, and Alaska. Other subjects about which I supplied information were the inflammability of Japanese buildings in the Tokyo earthquake, the handling of earthquake and volcano catastrophes and our material from journals on many places of volcanic danger in the Pacific.

Then W. H. Hammond, physicist in charge of a testing laboratory at Pearl Harbor, suggested that I revive my 1897–1908 testing of steel for abrasion hardness, later continued by Boynton, for his laboratory of the Navy. Thus I started hardness testing at the University and carried it on for ten years. I used diamond and other abrasives in instruments to show directly on a dial the rate of wear of metals or minerals under standardized conditions, with a constant and reproducible motor tool. Abrasion hardness turned out to be as tricky a problem as my range finders and shock recorders. This activity brought together in the University laboratory and in the laboratory at Hawaii National Park many records, manuscripts, and

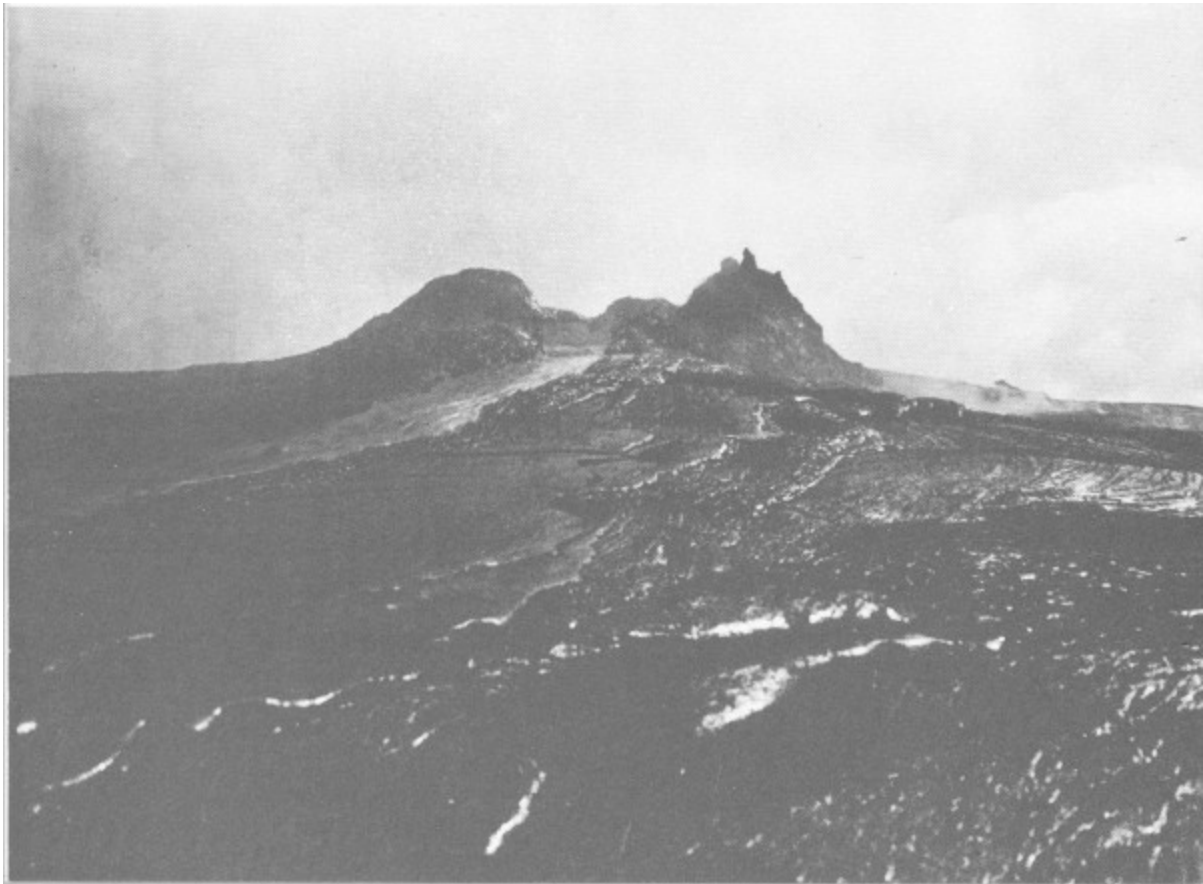
specimens. Ruth Baker, who succeeded Sato as secretary, did valiant work sorting out materials from many expeditions which had been dumped in disorder because of war and fire at the Kilauea Observatory. Though the Park had built a new house for naturalists, and for the seismographs, shops, and records, it was taken over by the Commanding General on Hawaii, imposing considerable hardship on Finch and his assistants. One assistant was Burton Loucks, instrument maker, who married Miss Baker. Another, Austin Jones the seismologist, was transferred to care for seismographs set up to measure faulting and tilt around Boulder Dam. Dr. Howard Powers, after work for the Geological Survey and the Territory on the island of Maui, joined Jones eventually to enter into a new section of volcanology, established in Denver under the Geological Survey, especially to assist the Army and Navy studies of Aleutian volcanic eruptions, wherefrom harbors and airfields were sometimes endangered.



27. Fountain in Halemaumau lava lake, May 23, 1917



28. Rare dome fountain during eruption of Kilauea Crater, March 20, 1921



29. *Lava stream issuing from a spatter cone near rim of Halemaumau, February 9, 1921*

Three events of volcanic and seismic importance to Hawaii during the 1940's were the eruptions of Mauna Loa in 1940 and 1942 and the 1946 tidal wave caused by a submarine earthquake south of Unalaska. The wave engulfed the wharves and shorefronts of Hilo and eastern Maui and caused considerable damage elsewhere.

We were familiar with the recording by our seismographs of earthquake centers under the sea of Alaska and Japan, and with the interval of hours that followed before dangerous water waves reached Hawaiian shores. We had also had a bad tidal wave in Kona, originating off Japan; and two or three such waves which damaged Kahului and Hilo had originated in big submarine earthquakes off the Alaskan Peninsula. The Japanese fishermen, from our published warnings, always took their sampans to deep water, and the Navy had instructed me to let them know right away if the seismographs recorded a distant earthquake capable of making a tidal wave.

I earlier had had one unhappy experience with warning the Navy, when we registered a seismogram of a big earthquake in Alaska, which if submarine, would send us a tidal wave. I notified Pearl Harbor of the probable time of arrival of the wave, should the quake be submarine. It happened a big Army and Navy dinner party at Waikiki was set for just that time, but orders went out calling officers back to their posts and the party was disrupted. No tidal wave came, as the earthquake proved to be on the mainland of Alaska. The newspapers unmercifully jeered at me, but the Commanding Admiral told me not to change my policy.

The 1946 wave was very large and the water rose in pulsations until it swept away the railroad bridge and washed out the whole waterfront of Hilo. The earthquake seismogram came at 2 A.M. when no one was watching, and the water wave at 8 A.M. came just when the Observatory workers went on duty. When the flood of ocean destroyed the Hilo breakwater and leaped over it to damage the principal wharves, many people were drowned. Considerable damage was done on Oahu and Maui. The disaster came when Dr. F. P. Shepard, oceanographer of La Jolla, was occupying a summer cottage on the north shore of Oahu; and he was delighted to experience a big tidal wave. Collaborating with geologists in Hawaii, Shepard compiled a most thorough report on height of waves in all bays of the Territory. Seismographs and tide gauges got to work all around the Pacific Ocean, the place on the sea bottom which had jolted was exactly located, and the Coast Survey and Navy started far-reaching precautions for predicting against future combinations of earthquake and water. This included seismographs that ring alarm bells at night. The object of science is always prediction and assisting humanity; and the need is always for more men.

Another significant event of 1947 was the visit of Hans Pettersson of the Oceanographic Institute of Sweden who was conducting an expedition which followed the path of the *Challenger*. The object of the project was to study the oceanography of the sea bottom around the equator. Thus Pettersson was enthusiastic about my paper in *Natural History* and its emphasis on studying sea bottoms. With him was inventor Kullenberg who had made a device for boring into the mud of the sea bottom and taking longer cores than had been dug previously. His apparatus consisted of a core barrel, tripped with valves close to the sea bottom under a heavy

weight, which would allow it to sink sixty feet in suitable bottom ooze while the core rose inside the pipe without being compressed.

Pettersson had a skilled staff consisting of biologist, physicist, chemist, and geologist; and they had laboratories on board the *Albatross* for study of the collected bottom materials. They also took echo data of explosions near sea bottom, giving depths of soft materials over hard rock. This place of transition was found to be shallower under the Pacific Ocean than under the Atlantic. They discovered hard lava flows in many places between Tahiti and Hawaii and under the Indian Ocean, indicating extensive submarine volcanic eruption. An attempt was made to measure the temperature of a core, and this suggested that the bottom of the boring was warmer than the top, meaning a thermal gradient of sea bottom. A core of volcanic agglomerate was obtained in the deep trench opposite the East Indies.

It was during this period that President Gregg Sinclair of the University of Hawaii urged a plan for geophysics of the Pacific, and Professor R. W. Hiatt of that institution succeeded in advancing interest in organic oceanography. I wrote an appeal, based on such work as that of Pettersson, Perret, and others urging the Regents of the University to plan a large geophysical institute in Hawaii, to make a science of the rock bottom of the Pacific Ocean.

Thousands of soundings made in the Gulf of Alaska and in the central Pacific had shown seamounts, or guyots, shaped like high volcanoes on the sea floor, some of them with flat tops, but having characteristics of ancient isolated volcanoes. New soundings revealed mountain ranges on the sea floor, probably volcanic, one of them right across the middle of the Hawaiian chain. No one had yet discovered fiery eruption in deep water, but oceanographers were beginning to use boring machines, cameras, electric lights, and devices for determining radioactivity of the muds. As sea bottom occupies three-quarters of the globe, it is inconceivable, when compared with the continents, that it has no hot solfataras, hot springs, and hot volcanoes. In fact, we know some of the latter in shallow water. It is only a question of scientific organization to locate the sources of Pettersson's submarine lava flows. President Sinclair took to the chiefs of the Rockefeller and Carnegie Foundations a proposal for a five million dollar Geophysical Institute at the University of Hawaii, to utilize the advantages of its central Pacific position.

As for my own experiments, my Department of Volcanology at the University was moved into a concrete basement room a thousand square feet in area in the Home Economics building, and the expense was shared with the Hawaiian Volcano Research Association. Here I had office and shop and collections of the Research Association, and the assistance of a secretary and a junior researcher who is an instrument maker. Thus were assembled in a fire resistant location my petrographic and mineral collections from Europe, the Caribbean, Central America, and the Pacific lands, together with manuscripts from my days of Harvard and Massachusetts Tech to the middle of the century and classified accumulations of my Navy monographs, lantern slides, negatives, photographs, maps, drawings, correspondence, and instruments, including material obtained by the Research Association for experiments still continuing on the hardness of minerals.

One objective of this hardness measurement was an instrument for machine shops which would give in half a minute the length of a standard scratch made by a standard dental disk of silicon carbide. I called this the "Jaggar Scratch Tester" and Mr. Paul Rushforth, a Honolulu optician, made improved models of the instrument. When a book was published on the experiments with some three hundred woods, minerals, metals, and plastics, Dr. Grodzinski of the commercial diamond establishments in London became interested and reproduced the paper in a review dealing with industrial diamonds, which have become of great importance in the world of grinding machinery. This made a new contact with England, similar to that made by Boynton with my microsclerometer in 1908, when he applied it to the microscopic constituents of steel under the British Iron and Steel Institute. I sent a copy of my new report to the Pearl Harbor industrial laboratory, along with one of the instruments. Endorsers of this report were Mr. W. H. Hammond and Dr. Earl Ingerson, director of the mineral laboratories of the U.S. Geological Survey.

A result of the experiments on hardness is the knowledge that the important quality is softness, or abrasability, and speed of removal of material in any uniform mechanical cutting process. It was formerly thought that the big intervals in values were between the hard substances. It turns out that the biggest gaps in value are in soft substances like coals and clays

and plasters. Hardness is purely a negative quality of resistance, and measurements are of yielding, not of resisting.

Other experiments on which I worked dealt with location of the Zenith in the sky for quick determination of latitude and longitude from stars and telescopic studies of the moon, an old hobby of my master, Shaler. I have long been convinced that Kilauea lava resembles moon lava in the craters it builds, and my special interest is that Mauna Loa and Kilauea build structures of basalt, small and large, which are earth experiments imitating the moon on a smaller scale. The astronomers say their field is the stars, the geologists must explain the moon. As a matter of fact, one geologist has made a start. My classmate J. E. Spurr, who after retirement to Florida from work as U.S. Geological Survey geologist among the faults and lavas of the far West, published books on the comparison of the moon with geology. In view of increasing attempts to explain moon craters by impact (Baldwin), I feel that experienced volcanologists should also take a hand in moon science. Larger arcs of circles on the globe, the Aleutian Islands for instance, resemble moon features and are deeply volcanic. Furthermore, magnificent detailed photographs of the moon from modern telescopes are available to volcanology.

I spent my summers at Hawaii National Park, becoming consulting geophysicist. Dr. Chester K. Wentworth of the Board of Water Supply became geologist. The laboratories were extended to a seismograph station seven miles up the northeast flank of Mauna Loa, but operation of the original cellar adjacent to the Volcano House was continued. A basement under the Natural History building of the Park held seismographs, Finch's office and library, and Loucks' shop.

In 1948 Observatory work was returned to the administration of the Geological Survey, and a volcanologic branch in Denver took over Dr. Powers to make airplane studies of the Aleutian Islands. This was under Mr. Walter Frederick Hunt, in charge of geology, U.S. Geological Survey.

When Hawaii National Park was reorganized, Frank Oberhansley, superintendent, the Natural History building was adopted as Park Headquarters, and the Uwekahuna buildings, with their magnificent view in all directions, were reconstructed as the Hawaiian Volcano Observatory. A new seismograph cellar was dug, away from disturbances of Uwekahuna

cliff, and modern instruments were installed. Mr. John Forbes became assistant machinist; and on Mr. Finch's retirement in 1951, Dr. Gordon Macdonald became volcanologist in charge. C. K. Wentworth moved from Honolulu to the National Park region and took charge of magnetic measurement, which had been established at numerous stations by physicists of the Geological Survey. During past decades physicists and chemists had visited the Observatory, among them Dr. Stanley Ballard, who equipped the laboratories with a Gaertner spectrograph; Dr. Harvey White of Berkeley, who found no radioactivity in Hawaiian lavas; and Dr. J. J. Naughton, who found a critical isotope of carbon in the emanations of Sulphur Bank. Modern chemistry was beginning to be applied to volcanology in the field, and this was what Hovey, Perret, and I hoped for fifty years ago. So much for dry facts of organization.

In 1949 the summit crater of Mauna Loa erupted, with fracture and outflow of its south end toward Kona. This was followed in 1950 by lengthy rupture of the southwest rift with the most voluminous and rapid outflows of history, three of them going into the ocean and wreaking destruction in South Kona.

The sequence of these outflows was from high sources first, with others opening farther south, and the most conspicuous flows following the steep Kona slope into the ocean, beginning at Hookena. Macdonald and National Park naturalists photographed and recorded everything. The old Hookena post office on the upper road at the home of the veteran Mr. Lincoln was carried away, and this occasioned much drama, for the old man didn't wish to leave his home. The next house destroyed, an old landmark, was the Magoon Ranch. The third was the attractive and modern Ohia Lodge, a resort built of native logs in the wilderness.

A separate large flow forked away from the rift, to the eastern side of the mountain, reaching the lowest landward elevation in the forest of Kahuku, and short flows spilled over the southwest rift on the east side.

Several persons approached the flows in South Kona from the ocean. The early photographs of the first flows, where the hard sprouts and boulders of stiff aa partially cooled entered the ocean, showed big columns of vapor from contact with sea water. Not so with the third flow farthest south, explored from a canoe by Jack Matsumoto and a companion,

equipped with motion picture cameras. The pictures were good color photographs, and the torrent of lava flowed down a steep bank of its own substance, hemmed in by hardened ridges at the sides, the stream intensely liquid and flowing directly into the ocean.

The result was most remarkable. The yellow liquid lava went into the salt water without making any column of steam at all; the sea bottom simply received it with its rush downward, the water boiling superhot, and the lava taking the water vapor into itself. The phenomenon was not due to the rising of dry steam, for there was no condensation cloud above. Scientists explained it by assuming a shell of lava making a tunnel under the ocean, with the crust ending just at sea level.

Such a submarine arch was definitely not present, for the waves surged back and forth and, Matsumoto states, there was no sign of a submerged reef. The motion picture bears this out. What was probably going on is what happens to slag in a patented process of the steel mills, where the glow liquid is flowed over a perforated surface emitting hundreds of water jets, and the melt at 1300° Centigrade absorbs the water without making visible steam. The slag turns into a myriad of microscopic glassy spheres, becoming a kind of pumice. A peculiarity of this substance is that if it is cooled at 700° Centigrade it will pass a critical point and give up the absorbed water with explosive effects. It seems likely that Matsumoto's golden torrent sweeping into the ocean was so excessively hot that it took up the water and continued to flow down the sea bottom as a water-charged product. The snapping and crackling effects, and the submarine earthquakes, making localized tidal waves such as those noted in 1919 when such a torrent entered the sea, may be due to the explosive cooling when the slag gives up its water.

The use of color motion pictures is one of the many improvements owed to modern science, and the mapping of lava flows by airplane photography. This gave Macdonald a new weapon for surveying the volume accumulation at the time of the 1950 outflows, for from air photographs he got exact outlines of the flows. These, checked against calculated thicknesses, gave him volumes which could be compared with volumes of older flows proportionate to areas. These calculations showed that nothing since 1868 has yielded such large volumes of lava, per days of outflow.

Mrs. Jaggar and I were returning to Hawaii from a trip to Nova Scotia, and the Matson steamer *Lurline* took us to the Kona coast toward the end of the 1950 eruption, for inspection of the glowing flows late at night. They looked like hot coals extending far up the mountainside under the clouds, with occasional bright flares where trees burst into flame. Visible motion there was none, as we were too late for the rapid flowing and too far away to see detailed motion. This eruption resembled the voluminous flow of Mauna Loa in 1868, from a low vent at the south end of the mountain, and lasted only a short time after preliminary summit outbursts. The similarity was a big earthquake series, and this was to happen again in Kona in 1951. The cataclysmal opening of the southwest rift in the nineteenth century eruption followed a quarter century of northern outflows, those from 1843 to 1859. Next came those from 1929 to 1952 in the twentieth century. The 1929 earthquakes subterraneously began the northern series.

The same argument applies to the twenty-six years of summit and southern outflows, from 1903 to 1929, which followed a quarter century of alternations north and south. None of this takes account of all the summit crater outbreaks, the hinge line between the jostlings of the north and south rift sectors. Roughly the whole argument centers about a supposed rocking of the Mauna Loa mountain sectors, northward and southward from the crater. The two rifts become stiff and seal up for twenty-five years, and then break open for a new period of looseness. The summit well is somehow full always.

A remarkable event, namely repose of Kilauea for eighteen years after 1934, may be another reaction. The previous excitement was the buildup, collapse, and recovery of the mountain for the quarter century preceding 1934, with its culmination the steam blast in 1924 of underground water, the dormancy of Kilauea beginning ten years later. Kilauea in 1790 had a bigger explosive eruption, and was in repose for eighteen years beginning ten years thereafter, namely in 1800. Thus it seems likely that Kilauea executes quarter centuries of crisis in its own right. These times are not exact, but are approximations of scientific search for order in a big machine, the Hawaiian volcanic system, where rhythmic pulsations exist wherever gravity operates. A third of a century may prove more exact than the estimate of a quarter century.

The end of this 1940 decade completes a half century of my experience of volcanoes and earthquakes, dwelling with a single crater, and learning that volcanoes and earthquakes are tied together. They appear tied to deep ruptures 2,000 miles long, in the thick shell of the earth over a white hot liquid core.

I have recently started an experiment with a thick globe of cement, made with a shell, proportional in thickness to the earth's crust, which is 1,800 miles deep, as all seismologists agree. Striking this shell with a sledge hammer, I find it breaks in straight lines at right angles to each other. Theory is bound to be influenced by the observational answers derived from watching lava emerging from the mountain rifts, at the end of the long straight belt of rifts of the whole Hawaiian chain.

I continually review my own geological muddles, the controversies over steam, flames, volcano swelling, explosion craters, layers in the crust, weighting and underflowing, continental uplift, the globe's armor plate, contraction wrinkling of basins of sediment, submarine volcanoes, linear chains of volcanoes, siliceous shell, blocks lifted or sunk, planets solar or from the sun's binary twin, original heat or radioactive heat, thick crust or thin shell, lava reservoirs or lava core, pregeology ancestors of volcanoes, and craters on the moon. The only way to calculate from observations on Hawaiian volcanoes is to copy the mathematicians; namely, to guess at the answers.

CHAPTER VII

Envoi

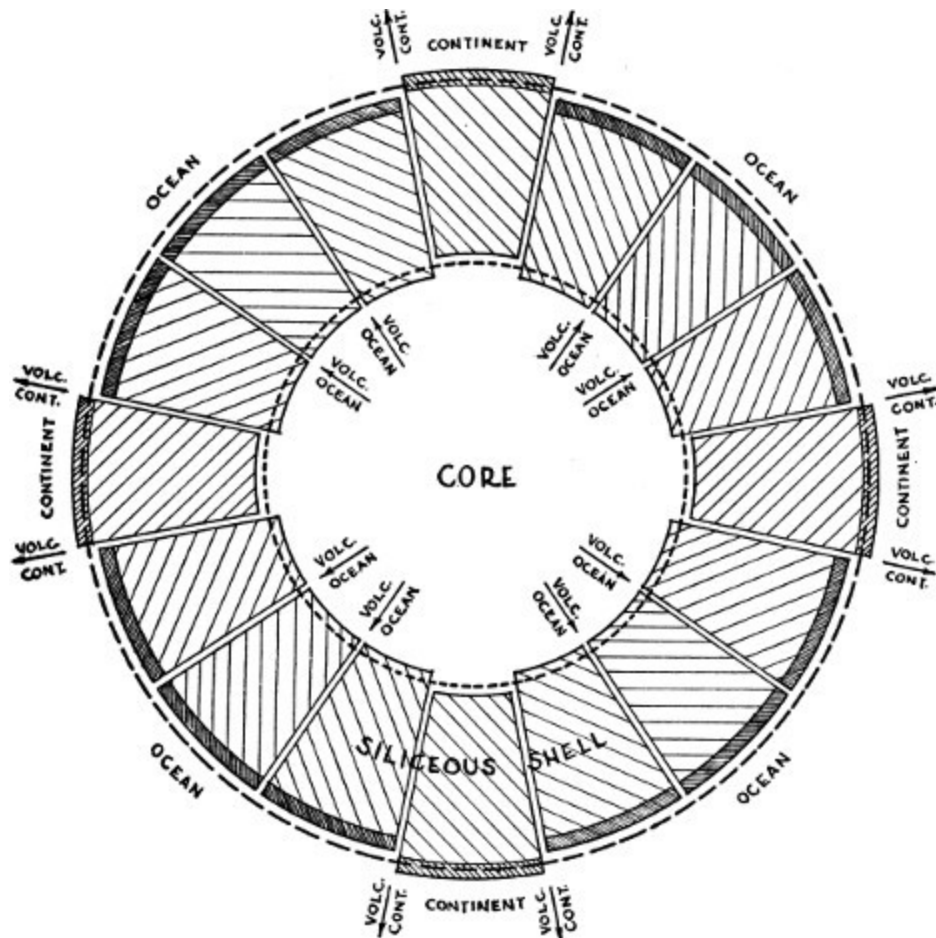
*“Tho’ world on world in myriad myriads roll round us
Each with different powers and other forms of life than ours.”*

I have spent sixty years in qualitative experiments in geology. I began with old volcanoes and geysers in the Yellowstone and the far west, and ended with experiments on the active Hawaiian volcano, Kilauea. Based on these experiments, I have written books about evolution of craters, and about distinct peculiarities of explosive eruptions from underground water.

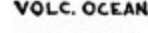
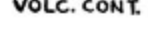
The accusation that I am not orthodox in professional geology is false. Professional geology is largely continental because its field work has been on continents. My work has been oceanic; my field, seventy percent of the earth’s surface, extending over a thick crust down to the earth core. The earth core is fluid and massive and hot, as all geology agrees. Isostasy, which postulates a thin flexible shell, is violated by the ocean deeps and the volcanic ridges. Volcanic rift echelons like the Cordillera and the Hawaiian ridge are too long to be generated as the fracture of a crust fifty miles deep. The circularity and graduated size in linear stretches of the Pacific arcs are functions of a fractured thick-shell sphere. Similar gradation of arcs is on the lunar surface. Arguments, based on the knowledge of meteors, for an iron core and for large lunar craters are without analogy. Substratum theories, from Stübel to Daly, do not agree with oceanic volcanism. Gravitational crust balance applies better to a primitive thick fault block crust than to a thin shell. Earth lavas, as natural experimental models, imitate lunar features on small and large scales. Both make consistent history for two similar globes. Volcanology has to stand as global and ancient, and any geologist may accept the reasonings here enumerated without being unorthodox.

The unquestioned certainties of modern seismology, the transmission of elastic waves through the globe to sensitive recording pendulums, are that the crust is 1,800 miles thick, that the core is a heavy ball of white hot fluid, and that its temperature at crustal contact has been estimated by Verhoogen

at 2200° Centigrade. The deep crust is less dense than the core, and is commonly conceded to be basic heavy rock not unlike stony meteorites. The outside shell under oceans, and over three-quarters of the earth, is covered by basaltic lava, and wherever igneous rock has been formed by volcanic action, intrusive or extrusive black basic lava recurs as dikes and outflows.



LEGEND

-  ARMOR PLATE AT OCEANS
-  ARMOR PLATE AT CONTINENTS
-  EXTERIOR OF FUNDAMENTAL GLOBE TO WHICH ADJUSTMENT TENDS
-  CORE LIMIT TO WHICH ADJUSTMENT TENDS
-  VOLC. OCEAN CHAINS OF OCEAN VOLCANOS
-  VOLC. CONT. CHAINS OF CONTINENTAL BORDER VOLCANOS

ARMOR PLATE AT OCEANS

ARMOR PLATE AT CONTINENTS

EXTERIOR OF FUNDAMENTAL GLOBE TO WHICH ADJUSTMENT TENDS

CORE LIMIT TO WHICH ADJUSTMENT TENDS

VOLC. OCEAN CHAINS OF OCEAN VOLCANOES

VOLC. CONT. CHAINS OF CONTINENTAL BORDER VOLCANOES

ON THE PAGE OPPOSITE *is a diagram of a hypothetical globe section near the equator, showing oceans and continents in true surface ratio; fault block segments of rigid crust isostatically supported on a liquid core; sixteen volcanic partitions, oceanic and continental; and Stübel's "armorplate" from pristine volcanic eruption. Possibly the profile is tetrahedral. My argument for this globe section is based on the following:*

A globe of core, siliceous shell, and armorplate was formed by primitive volcanic eruptions.

The shell resulted from external aggregation of solids and gases and internal segregation about a molten core.

The fault blocks came from shrinkage of the shell over a liquid core, adjusted by luni-solar gravitation and rotation through the pre-geologic ages.

The continental and oceanic boundaries of the fault blocks were determined by elevated and sunken blocks with core volcanism of escaping gas melting walls and laying down an exterior siliceous armorplate on the earliest solidified globe. This in continents is the seismologists' lighter exterior layer underlaid by denser rock at the armorplate bottom.

Continental volcanism (VOLC.-CONT.) became differentiated from oceanic volcanism, by light atmospheric pressure over the raised blocks and much greater water pressure over three-quarters of the earth, the sunken blocks.

The subdivision is represented in the diagrammatic section on the globe by three-quarters of the section being ocean, namely twelve-sixteenths.

The section shows twelve-sixteenths as sunken blocks, four-sixteenths as raised blocks. The four sixteenths by the tetrahedral hypothesis of Lowthian Green and Michel-Lévy make the four continental protuberances.

Twelve-sixteenths of the surface is broken by fundamental rifts of irregular shapes, some of them north-south, controlled by centrifugal stresses and corresponding to the north-south deeps

and heaps and known rifts. These have persisted since the first volcanism of primitive time.

Circum-continental volcanism is represented on the diagram by VOLC.-CONT., oceanic volcanism is represented by VOLC.-OCEAN. Both are shown as interblock rifts, adjusted through the ages (exaggerated on the drawing) and always tensional over expansional core pressure, with exothermal heating agencies.

The sixteen fundamental block boundaries correspond approximately to sixteen fundamental volcanic fault blocks known vaguely on the globe. Something similar is known on the moon. The rifts are the boundaries of sixteen blocks, some polyhedral, some elongate. Some are oceanic like New Zealand-Tonga, some are ancient and continental like Arabia. The imperfectly mapped ocean deeps are boundary lines. The rifts of Africa and Chile-Patagonia are boundary lines. The great arcs of Himalaya, Java-Sumatra, and Aleutian ridges are boundary lines of circular blocks. Possibly they were circular calderas of engulfment on the primitive spheroid. The edge of Mare Imbrium on the moon shows fault rifts. The straight alignment of lunar calderas hints at moon rifts under an unmapped mosaic. The blocks of the theory of continental drift, are guesses at a mosaic of crust blocks. But the possibility drift theory omitted is that the blocks are deep. Except for the Lowthian Green, Wegener, Holmes, and Daly speculations, based on thin crust blocks of continents, no mapping of the shell mosaic exists. It is not feasible until we map the detail of ocean bottoms. Primitive blocks require acceptance of a thick crust and justify new speculation. The cracks between blocks are the volcanic partitions of the earth, which I call ignisepts.

There are many points for speculation, some of them subject to mathematical inquiry. Does surface water penetrate the partitions? Is it high pressure and saline under oceans? How do deep earthquakes stem from friction 300 miles down under the Cordillera and the west Pacific? Do earth and moon spheres as rounded tetrahedra crack similarly? Because of rotation are

north-south cracks dominant? Are the core fluids changing volcanism through the ages?

The twenty-eight percent of earth surface which lies above the sea in continents is made up of siliceous sediments of shallow water basins, with quartz as the dominant mineral, their strata wrinkled, and eroded into mountain ranges. Desert and lake or river bottoms make up most of the remainder. This material, when ancient, was changed by heat and infiltration into what are called gneisses, schists, and granites; and the process of granitization is among the metamorphic processes. It is a process of deep burial, heat, gases, and water which has always been a puzzle, and may affect ancient volcanic lavas wherever they have covered the land. It is a process of solution of silica, and its deposition is by steam and other vapors.

In the same way volcanic action by the outpouring of lava through cracks is a process of solution of the deeper crust of the earth by hot gases, largely burning hydrogen. Lavas emerging from Etna or Mauna Loa are melted earth crust, dissolved and brought up by this same hydrogen and by other gases from the walls of profound cracks leading down to the earth core. Volcanism and metamorphism are thus the same process, namely the action of gases up cracks through deep earth crust. But metamorphism acts on continental sediments, whereas modern volcano eruption acts through sea bottom and sea shore faulting, very ancient features of the earth and distinct from continents. In Hawaii no metamorphic rock fragments have been found.

Such primitive oceanic fault fissures extend under continents remnant from the time of evolution of continents. They bring up the metamorphic hot gases, which in siliceous sediments, make granites and gneisses and schists with the aid of groundwater. Geology has no knowledge whatever of whether this metamorphic process affects the hard rock under the oceanic muds, because geology has never collected a piece of that rock. Geology however knows inclusions and explosive fragments from oceanic volcanoes, and it does not find there granite and gneiss and schist. However, generalization does not apply to continental volcanoes like those of Italy and Africa.

The beginning of fossils on continents is commonly considered to have been 500 million years ago, and this may be extended another 1,500 million years for the most ancient identifiable continental rocks, and an estimated total thickness of 120,000 feet to the bottom of the most ancient sediments on earth. We know nothing of thickness of most ancient volcanic deposits under the oceanic mud.

This brings us to the great German explorer Stübel, who mapped volcanoes of the Andes, founded a museum of his work in Leipzig, and published monographs on the Andes. He wrote a final book, including material on Mount Pelée, on the “genetic differences of volcanic mountains.” But such modern continentalists as Daly and Bucher in America have disregarded Stübel. Daly is the authority on a shallow earth shell and substratum of basalt, and Bucher of Columbia University is a specialist on continental sediments and granitization.

The point is that Stübel made a profound generalization which nobody has proved wrong. The earth is at least 3,000 million years old, and when oceanic fault blocks sank and received condensing atmospheric water and continental fault blocks remained high and became eroded, there was already a thick shell of volcanic lavas. For volcanism was the most ancient process on the earth's surface. It had always brought gases up cracks from the core, making atmosphere, water, and extrusions. Stübel, called the extrusive shell on the outside of the primitive crust the globe's armorplate. The primal gas escape, whatever the ancestral turbulence inside, had to come up cracks and make volcanic deposits. It is commonly presumed that the very thick inside crust formed rapidly by cooling and solidifying from outside the core inward, and from inside the atmosphere outward. The latter surface was eventually under water cooling over most of the earth and under air cooling over the small continental area, a marked difference of temperature and pressure for the two areas.

Seismometry teaches that most of the crust is of fairly uniform density. Therefore, presumably, a thick crust was arrived at early. There was obviously a time of conflict between the weighting of the crust by its heavier accumulations next to the core, by its lighter accumulations exteriorly under water and air, and finally by its external armor plate of unknown comparative weight, made of volcanic lava. For all we know, this

might have been volcanic pumice. Rapidity of crust thickening is speculative.

Right here there is an element of mystery in speculation as to which has to accommodate comparison with the moon, the merging of atmospheric condensation with volcanism, and the merging of suboceanic condensation of lava with pristine eruption. This is too hard a nut to crack, in our current ignorance of rock under sea bottom muds. But Stübel's insistence on a coating of lava armor plate over both continents and sea bottoms as the earlier volcanism, and an external veneer on the earth, is unavoidable. If it were all basalt like the present oceanic volcanoes, we should find basalt in continents underneath the granites. We do not do so. If it were all light weight granitizing by segregation of silica, we should find commonly granite and obsidian fragments within oceanic lavas. We do not do so. We have to conclude then that our sections, topographic and geologic, do not go deep enough. And as for the ocean bottoms, we have no sections at all. But Stübel was right. An unknown volcanic eruption period had to precede geologic volcanoes.

The question of ancient greenstones in Africa, Scandinavia, and Canada is much discussed, for there were old volcanic lavas in many places; mixed with gneisses, schists, and granites. They were not a deep layer, but presumed to be ancient remnants of interspersed lavas among sediments. They are one more evidence that volcanic eruption goes back to the time of the most ancient rocks on continents and that its lavas were affected by metamorphism. But no continuous deep stratum of greenstones is known. At depths of fifty miles, under continents only, is the Mohorovicic change to denser rock. This is an echo surface in earthquake waves, but it is absent over the whole Pacific. It may be the top of the armor plate.

Justice Holmes wrote that the Constitution of the United States was an experiment. That all law of the nation works salvation by prophecy based on experiment. The experiments were extended to the Bill of Rights and all the amendments to the Constitution. I feel that geology—in view of its extreme ignorance of submarine rocks, ores, metals, oils, spring waters, temperatures, magnetism, gravity, and gases for most of the earth—needs a bill of rights and numerous amendments to its constitution. Its salvation by prophecy needs to be based on experiments with instruments, drill rigs, and anchored laboratories in this vast area. These experiments, superficially,

have been conducted by oceanographic sampling of bottom materials, by gravity pendulums operated in submarines, by cameras on sea bottoms, and collections of bottom waters, by tests of radioactivity of bottom materials, by echo sounding to determine thickness of muds, by volcanology on oceanic islands, by topographic surveying of the bottom, and by all the excellent work of the oceanographic and geologic stations and their seismographs, with some studies of marine chemistry, physics, and biology. The conclusions in this book amount to only one small prophecy based on experiments with volcanoes. But the rock under deep ocean mud is still uncollected.

My volcano experiments are not influenced by any consensus of text books. I was educated on textbook opinions and found geologic science deficient in experimental measurement of the field progress of erosion, sedimentation, deformation, and eruption. I expended most of my teaching in a plea for field observatories of time measurement of these four processes. The plea has done some good, and in this century we have seen grow up the International Geophysical Union. Experiment stations have multiplied, to make geophysics and geochemistry pure quantitative sciences. But they are generally commercial and have not extended to deep boring under oceans.

While working from volcano observatories for the extension of geology in Alaska, Japan, Hawaii, Tonga, the Caribbean and Italy, and on the mainland of California, Central America, and New Zealand, I have found myself on the outskirts of vast oceans, engaged in a science almost as unsatisfactory as the textbook science of historical and continental geology. It is always a compromise, for we are up against a crying need for maps of the bedrock under the muds of the vast oceans. Volcanism cries out for a knowledge of the globe, and it is helped by such work as that of Gutenberg and Richter. These men compiled critical maps of earthquakes, measured by elastic theory the world over. Their work necessarily made many contacts with volcanoes. The same may be said of the geophysical summaries of gravity, magnetism, climatology, hydrology, and oceanography. But all our sciences stop at the immense sea bottoms, and need salvation through experiment.

Science is not doing all it can. Finances and engineering are competent to contact sea bottom directly with expensive machines not yet invented and

to create oceanic rock science. Offshore boring for oil is not enough. Pure science needs an example by financiers like Carnegie and Rockefeller who are not seeking profit. Engineering advice positively can reach under the few hundred feet of mud, find the rock, and bore into it in 2,000 fathoms. The first man who does it will open a new frontier. All honor to Shepard, Ewing, Piggot, Pettersson, and Kullenberg, men who have barely broken ground in this science. The whole of volcanology depends on collecting the crustal rock under the mud.

Hoyle's book "The Nature of the Universe" takes us one step farther. It shows that all science is essentially cosmology, and science deals with the origin and progress of all nature. I would go farther than the universe. I would include the science of life and of our brains. We need an imaginative picture starting with the outer universe. We end on the earth with volcanoes and the birth of life.

Hoyle and Lyttleton of Cambridge have presented a condensation of current astrophysics, which includes earth, moon, and planets; sun and stars; origin and future of stars; and origin of solar systems. A most gratifying conclusion is that the background material of space creates hydrogen. This is proved by precise mathematical equations. This accounts for the expanding universe under the pressure of such creation. The outermost nebulae continually pass beyond the speed of light. The galaxies move out into infinite space endlessly. They are renovated endlessly by gravitation from hydrogen eternally created.

The sun, by knowledge built up from the days of Jeans and Eddington, contains more than ninety percent of hydrogen, and the small remainder is helium, oxygen, nitrogen, carbon, and iron. It maintains its surface temperature by nuclear reactions from within outward, at a rate suitable to make helium out of hydrogen, so as to compensate for the energy which the sun radiates.

This dominance of hydrogen inside the solar star makes it impossible that the earth should be solar. Rather, it was a product of a companion star, a supernova which exploded and, with excessive heat, created elements atomically. The sun was a binary pair of stars, and the companion occupied the place of the four greater planets. The remnant body, after explosion, moved away.

A gaseous ring formed around the sun condensing from many molecules to rotating superplanets. These broke up many hundred million years ago into Jupiter, Saturn, Uranus, and Neptune. Small blobs escaped to become the inner planets including the earth. The earth captured small solids and acquired the moon as a satellite. It got radioactive matter exteriorly, plus nitrogen, water, oxygen, and carbon dioxide.

There is a hundred times more hydrogen per unit of mass in the sun than in the planets. Its supply will last for 50,000 million years. The solar system is tunneling through variable interstellar gas. It picks up more or less material, and so changes climates occasionally. This makes such episodes as the ice ages on earth. Lyttleton estimates that the dust clouds encountered form bundles of particles captured by the sun to make comets.

The mathematics of the interior of the sun, applied by Bethe to the use of carbon and nitrogen as catalysts and changing hydrogen to helium, is a model of experimentation. It should be imitated to explain Hawaiian basalt. The core of the earth produces gas reactions up cracks. The gases act on deep crust. The surface product is olivine basalt. What are the reactions between gas and crust to make Mauna Loa foam fountains? This problem has not been tackled. Geologists have clung to a theory of shallow reservoirs.

The astronomers of Cambridge, successors of the American experimenter George Ellery Hale and of Eddington and Jeans, are not the final word in cosmology. There will be a final word. The picture created from background material to gas, from gas to galaxies, and from galaxies to solar systems ends for us in our planet with a white hot liquid core. Nuclear reactions created this from the superheat of an exploding supernova. Our erupting volcanoes are the end product. We can sit beside erupting lava fountains and watch hydrogen flames, the same gas that was made of the background material in the universe.

All this is outcome of gravitation. It extends from the first eddies of hydrogen in outer space to the final rotation of the earth. The final hydrogen, with carbon, made life on the earth. The five elements of volcanic gas are identical with the five elements of organic chemistry. Dr. Hoyle mistakenly concludes that we have no clue to our own fate. But he points out that the universe is continuous creation. Our picture is one instant

of time in an everlasting now. Mind is an everlasting unit beyond which we cannot go.

It is illogical to pay any attention to existence after death unless we pay equal attention to existence before birth. All is continuous creation. The making of hydrogen is just as true within the creation of life as within the universe. Life is under gravitation. Gravitation controls the instantaneous moving picture, even the emergence of life from volcanic gases under enormous water pressure at sea bottom. It is just as much subject to experiment as the outer boundary of the universe.

Life is an end product; and it thinks, worships, and experiments. Treating life and volcanoes as end products of Hoyle's universe makes science fundamentally cosmology.

One final comment, after looking at sea bottom eruptions through all the ages. Continental life came out of the sea, and original life comes continually from the earth core. This gives new dignity to the future search for global action on the sea bottom.

The "emergent evolution" of Lloyd Morgan makes much of mutation as accounting for progress from unconscious life to consciousness, consciousness to memory, memory to reasoning, and reasoning to spirituality. Each one of these is a new mutation, in the same sense as a new fruit by Burbank. The first unconscious life may be considered a mutation from the inorganic of the globe. The totally unknown pressure-temperature conditions of volcanic eruption through the cracking earth of ocean bottom, and the ground waters under the ocean, lend a final dignity to exploration of that frontier.

Hoyle writes that the ultimate goal of the New Cosmology is continuous creation in outer space. The ultimate goal of the New Volcanology is continuous creation in oceanic depths.

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